

Indian Institute of Technology, Jodhpur

IMAT QUALIFIER - COMPILED QUESTION BANK

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Summary

- Total: **334 questions** across 5 qualifier papers (3 BS, 2 MBA Technology).
- By type: 306 multiple-choice (MCQ) + 28 Type-in-the-Answer (TITA).
- By section: VARC 120 | DILR 98 | QA 116.
- Marking (each paper): MCQ +3 correct / -1 wrong; TITA +3 / no negative; 2 hours per paper.

Each paper below retains its own cover page, marking scheme, questions with the correct option marked in green, and full worked solutions. Data-interpretation charts are embedded. Use the PDF bookmarks / outline panel to jump to any paper. The five papers contain no duplicate questions across them.

IIT Jodhpur Qualifier - BS

QUALIFIER EXAMINATION | Time: 2 Hours | Maximum Marks: 198

Question Distribution

- **VARC:** 24 questions (suggested time 40 min)
- **DILR:** 18 questions (suggested time 40 min)
- **QA:** 24 questions (suggested time 40 min)
- Total: 66 questions | MCQ 60 | TITA (non-MCQ) 6

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Total duration is 2 hours; each section has a suggested 40-minute window.
- Sections may be attempted in any order.

This is the answer-marked verification copy (correct option in green, with solution) and mirrors the content of the upload CSV.

Section: VARC (suggested 40 min)

Instructions (VARC-RC1): study the shared passage below and answer its 4 question(s).

For most of human history, the night sky was a shared inheritance: a band of stars visible to anyone who looked upward. That inheritance is quietly dissolving. The spread of electrified cities has scattered artificial light into the atmosphere, where dust and moisture scatter it further, producing a diffuse glow that astronomers call skyglow. Under this veil, the faint light of distant stars is simply drowned out. Surveys now suggest that a large share of people alive today have never seen the Milky Way from where they live, and that the brightness of the night sky over inhabited regions is rising by a few per cent each year.

The consequences reach well beyond stargazing. Many organisms time their behaviour by natural cycles of darkness. Migrating birds, which navigate partly by the stars, can be disoriented by illuminated buildings; insects exhaust themselves circling lamps; hatchling sea turtles, which instinctively crawl toward the brightest horizon, may turn inland toward roads instead of out to sea. In each case a cue that was reliable for millions of years has been overwritten by a human signal.

What makes light pollution unusual among environmental problems is how readily it can be reversed. Carbon dioxide lingers in the atmosphere for centuries; a poorly aimed streetlamp stops polluting the moment it is switched off or shielded. Lighting that points downward, that is no brighter than a task requires, and that is dimmed when no one is present can preserve both safety and darkness. The obstacle is not technical but a habit of mind that equates more light with more progress. Recovering the night sky, then, asks less for new invention than for a willingness to use, more thoughtfully, the light we already command.

Q1. [VARC-RC1 · mcsc · diff 0.5]

Which of the following best expresses the main idea of the passage?

- A) Artificial light has now made it completely impossible for most people on Earth to see any stars at all from any of the places where they currently live and work.
- B) The disorientation of migrating birds, circling insects, and stranded sea turtles is now the single most urgent environmental crisis facing the entire planet.
- C) Light pollution is a serious but unusually reversible problem, and addressing it depends more on changing our lighting habits than on inventing new technology. ✓
- D) Astronomers, rather than elected officials, should be given final authority over urban lighting decisions because they understand skyglow better than city planners.

Answer: C) Light pollution is a serious but unusually reversible problem, and addressing it depends more on changing our lighting habits than on inventing new technology.

Solution: The key captures both halves of the passage: light pollution is harmful yet uniquely reversible, and the fix is mainly a matter of changing habits rather than inventing new technology. The 'cannot see any stars at all' choice overstates the claim, since the passage says many cannot see the Milky Way, not all stars. The 'wildlife disorientation is the single most urgent crisis' choice elevates one example into the central thesis, which the passage never does. The 'astronomers should have final authority' choice introduces a policy recommendation the passage does not make.

Q2. [VARC-RC1 · mcsc · diff 0.5]

According to the passage, the author makes all of the following claims EXCEPT:

- A) Skyglow is produced partly because dust and moisture in the atmosphere scatter artificial light.
- B) Reducing light pollution requires expensive new lighting technologies that have not yet been invented. ✓
- C) Some animals rely on natural darkness or starlight as a behavioural cue that artificial light can disrupt.
- D) The brightness of the night sky over inhabited regions has been increasing year by year.

Answer: B) Reducing light pollution requires expensive new lighting technologies that have not yet been invented.

Solution: The claim that reducing light pollution needs expensive uninvented technology is the EXCEPT answer, since the passage argues the opposite: the fix needs not new invention but more thoughtful use of light already available. The statement about dust and moisture scattering light restates the stated mechanism of skyglow. The claim that some animals rely on natural darkness paraphrases the birds, insects and turtles examples. The statement that night-sky brightness has been rising directly matches the claim that it grows a few per cent each year.

Q3. [VARC-RC1 · mcsc · diff 0.5]

Why does the author contrast light pollution with carbon dioxide?

- A) To argue that carbon dioxide is, in the end, a far less serious threat to migrating wildlife than artificial light is.
- B) To suggest that the two problems are caused by exactly the same kind of polluting industrial activity in modern cities.
- C) To prove that most environmental problems generally cannot be solved at all without first achieving major scientific breakthroughs.
- D) To highlight that light pollution can be reversed almost at once, whereas carbon dioxide lingers for a very long time. ✓

Answer: D) To highlight that light pollution can be reversed almost at once, whereas carbon dioxide lingers for a very long time.

Solution: The 'reversed almost at once while carbon dioxide lingers' choice matches the explicit contrast: CO₂ lingers for centuries while a lamp stops polluting the instant it is shielded or switched off, illustrating reversibility. The 'carbon dioxide is less serious for wildlife' choice invents a ranking of threats not made in the text. The 'same polluting industrial activity' choice misreads the contrast as a shared cause. The 'breakthroughs needed to solve problems' choice contradicts the passage, which says the fix needs willingness, not breakthroughs.

Q4. [VARC-RC1 · mcsc · diff 0.5]

It can be inferred that the author would most likely agree with which of the following statements?

- A) A community could cut its light pollution without sacrificing safety by aiming lamps downward and dimming them when no one is around. ✓
- B) The loss of the visible Milky Way is, regrettably, an entirely unavoidable cost of building any modern, fully electrified society.
- C) Cities ought to abandon their street lighting almost entirely in order to recover the lost visibility of the dark and starry night sky.
- D) Concern about the fading night sky is mainly a private preoccupation of professional astronomers rather than a truly broad public issue.

Answer: A) A community could cut its light pollution without sacrificing safety by aiming lamps downward and dimming them when no one is around.

Solution: The 'cut light pollution without sacrificing safety' choice follows directly from the claim that downward-pointing, appropriately bright, motion-dimmed lighting can preserve both safety and darkness. The 'unavoidable cost of an electrified society' choice is too extreme, since the passage seeks thoughtful use, not abandonment, of light. The 'abandon street lighting almost entirely' choice contradicts the framing of the sky as a shared inheritance and the wildlife effects. The 'mainly a preoccupation of astronomers' choice conflicts with the central point that the loss is reversible and broadly relevant.

Instructions (VARC-RC2): study the shared passage below and answer its 4 question(s).

When a result in science is described as 'replicated,' it means that an independent team, following the same procedure, has obtained a similar finding. Replication is often called the gatekeeper of reliable knowledge, because a single study, however carefully done, can be misled by chance, by an unusual sample, or by an undetected flaw in its design. Only when a result reappears under different hands does the community gain confidence that it reflects something real rather than an accident of one laboratory.

In recent years, however, attempts to repeat well-known studies have frequently failed to reproduce the original effects. Several factors contribute. Researchers face strong incentives to publish striking, novel results, while careful repetitions of someone else's work attract little reward and are seldom published. Small samples can yield impressive-looking outcomes that vanish when the study is enlarged. And the freedom to analyse data in many ways allows a researcher, even unintentionally, to keep adjusting the analysis until a pattern emerges.

None of this means that science is broken. The very fact that the failures are being documented is itself a product of the scientific method turning its scrutiny on its own practices. Reforms are already spreading: some journals now invite researchers to register their hypotheses and planned analyses before collecting data, and commit to publishing the results whether or not they are striking. Others reserve space specifically for replication studies. What looks at first like a crisis of confidence may be better understood as a discipline learning, in public, to hold itself to its own standards.

Q5. [VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the central argument of the passage?

- A) Scientific journals are the primary cause of unreliable published findings and therefore should no longer be trusted at all by readers.
- B) Although many well-known studies have failed to replicate, this reflects science correcting itself rather than a breakdown of the method. ✓
- C) Small samples are, on their own, the single underlying reason that so many famous studies have proven difficult to reproduce reliably.
- D) Replication is essentially impossible in actual practice because no two separate laboratories ever follow exactly the same detailed procedure.

Answer: B) Although many well-known studies have failed to replicate, this reflects science correcting itself rather than a breakdown of the method.

Solution: The 'science correcting itself rather than a breakdown' choice captures the thesis: replication failures are real but signal a self-correcting discipline, not a broken one, supported by the closing paragraph. The 'journals are the primary cause and should not be trusted' choice overstates the role of journals, which the passage treats as one factor and a source of reform. The 'small samples are the single reason' choice isolates one of several listed causes as the only one. The 'replication is essentially impossible' choice contradicts the definition of replication offered in the first paragraph.

Q6. [VARC-RC2 · mcsc · diff 0.5]

The author identifies all of the following as factors contributing to replication failures EXCEPT:

- A) The freedom to analyse data in many ways until a pattern happens to appear.
- B) A deliberate effort by most researchers to fabricate data in order to deceive their peers. ✓
- C) Incentives that reward novel findings while offering little reward for repeating others' work.
- D) Small samples that can produce striking results which disappear when the study is enlarged.

Answer: B) A deliberate effort by most researchers to fabricate data in order to deceive their peers.

Solution: The 'deliberate effort to fabricate data' choice is the EXCEPT answer: the passage attributes problems to incentives, sample size, and analytic flexibility, and explicitly notes that adjusting analyses can happen 'even unintentionally' rather than via deliberate fraud. The 'freedom to analyse data many ways', 'incentives rewarding novel findings', and 'small samples that vanish when enlarged' choices each restate one of the three factors the passage names directly.

Q7. [VARC-RC2 · mcsc · diff 0.5]

What is the main purpose of registering hypotheses and planned analyses before collecting data, as described in the passage?

- A) To guarantee that each and every registered study will reliably end up producing some kind of novel and publishable finding.
- B) To replace careful replication studies entirely, so that any repetition of other people's earlier work is no longer needed at all.
- C) To allow researchers to choose freely, after they have seen the data, which analysis presents their results most favourably.
- D) To limit the freedom to adjust the analysis until a desired pattern appears, and to ensure results are reported either way. ✓

Answer: D) To limit the freedom to adjust the analysis until a desired pattern appears, and to ensure results are reported either way.

Solution: The 'limit the freedom to adjust analysis and report either way' choice follows from the passage: pre-registration commits researchers to a plan and to publishing results whether striking or not, curbing the analytic flexibility named earlier. The 'guarantee a publishable novel finding' choice misstates the goal as guaranteeing novelty. The 'replace replication studies entirely' choice wrongly claims it replaces replication, which the passage lists as a separate reform. The 'choose freely after seeing the data' choice describes the very practice pre-registration is meant to prevent.

Q8. [VARC-RC2 · mcsc · diff 0.5]

It can be inferred from the passage that the author regards the documentation of replication failures as:

- A) definite proof that most published scientific results will, sooner or later, be shown to be entirely false.
- B) an unfortunate and costly distraction that needlessly diverts attention away from producing genuinely novel discoveries.
- C) a serious problem that can only ever be solved by abandoning the basic requirement of replication altogether.
- D) evidence that science is applying its own critical methods to itself rather than a sign of fundamental decay. ✓

Answer: D) evidence that science is applying its own critical methods to itself rather than a sign of fundamental decay.

Solution: The 'science applying its own critical methods to itself' choice reflects the statement that documenting failures is itself the scientific method scrutinising its own practices, and the reframing of the 'crisis' as healthy self-correction. The 'proof that most published results are false' choice overgeneralises to most results being false, which the text does not claim. The 'unfortunate distraction from novel discoveries' choice contradicts the positive framing of self-scrutiny. The 'solved only by abandoning replication' choice reverses the argument, which treats replication as the gatekeeper to be strengthened, not abandoned.

Instructions (VARC-RC3): study the shared passage below and answer its 4 question(s).

A city map is never a perfect copy of the city it represents. If it were, it would have to be exactly as large and as complicated as the city itself, and so would be useless to a traveller. The value of a map lies precisely in what it leaves out. By dropping countless details and keeping only roads, rivers, and a few landmarks, a map becomes small enough to hold in the hand and clear enough to guide a journey.

The same principle applies to scientific models. A model of the weather, or of a population of animals, does not try to capture every raindrop or every individual creature. Instead it keeps the features that matter for the question being asked and ignores the rest. A model built to predict tomorrow's temperature can safely leave out the colour of people's umbrellas; a model of how a disease spreads can ignore the brand of car that infected people drive. Choosing what to leave out is not a weakness of a model but the very thing that makes it useful.

This has a practical lesson. When we judge a model, the right question is not 'Is it complete?' but 'Is it useful for its purpose?' A simple model that answers the question at hand may be far more valuable than an elaborate one that tries to include everything and so becomes impossible to understand. As one saying among scientists puts it, all models are wrong, but some are useful. Wisdom lies in matching the simplicity of the model to the job it must do.

Q9. [VARC-RC3 · mcsc · diff 0]

Which of the following best captures the main idea of the passage?

- A) Scientific models should always try to include as much fine-grained detail as they possibly can in order to be truly accurate.
- B) Maps are, in the end, far more reliable guides to a city than scientific models can ever be to the natural world.
- C) A good model, much like a good map, is valuable precisely because it leaves out details that do not matter for its purpose. ✓
- D) The colour of people's umbrellas and the brand of a car are in fact important variables in both weather and disease models.

Answer: C) A good model, much like a good map, is valuable precisely because it leaves out details that do not matter for its purpose.

Solution: The 'valuable because it leaves out details that do not matter' choice states the central analogy: usefulness comes from selective omission, true of both maps and models. The 'include as much detail as possible' choice contradicts the passage, which praises leaving things out. The 'maps are more reliable than models' choice invents a comparison of reliability the passage never makes. The 'umbrella colour and car brand are important variables' choice reverses the examples, which are given precisely as details that can safely be ignored.

Q10. [VARC-RC3 · mcsc · diff 0]

Why does the author say that a perfectly complete map would be useless?

- A) Because maps are always rather less accurate than the kind of scientific models that are discussed later on in the passage.
- B) Because it would have to be as large and as complicated as the city itself, and so could not guide a traveller. ✓
- C) Because real cities change far too quickly for any printed map of them to remain fully correct for very long.
- D) Because no ordinary traveller is ever genuinely interested in the roads, the rivers, or the landmarks of a city.

Answer: B) Because it would have to be as large and as complicated as the city itself, and so could not guide a traveller.

Solution: The 'as large and complicated as the city itself' choice paraphrases the explicit reason: a complete map would match the city in size and complexity and so help no one. The 'maps less accurate than models' choice introduces an accuracy comparison not present in the text. The 'cities change too quickly' choice raises the rate of urban change, which the passage never mentions. The 'travellers not interested in roads or rivers' choice contradicts the passage, which says maps keep roads, rivers, and landmarks because they matter.

Q11. [VARC-RC3 · mcsc · diff 0.5]

It can be inferred that the author would most likely judge a scientific model by asking:

- A) whether it serves its intended purpose well, even if it leaves out many features of the real situation. ✓
- B) whether it manages to include every single variable that could conceivably affect the final outcome in any way.
- C) whether it is noticeably more detailed and elaborate than any competing model that addresses the same question.
- D) whether it perfectly reproduces the whole of reality without leaving out or omitting anything at all whatsoever.

Answer: A) whether it serves its intended purpose well, even if it leaves out many features of the real situation.

Solution: The 'serves its intended purpose well even if it omits features' choice follows from the stated criterion 'Is it useful for its purpose?' and the praise of simple, fit-for-purpose models. The 'includes every variable that could affect the outcome' choice contradicts the argument against trying to include everything. The 'more detailed than any competing model' choice equates value with maximum detail, which the passage rejects. The 'perfectly reproduces all of reality' choice describes the impossible complete model the passage dismisses as useless.

Q12. [VARC-RC3 · mcsc · diff 0.5]

What is the main function of the saying 'all models are wrong, but some are useful' in the passage?

- A) To warn readers firmly that scientific models really should not be trusted when making any practical decision.
- B) To introduce a fresh objection that the author then goes on to refute carefully and in some detail.
- C) To claim that maps, quite unlike scientific models, are essentially never wrong about the cities they represent.

D) To reinforce the author's point that a model need not be complete in order to be valuable for its purpose. ✓

Answer: D) To reinforce the author's point that a model need not be complete in order to be valuable for its purpose.

Solution: The 'reinforce that a model need not be complete to be valuable' choice fits the saying's placement: it summarises and reinforces the thesis that usefulness, not completeness, is the right standard. The 'warn that models should not be trusted' choice misreads the saying as a blanket distrust of models. The 'introduce an objection the author refutes' choice wrongly treats it as something the author rebuts, when it supports the argument. The 'claim maps are never wrong' choice invents a contrast in which maps are infallible, contradicting the opening that no map is a perfect copy.

Q13. [VARC-ODD-1 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Coral reefs occupy less than one percent of the ocean floor, yet they shelter roughly a quarter of all marine species.
- (2) The dense limestone structures act as nurseries where countless fish spawn and mature.
- (3) They also buffer nearby coastlines by absorbing much of the energy carried by incoming waves.
- (4) Many tropical resorts now offer guided diving tours that bring tourists close to these formations.
- (5) Because of this ecological richness, the loss of a single reef can ripple through an entire marine food web.

Which sentence is the odd one out?

- A) 3
- B) 5
- C) 4 ✓
- D) 1
- E) 2

Answer: C) 4

Solution: Sentences 1, 2, 3 and 5 all describe the ecological importance of coral reefs (biodiversity, nursery role, coastal protection, food-web impact). Sentence 4 shifts to tourism and resort marketing, an unrelated commercial angle, so it is the misfit. The other numbers are wrong because each of those sentences contributes to the single ecological theme.

Q14. [VARC-ODD-2 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) The printing press, introduced in fifteenth-century Europe, dramatically lowered the cost of reproducing texts.
- (2) Books that had once taken months to copy by hand could now be produced in days.
- (3) As prices fell, reading spread beyond monasteries and universities to ordinary townspeople.
- (4) This wider circulation of ideas helped fuel movements in science and religion.
- (5) Modern libraries increasingly lend electronic books that readers download to personal devices.

Which sentence is the odd one out?

- A) 5 ✓
- B) 1
- C) 2
- D) 3
- E) 4

Answer: A) 5

Solution: Sentences 1 to 4 trace a single historical narrative about the printing press and its consequences for the spread of reading and ideas. Sentence 5 jumps to present-day e-book lending, a different era and technology, breaking the chain, so it is the odd one out. Each of the other sentences advances the printing-press story.

Q15. [VARC-ODD-3 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Honeybees communicate the location of food through a movement known as the waggle dance.
- (2) Beekeepers harvest surplus honey by carefully removing frames from the hive.
- (3) A returning forager runs in a figure-eight pattern on the vertical comb.
- (4) The angle of the run relative to gravity encodes the direction of the food source.
- (5) The duration of the waggle phase signals how far away the food lies.

Which sentence is the odd one out?

- A) 5
- B) 2 ✓
- C) 1
- D) 4
- E) 3

Answer: B) 2

Solution: Sentences 1, 3, 4 and 5 form a tight explanation of how the honeybee waggle dance conveys both the direction and distance of food. Sentence 2 turns to human honey harvesting, an unrelated beekeeping activity, so it does not belong. The remaining sentences each describe a feature of the communicative dance.

Q16. [VARC-JUMBLE-1 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. The result was a network of stone roads stretching across the whole empire.
2. Ancient Rome needed a fast way to move its armies between distant provinces.
3. To meet this need, engineers laid down durable paved routes wherever the legions marched.
4. Centuries later, many of these same roads still guide the paths of modern highways.

Select the correct sequence of sentences.

- A) 2341
- B) 3214
- C) 2314 ✓
- D) 2134

Answer: C) 2314

Solution: Sentence 2 states the problem (Rome needed fast troop movement), 3 gives the engineers' response, 1 reports the result (a road network), and 4 closes with the long-term legacy. The order 2-3-1-4 is the only sequence in which cause, action, outcome and aftermath follow naturally; the distractors break this problem-solution-result chain.

Q17. [VARC-JUMBLE-2 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. Over time, this trapped heat raises the planet's average surface temperature.
2. Certain gases in the atmosphere allow sunlight to pass through but absorb the heat that the ground radiates back.
3. This natural process is what scientists call the greenhouse effect.
4. When the concentration of these gases rises, more heat is retained than before.

Select the correct sequence of sentences.

- A) 2341
- B) 3241
- C) 2314 ✓
- D) 2431

Answer: C) 2314

Solution: Sentence 2 introduces the gases and what they do, 3 names the process just described (the greenhouse effect), 1 states the gradual consequence of trapped heat, and 4 adds the amplifying condition when gas levels rise. The 2-3-1-4 order moves from mechanism to label to effect to intensification; the alternatives misplace the naming sentence or the consequence.

Q18. [VARC-JUMBLE-3 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. She began by sketching rough thumbnails to test different compositions.
2. Maria had been asked to design the cover for the school magazine.
3. Once a layout felt right, she refined it into a detailed digital drawing.
4. Finally, she added colour and sent the finished file to the printer.

Select the correct sequence of sentences.

- A) 1234
- B) 2143
- C) 2134 ✓
- D) 2314

Answer: C) 2134

Solution: Sentence 2 sets up the task (Maria asked to design a cover), 1 starts the work with thumbnails, 3 refines a chosen layout, and 4 completes the process with colour and printing. The 2-1-3-4 sequence follows the natural order of a design workflow from assignment to delivery; the distractors either open mid-process or reverse the refining and finishing steps.

Q19. [VARC-JUMBLE-4 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. At first the seedlings were too fragile to survive outdoors.
2. The gardeners started the tomato plants indoors during early spring.
3. So they kept the young plants under warm lamps for several weeks.
4. Only when the stems had thickened did the gardeners move them into the open beds.

Select the correct sequence of sentences.

- A) 2314
- B) 1234
- C) 2134 ✓
- D) 2143

Answer: C) 2134

Solution: Sentence 2 establishes that the plants were started indoors, 1 explains why they could not yet go outside (too fragile), 3 gives the response (kept under lamps), and 4 marks the transfer outdoors once strong enough. The 2-1-3-4 order links the indoor start, the problem, the remedy and the eventual move; the distractors break this cause-and-effect progression.

Q20. [VARC-SUMMARY-1 · mcsc · diff 0]

Public libraries today do far more than lend books. Many now offer free internet access, run job-search workshops, host story hours for children, and lend tools or musical instruments. For people without these resources at home, the library has become a community centre as much as a place to read.

Which of the following best summarises the paragraph?

- A) Modern public libraries have grown into broad community centres that now offer many services going well beyond simply lending out books. ✓
- B) Public libraries today mainly function as places that lend out musical instruments and household tools to the local residents who ask.
- C) Libraries have lost all interest in books and now devote themselves entirely to providing free internet access to their local visitors.
- D) The single most valuable thing libraries do now is to host the enjoyable weekly story hours for the young children of the community.

Answer: A) Modern public libraries have grown into broad community centres that now offer many services going well beyond simply lending out books.

Solution: The 'grown into broad community centres beyond lending books' choice captures the paragraph's main point: libraries now serve as community centres providing many services in addition to books. The 'mainly lend instruments and tools' choice is too narrow, naming only one detail. The 'lost all interest in books for internet access' choice distorts the text, which says libraries do more than lend books, not that they abandoned books. The 'story hours are the single most valuable thing' choice is a minor detail, not the overall idea.

Q21. [VARC-SUMMARY-2 · mcsc · diff 0.5]

Sleep researchers have found that pulling an all-nighter before an exam usually backfires. While extra waking hours seem to allow more study, the tired brain struggles to store new information, and fatigue slows recall the next day. Students who sleep a full night generally remember more, even though they spent fewer hours reviewing.

Which of the following best summarises the paragraph?

- A) Students who choose to stay awake all night before an exam manage to study far more material and so tend to recall it better.
- B) Sacrificing sleep to study longer tends to weaken memory and recall, so a full night of proper rest usually serves students better. ✓
- C) How much a student sleeps has no measurable effect at all on how well that student later performs on a difficult written examination.
- D) Sleep researchers now firmly advise students that they should never review any of their own notes during the evening before an exam.

Answer: B) Sacrificing sleep to study longer tends to weaken memory and recall, so a full night of proper rest usually serves students better.

Solution: The 'sacrificing sleep weakens memory so rest serves better' choice conveys the central trade-off: the extra study time gained by skipping sleep is outweighed by weaker memory, so resting is generally better. The 'staying awake lets you recall more' choice repeats only the surface claim the passage rebuts. The 'sleep has no measurable effect' choice contradicts the finding that sleep helps recall. The 'never review notes the evening before' choice overreaches into advice the paragraph never gives.

Q22. [VARC-SUMMARY-3 · mcsc · diff 0.5]

Electric cars produce no exhaust as they drive, which clearly improves air quality in cities. Yet whether they are truly cleaner depends on how their electricity is generated. Where power comes mostly from coal, the overall emissions can rival those of petrol cars; where it comes from renewables, the advantage is large.

Which of the following best summarises the paragraph?

- A) Coal-fired power stations still remain the main source of almost all of the electricity that is used to charge most electric cars today.
- B) Electric cars are always far cleaner than ordinary petrol cars in every situation, no matter how their electrical power happens to be generated.
- C) Electric cars release no pollution at all from their tailpipes while they are being driven along busy city streets and the open roads beyond.
- D) Electric cars cut local pollution, but their full environmental benefit ultimately depends on how the electricity charging them is produced. ✓

Answer: D) Electric cars cut local pollution, but their full environmental benefit ultimately depends on how the electricity charging them is produced.

Solution: The 'cut local pollution but full benefit depends on how electricity is produced' choice balances both halves of the paragraph: a clear local benefit, but overall cleanliness depends on the electricity source. The 'no tailpipe pollution while driving' choice states only one detail and misses the central condition. The 'always far cleaner in every situation' choice asserts a fact the passage does not claim. The 'coal stations supply almost all charging electricity' choice overreaches by treating the coal caveat as a universal fact.

Q23. [VARC-SUMMARY-4 · mcsc · diff 0.5]

Volunteering is often described as pure generosity, but studies suggest the giver gains too. People who regularly help others report lower stress, a stronger sense of purpose, and wider social networks. In this sense, the act of giving time quietly returns a benefit to the volunteer.

Which of the following best summarises the paragraph?

A) Although volunteering clearly helps other people in many ways, research indicates that it also brings the volunteer real personal benefits. ✓

B) Volunteering is best understood as a wholly selfless act of giving that quietly returns no benefit of any real kind to the person doing it.

C) People generally choose to volunteer mainly because doing so lets them expand and strengthen their own valuable personal social networks.

D) Regular volunteers consistently report that they now feel considerably less stressed and rather more relaxed than they did before they began.

Answer: A) Although volunteering clearly helps other people in many ways, research indicates that it also brings the volunteer real personal benefits.

Solution: The 'helps others yet also brings the volunteer real personal benefits' choice captures the paragraph's twist: volunteering helps others yet also rewards the giver. The 'wholly selfless act returning no benefit' choice contradicts the passage, which argues the giver does gain. The 'volunteer mainly to expand social networks' choice narrows the motive to one benefit and misstates intention. The 'feel less stressed and more relaxed' choice reports a single detail rather than the overall claim that giving returns several benefits.

Q24. [VARC-SUMMARY-5 · mcsc · diff 0.5]

When a new word enters a language, it rarely arrives by official decree. Instead, speakers begin using it, others copy them, and dictionaries record the word only after it has already taken hold. Language, in other words, is shaped from the bottom up by ordinary use rather than handed down by authorities.

Which of the following best summarises the paragraph?

A) Dictionaries hold the real authority here and are the bodies responsible for officially deciding which new words a language may properly adopt.

B) Most new words manage to enter a language only after some official government body has formally reviewed and then finally approved each one.

C) New words spread through everyday use among ordinary speakers, and dictionaries follow rather than simply lead this gradual change over time. ✓

D) Speakers occasionally invent striking new words that many other people across the wider language community then choose to copy and reuse.

Answer: C) New words spread through everyday use among ordinary speakers, and dictionaries follow rather than simply lead this gradual change over time.

Solution: The 'new words spread through everyday use while dictionaries follow' choice states the main idea: language changes from the bottom up through ordinary use, with dictionaries recording words after the fact. The 'dictionaries hold the real authority to decide' choice reverses this by giving dictionaries the deciding role. The 'words enter only after government approval' choice captures only a fragment of the process and omits the role of dictionaries. The 'speakers occasionally invent striking words others copy' choice contradicts the passage's denial of official decree by ignoring the dictionaries-follow point.

Section: DILR (suggested 40 min)

Instructions (DILR-1): study the shared data below and answer its 3 question(s).

Four teams, Alfa, Bravo, Comet and Delta, play a single round-robin tournament: every team plays every other team exactly once (6 matches in all). A win is worth 3 points, a tie is worth 1 point to each team, and a loss is worth 0 points. The following facts are known about the completed tournament:

- Alfa won every match it played. - Delta lost every match it played. - Exactly one match in the whole tournament ended in a tie. - That single tie was the match between Bravo and Comet.

Q25. [DILR-1 · mcsc · diff 0.5]

How many points did Comet earn in the tournament?

- A) 4 points ✓
- B) 7 points
- C) 3 points
- D) 6 points

Answer: A) 4 points

Solution: Alfa won all 3 of its matches, so Alfa beat Bravo, Comet and Delta. Delta lost all 3. The only tie was Bravo vs Comet, so every other match had a winner. Comet thus lost to Alfa (0), tied Bravo (1) and beat Delta (3): total $1 + 3 = 4$ points. Distractor 3 = only the win; distractor 6 = double-counting the tie as 3+3; distractor 7 = treating the Bravo match as a win (3+3+1).

Q26. [DILR-1 · mcsc · diff 0.5]

How many of the 6 matches ended with a winner (that is, did NOT end in a tie)?

- A) 6 matches
- B) 4 matches
- C) 5 matches ✓
- D) 3 matches

Answer: C) 5 matches

Solution: There are 6 matches in a round-robin among 4 teams. Exactly one match (Bravo vs Comet) ended in a tie, so the remaining $6 - 1 = 5$ matches ended with a winner. Distractor 6 ignores the tie; distractors 4 and 3 over-subtract.

Q27. [DILR-1 · mcsc · diff 0.5]

Which statement about the final points table is correct?

- A) Comet finished ahead of Bravo on points, with Alfa on top.
- B) Bravo and Comet finished level on points, behind Alfa and ahead of Delta. ✓
- C) Delta finished level on points with Bravo, behind Alfa and Comet.
- D) Alfa and Bravo finished level on points at the top, ahead of Comet and Delta.

Answer: B) Bravo and Comet finished level on points, behind Alfa and ahead of Delta.

Solution: Final points: Alfa 9 (three wins), Delta 0 (three losses). Bravo = loss to Alfa (0) + tie with Comet (1) + win over Delta (3) = 4; Comet = loss to Alfa (0) + tie with Bravo (1) + win over Delta (3) = 4. So Bravo and Comet are level on 4, below Alfa's 9 and above Delta's 0. The claim that Comet finished ahead of Bravo is false (they are equal); the claim that Delta is level with Bravo is false (Delta has 0); the claim that Alfa and Bravo are level at the top is false (Alfa alone leads on 9).

Instructions (DILR-2): study the shared data below and answer its 3 question(s).

Three friends, Mira, Nina and Owen, each take the same short quiz of 5 questions. Every correct answer is worth 3 points; a wrong or blank answer scores 0 points. So each person's score equals 3 times the number of questions they answered correctly. The following facts are known:

- Between them, the three friends answered exactly 9 questions correctly. - Mira answered more questions correctly than Nina, and Nina answered more correctly than Owen. - Owen answered exactly 2 questions correctly.
- No one answered all 5 questions correctly.

Q28. [DILR-2 · mcsc · diff 0.5]

How many points did Mira score?

- A) 9 points
- B) 6 points
- C) 12 points ✓
- D) 15 points

Answer: C) 12 points

Solution: Owen got 2 correct. The three totals must sum to 9, with $Mira > Nina > Owen = 2$ and no one above 4 correct (since nobody got all 5). So $Nina = 3$ and $Mira = 4$ ($4 + 3 + 2 = 9$). Mira's score = $3 \times 4 = 12$. Distractor 9 = Nina's score; distractor 6 = Owen's score; distractor 15 wrongly assumes Mira got 5 correct.

Q29. [DILR-2 · mcsc · diff 0.5]

How many questions did Nina answer correctly?

- A) 4 questions
- B) 3 questions ✓
- C) 2 questions
- D) 5 questions

Answer: B) 3 questions

Solution: With Owen = 2 correct, totals summing to 9, the strict order Mira > Nina > Owen, and nobody reaching 5, the only fit is Mira = 4, Nina = 3, Owen = 2. So Nina answered 3 correctly. Distractor 4 = Mira's count; distractor 2 = Owen's count; distractor 5 is impossible since no one got all 5.

Q30. [DILR-2 · mcsc · diff 0.5]

What was the combined total score of all three friends?

- A) 27 points ✓
- B) 30 points
- C) 9 points
- D) 18 points

Answer: A) 27 points

Solution: The three answered 9 questions correctly in total, each worth 3 points, so the combined score is $3 \times 9 = 27$ points (Mira 12 + Nina 9 + Owen 6 = 27). Distractor 9 = total correct answers, not points; distractor 18 = scoring only 2 points per correct; distractor 30 misadds the individual scores.

Instructions (DILR-3): study the shared data below and answer its 4 question(s).

Units sold during the month (number of packets)

Product	Aria	Bani	Cera	Dovi
Tea	120	200	150	180
Coffee	80	150	200	120
Cocoa	60	100	50	90

Selling price per packet (in rupees)

Product	Price
Tea	20
Coffee	30
Cocoa	40

Q31. [DILR-3 · mcsc · diff 0.5]

How many packets in total did store Bani sell across all three products during the month?

- A) 350
- B) 400
- C) 450 ✓
- D) 260

Answer: C) 450

Solution: Store Bani's column: Tea 200 + Coffee 150 + Cocoa 100 = 450 packets. Distractor 400 is Cera's column total, 260 is Aria's column total, and 350 drops the cocoa figure.

Q32. [DILR-3 · mcsc · diff 0.5]

What was the total sales revenue (in rupees) earned by store Cera from all three products?

- A) 12500
- B) 11000 ✓
- C) 10800
- D) 7200

Answer: B) 11000

Solution: Cera: Tea $150 \times 20 = 3000$, Coffee $200 \times 30 = 6000$, Cocoa $50 \times 40 = 2000$; total $3000 + 6000 + 2000 = 11000$. Distractor 12500 is Bani's revenue, 10800 is Dovi's, and 7200 is Aria's.

Q33. [DILR-3 · mcsc · diff 0.5]

What was the combined revenue (in rupees) from Tea across all four stores?

- A) 650
- B) 16500
- C) 11000
- D) 13000 ✓

Answer: D) 13000

Solution: Tea packets sold: $120+200+150+180 = 650$; at 20 rupees each the revenue is $650 \times 20 = 13000$. Distractor 650 forgets to multiply by price, 16500 multiplies by coffee's price 30 (wrong rate), and 11000 is Cera's total revenue.

Q34. [DILR-3 · mcsc · diff 0.5]

In how many of the four stores did the total number of packets sold (all products combined) exceed 350?

- A) 4
- B) 3 ✓
- C) 1
- D) 2

Answer: B) 3

Solution: Store totals are Aria 260, Bani 450, Cera 400, Dovi 390. Three of them (Bani, Cera, Dovi) exceed 350; only Aria does not. Distractor 2 miscounts Dovi, 4 wrongly includes Aria, and 1 counts only Bani.

Instructions (DILR-4): study the shared data below and answer its 3 question(s).

Three investors, Priya, Rohan and Sara, each put a whole number of lakhs of rupees into a separate fund at the start of the year. The following facts are known:

- Together the three invested a total of 24 lakh. - Rohan invested exactly twice as much as Priya. - Sara invested 4 lakh more than Priya.

Over the year the funds gave these returns on the amount invested: Priya's fund returned 20%, Rohan's fund returned 10%, and Sara's fund returned 30%.

Q35. [DILR-4 · mcsc · diff 0.5]

How much did Rohan invest?

- A) 5 lakh
- B) 10 lakh ✓
- C) 9 lakh
- D) 12 lakh

Answer: B) 10 lakh

Solution: Let Priya invest P . Then Rohan = $2P$ and Sara = $P + 4$. The total gives $P + 2P + (P + 4) = 24$, so $4P + 4 = 24$, $P = 5$. Hence Rohan = $2 \times 5 = 10$ lakh. Distractor 5 = Priya's amount; distractor 9 = Sara's amount; distractor 12 solves the equation incorrectly.

Q36. [DILR-4 · mcsc · diff 0.5]

What was the combined amount invested by Priya and Sara together?

- A) 19 lakh
- B) 10 lakh
- C) 15 lakh
- D) 14 lakh ✓

Answer: D) 14 lakh

Solution: From $P + 2P + (P + 4) = 24$ we get $P = 5$, so Priya = 5 and Sara = $5 + 4 = 9$. Together they invested $5 + 9 = 14$ lakh. Distractor 10 = Rohan's amount; distractor 15 = Priya plus Rohan; distractor 19 = Rohan plus Sara.

Q37. [DILR-4 · mcsc · diff 1]

After the year's returns are added to the amounts invested, which investor ends with the largest total amount of money?

- A) Rohan and Sara end level, each with 11 lakh
- B) Sara, with 11.7 lakh ✓
- C) Rohan, with 11 lakh
- D) Priya, with 6 lakh

Answer: B) Sara, with 11.7 lakh

Solution: Investments: Priya 5, Rohan 10, Sara 9 lakh. Adding returns: Priya = $5 \times 1.20 = 6.0$; Rohan = $10 \times 1.10 = 11.0$; Sara = $9 \times 1.30 = 11.7$. The largest final amount is Sara's 11.7 lakh, just above Rohan's 11.0. Distractor 2 stops at Rohan, ignoring Sara's higher return rate; distractor 3 is the smallest total; distractor 4 wrongly treats them as tied (11.7 not equal to 11.0).

Instructions (DILR-5): study the shared data below and answer its 5 question(s).

Units produced by a factory in each quarter

Product	Q1	Q2	Q3
Steel	400	500	600
Cement	300	300	450
Glass	200	250	200
Wood	100	150	250

Production cost per unit (in rupees)

Product	Rate
Steel	50
Cement	40
Glass	60
Wood	30

Q38. [DILR-5 · mcsc · diff 0.5]

What was the total number of Steel units produced across the three quarters?

- A) 1500 ✓
- B) 1050
- C) 1100
- D) 1300

Answer: A) 1500

Solution: Steel row: $400 + 500 + 600 = 1500$ units. Distractor 1050 is Cement's three-quarter total, 1300 sums all four products in Q1+Q2 only, and 1100 drops one quarter.

Q39. [DILR-5 · mcsc · diff 0.5]

By what percentage did Steel production increase from Q1 to Q3?

- A) 20
- B) 200
- C) 33
- D) 50 ✓

Answer: D) 50

Solution: Steel rose from 400 in Q1 to 600 in Q3, an increase of 200; $200/400 \times 100 = 50$ percent. Distractor 20 uses the Q1-to-Q2 step over the wrong base, 200 reports the raw increase as a percentage, and 33 computes the increase over Q3 instead of Q1.

Q40. [DILR-5 · mcsc · diff 0.5]

What was the average quarterly production of Cement over the three quarters?

- A) 300
- B) 1050
- C) 350 ✓
- D) 375

Answer: C) 350

Solution: Cement: $(300 + 300 + 450)/3 = 1050/3 = 350$ units. Distractor 300 takes only the most frequent value, 1050 is the three-quarter sum (not the average), and 375 averages the wrong figures.

Q41. [DILR-5 · mcsc · diff 0.5]

What is the largest single value of units produced by any one product in any one quarter?

- A) 500
- B) 450
- C) 250
- D) 600 ✓

Answer: D) 600

Solution: Scanning every cell, the maximum is Steel in Q3 with 600 units. Distractor 500 is Steel in Q2, 450 is the top Cement figure, and 250 is the highest Wood figure.

Q42. [DILR-5 · mcsc · diff 0.5]

What was the total production cost (in rupees) of all four products in Q1?

- A) 47000 ✓
- B) 44000
- C) 1000
- D) 50000

Answer: A) 47000

Solution: Q1 costs: Steel $400 \times 50 = 20000$, Cement $300 \times 40 = 12000$, Glass $200 \times 60 = 12000$, Wood $100 \times 30 = 3000$; total $20000 + 12000 + 12000 + 3000 = 47000$. Distractor 1000 sums only the Q1 units, 44000 omits the Wood term, and 50000 applies a flat rate of 50 to every product.

Section: QA (suggested 40 min)

Q43. [QA-MCQ-1 · mcsc · diff 0]

A shopkeeper buys an article for 80 euros and sells it for 100 euros. What is the profit percentage?

- A) 20%
- B) 80%
- C) 125%
- D) 25% ✓

Answer: D) 25%

Solution: Profit = $100 - 80 = 20$; profit% = $20/80 \times 100 = 25\%$. The 20% value wrongly uses the selling price as the base (20/100). The 80% value just restates the cost figure as a percentage. The 125% value is the SP-to-CP ratio, not the profit rate.

Q44. [QA-MCQ-2 · mcsc · diff 0]

The ratio of boys to girls in a class is 3 : 5. If there are 20 girls, how many boys are there?

- A) 33 boys
- B) 15 boys
- C) 8 boys
- D) 12 boys ✓

Answer: D) 12 boys

Solution: One part = $20/5 = 4$ students; boys = $3 \times 4 = 12$. The 33 value sets girls equal to 5 parts of value 5 plus 20 wrongly. The 15 value takes 3/4 of 20 (treats total as 4 parts). The 8 value takes 2 parts, using a 3:5 difference incorrectly.

Q45. [QA-MCQ-3 · mcsc · diff 0]

What is the average of the five numbers 4, 8, 10, 12 and 16?

- A) 10 ✓
- B) 50
- C) 12
- D) 8

Answer: A) 10

Solution: Sum = $4+8+10+12+16 = 50$; average = $50/5 = 10$. The 50 value gives the sum without dividing by 5. The 12 value is the median-like middle pick, not the mean. The 8 value divides the sum by 6 instead of 5 (off-by-one count error).

Q46. [QA-MCQ-4 · mcsc · diff 0]

A circle has radius 7 cm. What is its area? (use $\pi = 22/7$)

- A) 44 cm^2
- B) 154 cm^2 ✓
- C) 616 cm^2
- D) 49 cm^2

Answer: B) 154 cm^2

Solution: Area = $\pi r^2 = (22/7)(7^2) = (22/7)(49) = 154 \text{ cm}^2$. The 44 value is the circumference $2\pi r$, not the area. The 616 value uses diameter 14 as the radius ($\pi(14^2)$). The 49 value gives r^2 alone, forgetting the factor π .

Q47. [QA-MCQ-5 · mcsc · diff 0]

Solve for x: $3x + 7 = 22$.

- A) $29/3$
- B) 9
- C) 5 ✓
- D) 15

Answer: C) 5

Solution: $3x = 22 - 7 = 15$, so $x = 15/3 = 5$. The $29/3$ value adds 7 instead of subtracting it ($22+7 = 29$). The 9 value subtracts 7 from 22 but then subtracts 3 instead of dividing. The 15 value stops at $3x = 15$ without dividing by 3.

Q48. [QA-MCQ-6 · mcsc · diff 0]

A car travels 150 km in 3 hours. What is its average speed?

- A) 50 km/h ✓
- B) 450 km/h
- C) 75 km/h
- D) 2 km/h

Answer: A) 50 km/h

Solution: Speed = distance/time = $150/3 = 50 \text{ km/h}$. The 450 value multiplies distance by time. The 75 value divides by 2 instead of 3. The 2 value computes time/distance ($3/150$ scaled), inverting the formula.

Q49. [QA-MCQ-7 · mcsc · diff 0.5]

An article marked at 500 euros is sold after a 20% discount, and the seller still makes a 25% profit on cost. What was the cost price?

- A) 400 euros
- B) 333 euros
- C) 320 euros ✓
- D) 300 euros

Answer: C) 320 euros

Solution: Selling price = $500 \times 0.80 = 400$. With 25% profit, $SP = 1.25 \times CP$, so $CP = 400/1.25 = 320$. The 300 value divides 400 by 1.2 (uses the discount rate as the profit rate). The 400 value reports the selling price, not the cost. The 333 value takes $400/1.2$ rounded, again misusing the rate.

Q50. [QA-MCQ-8 · mcsc · diff 0.5]

A sum of 8000 euros is invested at 10% per annum compound interest. What is the amount after 2 years?

- A) 1680 euros
- B) 9700 euros
- C) 9600 euros
- D) 9680 euros ✓

Answer: D) 9680 euros

Solution: Amount = $8000 \times (1.10)^2 = 8000 \times 1.21 = 9680$. The 9600 value uses simple interest ($8000 + 2 \times 800$). The 1680 value gives only the compound interest, not the total amount. The 9700 value adds an extra rounding bump, miscomputing 1.21.

Q51. [QA-MCQ-9 · mcsc · diff 0.5]

Pipe A fills a tank in 12 hours and pipe B fills it in 6 hours. If both pipes are opened together, how long do they take to fill the tank?

- A) 4 hours ✓
- B) 9 hours
- C) 18 hours
- D) 3 hours

Answer: A) 4 hours

Solution: Combined rate = $1/12 + 1/6 = 1/12 + 2/12 = 3/12 = 1/4$ tank/hour, so time = 4 hours. The 9 value averages the two times. The 18 value adds the two times. The 3 value takes half of the faster pipe's time arbitrarily.

Q52. [QA-MCQ-10 · mcsc · diff 0.5]

When a positive integer n is divided by 7 the remainder is 4. What is the remainder when $3n$ is divided by 7?

- A) 4
- B) 12
- C) 2
- D) 5 ✓

Answer: D) 5

Solution: $n = 7k + 4$, so $3n = 21k + 12 = 7(3k+1) + 5$, giving remainder 5. The 4 value keeps the original remainder unchanged. The 12 value gives 3×4 without reducing modulo 7. The 2 value subtracts 7 twice from 12 instead of once.

Q53. [QA-MCQ-11 · mcsc · diff 0.5]

The HCF of two numbers is 6 and their LCM is 180. If one number is 36, what is the other number?

- A) 30 ✓
- B) 180
- C) 1080
- D) 5

Answer: A) 30

Solution: Product of numbers = HCF $\times$ LCM = $6 \times 180 = 1080$; other number = $1080/36 = 30$. The 180 value confuses the LCM with the missing number. The 1080 value gives the product itself, not the second number. The 5 value divides 180 by 36 (uses LCM instead of HCF $\times$ LCM).

Q54. [QA-MCQ-12 · mcsc · diff 0.5]

A bag contains 4 red and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both are red?

- A) $2/15$ ✓
- B) $4/25$
- C) $1/5$
- D) $2/9$

Answer: A) $2/15$

Solution: $P = (4/10)(3/9) = 12/90 = 2/15$. The $4/25$ value uses $(4/10)^2$, i.e. with replacement. The $1/5$ value is $C(4,2)/C(6,2)$ using the wrong denominator. The $2/9$ value takes $(4/9)(\dots)$ mishandling the first draw count.

Q55. [QA-MCQ-13 · mcsc · diff 0.5]

How many distinct arrangements can be made using all the letters of the word LEVEL?

- A) 120
- B) 30 ✓
- C) 60
- D) 20

Answer: B) 30

Solution: LEVEL has 5 letters with L repeated twice and E repeated twice: $5!/(2!2!) = 120/4 = 30$. The 120 value is 5! ignoring repeats. The 60 value divides by only one 2! (corrects a single repeat). The 20 value divides by 3! erroneously.

Q56. [QA-MCQ-14 · mcsc · diff 0.5]

In a group of 50 people, 30 like tea and 25 like coffee. If 10 like neither, how many like both tea and coffee?

- A) 5
- B) 15 ✓
- C) 55
- D) 40

Answer: B) 15

Solution: People liking at least one = $50 - 10 = 40$. By inclusion-exclusion, both = $30 + 25 - 40 = 15$. The 5 value forgets to subtract the 10 (uses total 50: $30+25-50$). The 55 value just adds $30+25$. The 40 value reports the union, not the intersection.

Q57. [QA-MCQ-15 · mcsc · diff 0.5]

The third term of an arithmetic progression is 11 and the seventh term is 27. What is the first term?

- A) 5
- B) 4
- C) 7
- D) 3 ✓

Answer: D) 3

Solution: $a + 6d = 27$ and $a + 2d = 11$, so $4d = 16$, $d = 4$; then $a = 11 - 2(4) = 3$. The 4 value gives the common difference d . The 7 value uses $a + 4d = 27$ (miscounts the term gap). The 5 value takes $11 - 6$ by guessing $d = 3$.

Q58. [QA-MCQ-16 · mcsc · diff 1]

Two trains start at the same time from stations A and B, 330 km apart, moving towards each other at 60 km/h and 50 km/h. How far from A do they meet?

- A) 180 km ✓
- B) 150 km
- C) 165 km
- D) 110 km

Answer: A) 180 km

Solution: Closing speed = $60 + 50 = 110$ km/h; meeting time = $330/110 = 3$ h; distance from A = $60 \times 3 = 180$ km. The 150 value is the distance the slower train covers (from B). The 165 value is the midpoint, ignoring the speed difference. The 110 value mistakes the closing speed for the distance.

Q59. [QA-MCQ-17 · mcsc · diff 1]

Vessel A holds 40 litres of a mixture with milk and water in the ratio 3 : 1, and vessel B holds 20 litres in the ratio 1 :

1. If both are poured into one vessel, what is the ratio of milk to water in the new mixture?

- A) 2 : 1 ✓
- B) 4 : 3
- C) 5 : 3
- D) 3 : 2

Answer: A) 2 : 1

Solution: A: milk = 30, water = 10. B: milk = 10, water = 10. Total milk = 40, water = 20, ratio $40:20 = 2:1$. The 3:2 value averages the two ratios. The 4:3 value mixes up B's split as 2:1. The 5:3 value errs by using 25 litres of milk from A.

Q60. [QA-MCQ-18 · msc · diff 1]

A rectangular garden is 40 m long and 30 m wide. A path of uniform width 2 m runs all around it on the inside. What is the area of the path?

- A) 280 m²
- B) 144 m²
- C) 264 m² ✓
- D) 536 m²

Answer: C) 264 m²

Solution: Outer area = $40 \times 30 = 1200$. The inner rectangle shrinks by 2 m on each side, so it is $(40 - 4) \times (30 - 4) = 36 \times 26 = 936$. Path area = $1200 - 936 = 264$ m². The 280 value subtracts width only once per side ($38 \times 28 = 1064$ misstep). The 144 value is just the corner squares plus a slip. The 536 value treats the path as a 2 m strip on the outside of a 40 x 30 frame, doubling the perimeter contribution.

Q61. [QA-TITA-1 · tita · diff 0]

Evaluate the expression $(2^5 \times 2^3) \div 2^4$. Type your answer as an integer.

Answer (TITA): 16

Solution: Using the laws of indices, the numerator $2^5 \times 2^3 = 2^{(5+3)} = 2^8$. Dividing by 2^4 gives $2^{(8-4)} = 2^4 = 16$. So the value of the expression is 16.

Q62. [QA-TITA-2 · tita · diff 0]

An item is priced at 80 euros. Its price is first increased by 25%, then the new price is decreased by 20%. What is the final price, in euros? Type your answer as an integer.

Answer (TITA): 80

Solution: A 25% increase multiplies the price by 1.25, giving $80 \times 1.25 = 100$. A subsequent 20% decrease multiplies by 0.80, giving $100 \times 0.80 = 80$. The successive +25% then -20% changes cancel exactly here, so the final price is 80 euros.

Q63. [QA-TITA-3 · tita · diff 0.5]

Using the digits 1, 2, 3, 4 and 5 without repetition, how many three-digit numbers divisible by 5 can be formed? Type your answer as an integer.

Answer (TITA): 12

Solution: A number is divisible by 5 only if its last digit is 0 or 5. Since 0 is not available, the units digit must be 5 (1 way). The hundreds digit can then be any of the remaining 4 digits, and the tens digit any of the remaining 3 digits: $4 \times 3 = 12$ numbers. Total = 12.

Q64. [QA-TITA-4 · tita · diff 0.5]

In a class, 30 students study French and 25 study German. Exactly 10 students study both languages. How many students study exactly one of the two languages? Type your answer as an integer.

Answer (TITA): 35

Solution: Students studying only French = $30 - 10 = 20$. Students studying only German = $25 - 10 = 15$. Those studying exactly one language = $20 + 15 = 35$. Equivalently, $|A| + |B| - 2|A \cap B| = 30 + 25 - 20 = 35$.

Q65. [QA-TITA-5 · tita · diff 0.5]

In an arithmetic progression the first term is 7 and the common difference is 5. What is the value of the 12th term? Type your answer as an integer.

Answer (TITA): 62

Solution: The n th term of an arithmetic progression is $a + (n-1)d$. With $a = 7$, $d = 5$ and $n = 12$, the 12th term = $7 + (12-1) \times 5 = 7 + 11 \times 5 = 7 + 55 = 62$.

Q66. [QA-TITA-6 · tita · diff 1]

A vessel contains 64 litres of pure wine. 16 litres are drawn out and replaced with water; this operation (drawing 16 litres of the mixture and replacing it with water) is performed two more times, for a total of three times. How many litres of pure wine remain in the vessel? Type your answer as an integer.

Answer (TITA): 27

Solution: After each replacement of 16 litres out of 64, the fraction of wine remaining is multiplied by $(64-16)/64 = 48/64 = 3/4$. After three operations the wine left = $64 \times (3/4)^3 = 64 \times 27/64 = 27$ litres. So 27 litres of pure wine remain.

IIT Jodhpur Qualifier - BS

Time: 2 Hours | Maximum Marks: 198

Question Distribution

- **VARC:** 24 questions (suggested time 40 min)
- **DILR:** 18 questions (suggested time 40 min)
- **QA:** 24 questions (suggested time 40 min)
- Total: 66 questions | MCQ 60 | TITA (non-MCQ) 6

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Total duration is 2 Hours; the suggested per-section times are shown above.
- Sections may be attempted in any order; manage your time across all sections.

This is the answer-marked verification copy (correct option in green, with solution) and mirrors the content of the upload CSV.

Section: VARC (suggested 40 min)

Instructions (VARC-RC1): study the shared passage below and answer its 4 question(s).

Before the middle of the nineteenth century, most towns kept their own local time, setting clocks by the moment the sun stood highest overhead. A traveller moving west by carriage might find each new town a few minutes behind the last, and nobody minded, because journeys were slow enough that small differences never mattered. Railways changed that arithmetic. A train that left one city at noon and reached another whose clocks also read noon had, on paper, taken no time at all. Timetables became puzzles, and collisions on single-track lines were a real danger when two crews disagreed about when a siding had to be cleared.

The response was not a government decree but a private one. Railway companies began publishing schedules based on a single reference clock, usually the time kept at their largest terminus, and asked stations along the line to follow it. Towns that resisted found themselves consulting two clocks: one on the church tower for daily life and one at the station for departures. In practice the station clock won, because it governed the arrival of mail, freight and visitors.

Only later did legislatures ratify what commerce had already settled, dividing territories into broad zones within which every clock agreed. The reform is often described as a triumph of science, and astronomers did supply the measurements, but the pressure came from the ledger rather than the observatory. What made local time intolerable was not that it was inaccurate; by the standard of the sun it was perfectly accurate. It was that accuracy of that kind could not be shared. A network needs its members to agree on a convention, and the value of the convention lies almost entirely in the agreement, not in the truth of the thing agreed. That is why the change, once begun, spread quickly: each town that joined made joining more attractive for the rest.

Q1. [VARC-RC1 · mcsc · diff 0.5]

Which of the following best captures the central point of the passage?

- A) Uniform clock time spread because travellers grew tired of resetting their watches in every town they entered.
 - B) Uniform clock time spread because networks needed a shared convention, not because local time was wrong. ✓
 - C) Uniform clock time spread because governments saw that public safety required identical clocks everywhere.
 - D) Uniform clock time spread because astronomers at last produced measurements accurate enough to trust.
- Answer: B)** Uniform clock time spread because networks needed a shared convention, not because local time was wrong.

Solution: The passage insists that local solar time was accurate yet unshareable, and that railway networks needed agreement, so the convention mattered more than the truth of any clock. The astronomers version is rejected explicitly, since the pressure came from the ledger; the safety and traveller versions name effects the passage treats as symptoms, not as the driving reason.

Q2. [VARC-RC1 · mcsc · diff 0.5]

The passage attributes all of the following to the arrival of railways EXCEPT:

- A) confusion in drawing up timetables that different lines could actually work with
- B) danger of accidents when two crews disagreed about when a track would be clear
- C) a legal rule obliging church towers to show the same hour as the station clock ✓
- D) schedules published from a single reference clock kept at a large terminus

Answer: C) a legal rule obliging church towers to show the same hour as the station clock

Solution: The passage says legislatures later ratified broad zones; it never mentions any law about church towers, which simply lost influence in practice. Timetable puzzles, the collision risk on single-track lines, and schedules based on the terminus clock are all stated directly as consequences of the railways.

Q3. [VARC-RC1 · mcsc · diff 0.5]

It can be inferred from the passage that a town's church clock lost its authority mainly because:

- A) astronomers had declared solar time an unscientific way of setting clocks
- B) residents faced penalties for keeping time that differed from the railway's
- C) the church clock was found to be measurably slower than the railway's own clock
- D) the station clock decided when mail, goods and visitors would actually arrive ✓

Answer: D) the station clock decided when mail, goods and visitors would actually arrive

Solution: The passage says the station clock won because it governed the arrival of mail, freight and visitors, so practical dependence rather than any ruling settled the matter. Nothing suggests the church clock was slow, that penalties existed, or that astronomers condemned solar time; indeed solar time is called perfectly accurate.

Q4. [VARC-RC1 · mcsc · diff 0.5]

The author calls local time 'perfectly accurate' in order to:

- A) suggest that the towns were right to resist the schedules imposed on them
- B) concede that early railway schedules were built on faulty measurements
- C) praise the astronomers who first measured the sun's position from each town
- D) stress that its defect was not error but that it could not be shared ✓

Answer: D) stress that its defect was not error but that it could not be shared

Solution: The phrase sets up the next sentence, that accuracy of that kind could not be shared, so the concession about accuracy sharpens the argument about convention. The passage never calls railway measurements faulty, never endorses the resisting towns, and treats the astronomers as suppliers of data rather than as the point being made.

Instructions (VARC-RC2): study the shared passage below and answer its 4 question(s).

Beneath the floor of most forests lies a dense mesh of fungal threads, thinner than hair and far more numerous than roots. Many of these fungi wrap themselves around, or grow into, the roots of trees. The arrangement suits both sides: the fungus takes sugars that the tree makes in its leaves, and the tree gains water and minerals that the fungal threads, with their enormous surface area, can gather from soil far beyond a root's reach.

Because a single fungus can be linked to several trees at once, the threads also form a route between plants. Experiments in which one tree is fed carbon labelled so that it can be traced show that a small share of that carbon later turns up in neighbours connected to the same fungal partner. This finding has been retold in popular writing as evidence that forests behave like cooperatives, with vigorous trees deliberately feeding shaded seedlings.

The evidence supports something narrower. That a substance moves between two organisms does not establish that either one intended the transfer, or that the recipient benefited enough to matter. Carbon may simply follow a gradient, flowing from where it is abundant to where it is scarce, much as water does. The fungus, moreover, is not a neutral pipe. It is an organism with interests of its own, and it may retain much of what passes through, trading resources on terms that favour itself rather than any particular tree.

None of this makes the network unimportant. A seedling that gains even a modest supply of minerals in its first seasons may survive a drought it would otherwise not survive, and that is a large effect from a small transfer. The caution concerns the story we drape over the measurement. Describing the forest as generous is a metaphor; the data describe an exchange whose accounting has not yet been done.

Q5. [VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the main argument of the passage?

- A) Fungal links between trees are real, but claims that trees deliberately share food outrun the data. ✓
- B) Fungal links between trees mostly injure the trees, because the fungus keeps nearly every resource.
- C) Fungal links between trees are so faint that the transfers recorded are probably measurement error.
- D) Fungal links between trees prove that a forest works as a cooperative rather than as a competition.

Answer: A) Fungal links between trees are real, but claims that trees deliberately share food outrun the data.

Solution: The passage accepts that carbon moves along fungal threads and that the effect on seedlings can matter, while rejecting the added story of deliberate sharing, so the caution is about interpretation rather than about the measurement. Calling the transfers error contradicts the drought example; the cooperative reading is the one being criticised; and the fungus is said to retain much, not nearly all.

Q6. [VARC-RC2 · mcsc · diff 0.5]

According to the passage, the fungal threads do all of the following EXCEPT:

- A) gather water and minerals from soil that lies beyond the reach of the roots
- B) produce sugars for themselves rather than take them from the tree's leaves ✓
- C) hold back for themselves a share of the resources that move along them
- D) connect one tree to another when a single fungus is joined to both of them

Answer: B) produce sugars for themselves rather than take them from the tree's leaves

Solution: The passage states that the fungus takes sugars the tree makes in its leaves, so making its own sugars instead is exactly what it does not do. Collecting water and minerals from distant soil, linking several trees through one fungus, and retaining part of what passes through are each stated in the text.

Q7. [VARC-RC2 · mcsc · diff 0.5]

It can be inferred that the author regards the carbon-tracing experiments as:

- A) flawed in method and therefore in need of repetition by other research teams
- B) sound in what they measure but weak support for the claims built on them ✓
- C) beside the point, since the quantities involved cannot affect survival
- D) conclusive proof that mature trees support shaded seedlings on purpose

Answer: B) sound in what they measure but weak support for the claims built on them

Solution: The author never questions the tracing itself; the objection is that movement of a substance does not establish intention or benefit, which makes the results reliable but thinner than advertised. Nothing calls the method flawed, the deliberate-feeding reading is the one resisted, and the drought example shows small transfers can matter.

Q8. [VARC-RC2 · mcsc · diff 0.5]

The comparison with water flowing down a gradient serves mainly to:

- A) show that the fungal threads carry mostly water rather than dissolved sugar
- B) argue that carbon moves too slowly to reach a neighbouring tree in a season
- C) suggest that the labelled carbon was washed through the soil by heavy rain
- D) offer a mechanism for the transfer that needs no intention by either tree ✓

Answer: D) offer a mechanism for the transfer that needs no intention by either tree

Solution: The gradient image supplies a plain physical reason for carbon to travel from abundance to scarcity, which removes the need to credit any tree with a purpose. The passage does not claim the threads carry mainly water, does not raise the speed of movement, and does not attribute the traced carbon to rainfall.

Instructions (VARC-RC3): study the shared passage below and answer its 4 question(s).

For much of the twentieth century, planners in growing cities treated the bicycle as a relic. Streets were widened, parking was added, and the space left for anyone on two wheels shrank to a painted strip beside fast traffic. Cycling did not disappear, but it became a choice for the brave.

In the last three decades many of those same cities have reversed course. The reason is mostly arithmetic. A lane of road carries far more people per hour on bicycles than in cars, because a bicycle needs about a tenth of the space and, when parked, almost none. Cities that could not afford to widen a single street found they could move more residents simply by changing what the existing width was used for.

Comfort turned out to matter more than exhortation. Where a city merely painted lines on the asphalt, ridership rose a little. Where it separated the cycle lane from moving traffic with a kerb or a row of parked cars, ridership rose sharply, and the new riders included many more children and older people. Surveys give a consistent explanation: most people are willing to cycle but unwilling to cycle beside lorries.

The change has limits. Bicycles work best for trips under about eight kilometres, so a city spread thinly over a wide area will always need other options, and hills and heavy rain reduce the numbers. Freight and emergency vehicles still need road space. What the experience of recent decades suggests is narrower than a slogan: for short journeys in dense districts, the cheapest way to add capacity is often to give a little room to bicycles, and the room only works if it feels safe.

Q9. [VARC-RC3 · mcsc · diff 0]

Which of the following best expresses the main idea of the passage?

- A) Cities have returned to cycling because residents demanded a healthier way of travelling each day.
- B) Cities have returned to cycling because protected lanes move many people cheaply on short trips. ✓
- C) Cities have returned to cycling because bicycles can replace cars on journeys of almost any length.
- D) Cities have returned to cycling because widening roads for cars proved technically impossible to do.

Answer: B) Cities have returned to cycling because protected lanes move many people cheaply on short trips.

Solution: The passage explains the reversal by space efficiency plus the sharp rise in ridership where lanes are separated from traffic, which is exactly capacity gained cheaply for short trips. Health is never mentioned; widening is described as unaffordable rather than impossible; and the text limits bicycles to trips under about eight kilometres.

Q10. [VARC-RC3 · mcsc · diff 0]

The passage mentions all of the following as limits on cycling EXCEPT:

- A) sloping ground and spells of heavy rainfall
- B) the price of buying and keeping up a bicycle ✓
- C) space that freight and emergency vehicles need
- D) journeys longer than roughly eight kilometres

Answer: B) the price of buying and keeping up a bicycle

Solution: Cost of ownership never appears in the passage, which discusses road space rather than household budgets. The final paragraph names distance beyond about eight kilometres, hills and heavy rain, and the road space still required by freight and emergency vehicles.

Q11. [VARC-RC3 · mcsc · diff 0.5]

It can be inferred that painted cycle lanes attract fewer new riders because:

- A) they are built only in districts where very few residents own a bicycle
- B) they are usually narrower than the width a bicycle actually needs to use
- C) they leave riders beside moving traffic, which most people will not take ✓
- D) they are marked in colours that drivers find difficult to notice in time

Answer: C) they leave riders beside moving traffic, which most people will not take

Solution: Paint leaves the rider next to moving vehicles, and the surveys quoted say most people are willing to cycle but unwilling to cycle beside lorries, so the missing element is separation. Lane width, paint colour and the location of the lanes are never discussed in the passage.

Q12. [VARC-RC3 · mcsc · diff 0.5]

The tone of the final paragraph is best described as:

- A) enthusiastic, urging cities to replace their car traffic with bicycles entirely
- B) sceptical, doubting that the recent changes to city streets achieved anything
- C) measured, marking out where cycle lanes help instead of stating a general rule ✓
- D) detached, refusing to take any position at all on the question under review

Answer: C) measured, marking out where cycle lanes help instead of stating a general rule

Solution: The paragraph lists limits and then states a deliberately narrow claim, calling it narrower than a slogan, which is qualified endorsement rather than advocacy. Enthusiasm is ruled out by the limits, scepticism by the endorsement, and detachment by the fact that a clear recommendation is offered.

Q13. [VARC-ODD-1 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Many plants depend on animals to carry their seeds well away from the parent tree.
- (2) Fleshy fruits attract birds and mammals, which eat the pulp and drop the seeds elsewhere.
- (3) Other seeds carry small hooks that catch on fur and are brushed off much later.
- (4) Wind speed over open grassland is recorded by instruments mounted on tall poles.
- (5) Such journeys reduce the competition between a seedling and the plant that produced it.

Which sentence is the odd one out?

- A) (4) ✓
- B) (2)
- C) (5)
- D) (1)
- E) (3)

Answer: A) (4)

Solution: Sentences (1), (2), (3) and (5) form one chain: plants need animals to move seeds, two mechanisms for doing so (fruit pulp and hooks), and the benefit of the resulting distance. Sentence (4) is about instruments for measuring wind speed; it introduces neither seeds nor dispersal, so it is the misfit. (1) is the natural topic sentence and (5) the conclusion, so neither can be dropped without leaving the paragraph headless or unfinished; (2) and (3) are the paired examples that (1) demands.

Q14. [VARC-ODD-2 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Before railways, every town kept its own clock, set roughly by the position of the sun.
- (2) Athletes often train at altitude in the hope of raising the oxygen capacity of their blood.
- (3) Timetables grew confusing once a train could cross many such local times in one day.
- (4) Railway companies therefore imposed a single standard time along the whole of their lines.
- (5) Governments later turned these private arrangements into the time zones still used today.

Which sentence is the odd one out?

- A) (5)
- B) (2) ✓
- C) (4)
- D) (1)
- E) (3)

Answer: B) (2)

Solution: The coherent paragraph runs (1), (3), (4), (5): local sun time, the problem trains created, the railways' fix, and its later official form. Sentence (2), about altitude training and blood oxygen, belongs to a different subject entirely and is the misfit. (3) supplies the problem that (4) solves, and (5) closes the sequence, so removing any of these breaks the causal chain instead of tightening it.

Q15. [VARC-ODD-3 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Trees planted along city streets do far more than soften the look of concrete.
- (2) Their canopies shade the pavement and bring summer temperatures down noticeably.
- (3) Their roots take up rainwater that would otherwise overwhelm drains during a storm.
- (4) Their leaves also trap much of the dust that passing traffic throws into the air.
- (5) Architects still disagree about whether glass facades suit historic city centres.

Which sentence is the odd one out?

- A) (2)
- B) (4)
- C) (5) ✓
- D) (1)
- E) (3)

Answer: C) (5)

Solution: Sentences (1) to (4) form a unit: a claim that street trees are useful, then three uses (shade, drainage, dust capture). Sentence (5) stays in the city but switches to a dispute about building facades, mentioning no trees and no benefit, so it is the misfit. The three example sentences all begin from the tree itself and cannot be dropped without leaving (1) unsupported.

Q16. [VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Riders who still feel unsafe on the road go on avoiding it for months afterwards.
- (2) Planners therefore judge such projects over a span of years rather than weeks.
- (3) When a city paints a cycle lane along a busy road, the effect is rarely immediate.
- (4) Only as the numbers slowly build does the lane start to feel ordinary and safe.

Select the correct order.

- A) 3412
- B) 1432
- C) 3142 ✓
- D) 1342

Answer: C) 3142

Solution: Sentence (3) is the only one that introduces the cycle lane as a new subject, so it opens. Its claim that the effect is not immediate is explained by (1), which says riders keep away for months. (4) then marks the turn, as numbers build the lane feels normal, and (2) draws the planning conclusion with 'therefore'. Orders beginning 1 fail because (1) starts with 'Riders who still feel unsafe on the road', which presupposes the lane already named in (3); 3412 puts the slow build before the avoidance it is supposed to follow.

Q17. [VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) After about a week it is lively enough to raise a loaf without any added yeast.
- (2) A sourdough starter is nothing more than flour and water left to attract wild yeasts.
- (3) Bakers who keep feeding it can go on using the same starter for decades.
- (4) Once bubbles appear, the mixture is fed every day with a little fresh flour.

Select the correct order.

- A) 2431
- B) 4231
- C) 2413 ✓
- D) 4213

Answer: C) 2413

Solution: Only (2) defines the starter, so it must come first; the pronoun 'the mixture' in (4) refers back to that flour and water. The sequence then follows the timeline: bubbles and daily feeding in (4), readiness after a week in (1), and continued use over decades in (3). Any order starting with (4) leaves 'the mixture' with no antecedent, and 2431 puts the decades-long habit before the starter is even usable.

Q18. [VARC-JUMBLE-3 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Rising concern about the emissions of air travel then changed that calculation.
- (2) Cheap flights had made the sleeper look slow and expensive by comparison.
- (3) Several operators have since restored night routes between major European cities.
- (4) Night trains had almost disappeared from Europe by the early years of this century.

Select the correct order.

- A) 2413
- B) 4213 ✓
- C) 4231
- D) 2431

Answer: B) 4213

Solution: Sentence (4) states the situation to be explained, the near disappearance of night trains, and (2) gives its cause, cheap flights, while also depending on (4) for the term 'the sleeper'. (1) reverses the trend with 'then changed that calculation', and (3) reports the result with 'since'. Beginning with (2) leaves the comparison hanging, and placing (3) before (1) makes the revival precede the reason for it.

Q19. [VARC-JUMBLE-4 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Grams simply remove the guesswork that volume measures quietly introduce.
- (2) Bakers who want repeatable results therefore reach for a kitchen scale instead.
- (3) Recipes in many countries still ask the cook to measure flour by the cupful.
- (4) The same cup, though, holds different weights depending on how tightly it is packed.

Select the correct order.

- A) 4321
- B) 3421 ✓
- C) 3241
- D) 4231

Answer: B) 3421

Solution: Sentence (3) introduces the cup as a measure, and (4) picks it up with 'The same cup, though' to state the problem, so this pair is fixed and must stand in that order. (2) draws the practical response with 'therefore', and (1) closes by explaining why weight settles the matter. Orders beginning with (4) leave 'the same cup' without a first mention, and 3241 places the scale before the flaw that motivates it.

Q20. [VARC-SUMMARY-1 · mcsc · diff 0]

Open-plan offices were introduced partly to encourage conversation between colleagues. In practice many workers find the constant background noise distracting and put on headphones in order to concentrate, which cuts off the very exchanges the layout was meant to create.

Which option best summarises the paragraph?

- A) Workers in open-plan offices wear headphones so that they can concentrate on their own tasks.
- B) Offices ought to be rebuilt with private rooms so that colleagues can talk to one another freely.
- C) Open-plan offices defeat their own aim, since noise drives workers to shut themselves off. ✓
- D) Quiet matters more to office productivity than the conversation between colleagues ever did.

Answer: C) Open-plan offices defeat their own aim, since noise drives workers to shut themselves off.

Solution: The paragraph has two halves, an aim (more conversation) and an outcome that undoes it (noise, headphones, no exchanges), and only the self-defeating-purpose statement captures both. Saying merely that workers wear headphones to concentrate reports one detail and drops the irony that is the point. The call to rebuild offices with private rooms is a recommendation the paragraph never makes. Ranking quiet above conversation distorts the text, which treats the loss of conversation as a cost, not a preference.

Q21. [VARC-SUMMARY-2 · mcsc · diff 0.5]

Delivery apps bring a restaurant orders it would never otherwise have received. They also take a share of every bill, and customers who once walked in now stay at home, so a busy kitchen can finish the month having sold more meals and kept less money.

Which option best summarises the paragraph?

- A) Restaurants that use delivery apps must pay a commission on every single order received.
- B) Customers order from home because delivery apps charge less than restaurants themselves do.
- C) Delivery apps lift a restaurant's order count while thinning the margin it earns on each. ✓
- D) Restaurants should abandon delivery apps and serve only customers who come in person.

Answer: C) Delivery apps lift a restaurant's order count while thinning the margin it earns on each.

Solution: The paragraph's shape is a gain in volume set against a loss in earnings, which is exactly what the statement about more orders but thinner margins preserves. The commission point is only one of the two forces named and ignores the lost walk-in trade, so it is too narrow. Telling restaurants to abandon the apps is advice the paragraph does not give. The claim that apps charge customers less is invented; the text says nothing about relative prices.

Q22. [VARC-SUMMARY-3 · mcsc · diff 0.5]

Where beavers return to a river they build dams, and parts of the valley flood. Farmers lose the use of some low-lying fields, but the new ponds slow floodwater heading downstream and hold water through dry spells, so the towns below gain what the fields lose.

Which option best summarises the paragraph?

- A) Beavers should be returned to every river, since their dams end both flooding and drought.
- B) Beaver dams cost nearby farmland while handing downstream towns a gain in water security. ✓
- C) Beavers build dams that flood low-lying fields and take useful farmland out of production.
- D) Farmers gain more from beaver ponds than the towns downstream do in long spells of dry weather.

Answer: B) Beaver dams cost nearby farmland while handing downstream towns a gain in water security.

Solution: The paragraph balances a loss borne by farmers against a benefit received downstream, so the statement naming both the farmland cost and the town's water gain is the only complete summary. Stopping at flooded fields records the cost and omits the benefit that the word 'but' introduces. Recommending beavers for every river, and claiming they end flooding and drought outright, goes far beyond 'slow' and 'hold'. Putting the farmers ahead of the towns reverses the direction of benefit the text states.

Q23. [VARC-SUMMARY-4 · mcsc · diff 0.5]

A translator of poetry cannot keep everything. Sound, rhythm and literal sense pull in different directions, and holding on to one of them usually means loosening the grip on another. Every finished translation is therefore a record of the choices its translator was willing to make.

Which option best summarises the paragraph?

- A) Skilled translators of poetry do manage to preserve sound, rhythm and literal meaning at once.
- B) Poetry is best read in the original, because any translation destroys what really matters in it.
- C) Translators of poetry find it hard to reproduce the rhythm of the original in another language.
- D) A poem's sound and sense compete, so translation records the trade-offs a translator accepts. ✓

Answer: D) A poem's sound and sense compete, so translation records the trade-offs a translator accepts.

Solution: The paragraph argues that the demands of a poem conflict and that the finished translation therefore shows what its translator chose to sacrifice, which the trade-off statement states directly. Naming rhythm alone as the difficulty covers one third of the tension and misses the conclusion about choices. The advice to read only originals adds a verdict the text avoids, since it never calls translation destructive. Claiming that skilled translators keep all three contradicts the premise that one must be loosened.

Q24. [VARC-SUMMARY-5 · mcsc · diff 0.5]

Cities that switch to bright white LED streetlights cut their electricity bills at once. Insects, however, are drawn to that part of the spectrum far more strongly than to the old orange lamps, and the moths gathering under the new lights are eaten or exhausted well before dawn.

Which option best summarises the paragraph?

- A) Moths keep away from white LED lights and so survive better than they did under orange ones.
- B) Cities ought to bring back orange streetlamps in order to protect their declining moths.
- C) The savings from white LED street lighting carry a hidden cost for night-flying insects. ✓
- D) White LED streetlights consume markedly less electricity than the orange lamps they replace.

Answer: C) The savings from white LED street lighting carry a hidden cost for night-flying insects.

Solution: The paragraph pairs an immediate financial benefit with a biological cost, and the statement about savings carrying a hidden cost to insects holds both halves together. Reporting only the lower consumption keeps the first sentence and discards the second. Urging a return to orange lamps is a policy proposal the text never advances. Saying moths keep away from the white lights inverts the paragraph, which has them gathering there and dying.

Section: DILR (suggested 40 min)

Instructions (DILR-1): study the shared data below and answer its 3 question(s).

Five runners, Farid, Lena, Mirko, Priya and Tomas, finished a 5 km club race in five different positions, from first to fifth. No two runners finished together.

- Priya finished ahead of Lena, and Lena finished ahead of Mirko. - Exactly one runner finished between Farid and Tomas, with Farid ahead of Tomas. - Priya did not finish first.

Q25. [DILR-1 · mcsc · diff 0.5]

Which runner finished third?

- A) Priya
- B) Tomas ✓
- C) Farid
- D) Lena

Answer: B) Tomas

Solution: The Farid-Tomas gap of exactly one runner allows the pairs 1-3, 2-4 or 3-5. Combining with Priya ahead of Lena ahead of Mirko and Priya not first leaves only Farid first, Priya second, Tomas third, Lena fourth, Mirko fifth. Priya is second, Lena fourth, and Farid first, so each of those choices misreads one clue.

Q26. [DILR-1 · mcsc · diff 0.5]

In which position did Lena finish?

- A) Fourth ✓
- B) Second
- C) Fifth
- D) Third

Answer: A) Fourth

Solution: The unique order is Farid, Priya, Tomas, Lena, Mirko, so Lena is fourth. Third belongs to Tomas and second to Priya; fifth comes from forgetting that Mirko must finish behind Lena.

Q27. [DILR-1 · mcsc · diff 0.5]

Which of the following statements is true?

- A) Lena finished immediately behind Tomas but ahead of Priya
- B) Tomas finished immediately ahead of Priya
- C) Mirko finished immediately behind Lena ✓
- D) Farid finished two places behind Priya

Answer: C) Mirko finished immediately behind Lena

Solution: In the order Farid, Priya, Tomas, Lena, Mirko, positions four and five are Lena and Mirko, so Mirko is immediately behind Lena. Tomas is behind Priya, not ahead of her; Farid is ahead of Priya rather than two places behind; and Lena finishes behind Priya, not ahead of her.

Instructions (DILR-2): study the shared data below and answer its 3 question(s).

Four members of a college club, Anya, Boris, Carla and Dev, earned points in a month-long fundraising drive. Together they earned 120 points.

- Boris earned 6 points more than Carla. - Anya earned exactly twice as many points as Dev. - Dev earned 16 points.

Q28. [DILR-2 · mcsc · diff 0.5]

How many points did Carla earn?

- A) 36
- B) 33 ✓
- C) 30
- D) 39

Answer: B) 33

Solution: Dev earned 16, so Anya earned 32 and the remaining 72 points are split between Boris and Carla with Boris 6 ahead: Carla 33 and Boris 39. The 30 figure comes from splitting 120 carelessly, 36 repeats Anya's total doubled wrongly, and 39 is Boris's total, not Carla's.

Q29. [DILR-2 · mcsc · diff 0.5]

Who earned the most points?

- A) Carla
- B) Dev
- C) Boris ✓
- D) Anya

Answer: C) Boris

Solution: The totals are Anya 32, Boris 39, Carla 33 and Dev 16, so Boris leads. Anya and Carla are close behind but neither tops 39, and Dev has the smallest total of all four members.

Q30. [DILR-2 · mcsc · diff 0.5]

By how many points does the highest total exceed the lowest total?

- A) 23 ✓
- B) 27
- C) 7
- D) 17

Answer: A) 23

Solution: Boris tops the list with 39 and Dev is lowest with 16, a gap of 23 points. The 7 comes from comparing Boris with Carla, 17 from using Anya as the top scorer, and 27 from subtracting Carla's total from Boris and Dev incorrectly.

Instructions (DILR-3): study the shared data below and answer its 4 question(s).

A beverage wholesaler reports last month's trade at its three depots. Table 1 gives the number of packs sold, in hundreds of packs. Table 2 gives the selling price of one pack, in euros, at that depot.

Table 1: Packs sold (in hundreds)

Category	Verona	Padua	Ferrara
Espresso blend	12	15	9
Herbal tea	8	6	10
Cocoa	5	9	11

Table 2: Price per pack (in euros)

Category	Verona	Padua	Ferrara
Espresso blend	20	20	25
Herbal tea	10	15	10
Cocoa	30	20	20

Q31. [DILR-3 · mcsc · diff 0.5]

How many hundreds of packs were sold in total at the Ferrara depot?

- A) 30 ✓
- B) 28
- C) 25
- D) 32

Answer: A) 30

Solution: The Ferrara column of Table 1 holds 9 packs of espresso blend, 10 of herbal tea and 11 of cocoa, so the depot sold $9 + 10 + 11 = 30$ hundred packs. The 25 figure is the Verona column total, 28 drops the cocoa row to 9, and 32 comes from reading the Padua espresso entry of 15 in place of 9.

Q32. [DILR-3 · mcsc · diff 0.5]

Which depot and category together generated the highest revenue?

- A) Padua, espresso blend ✓
- B) Ferrara, cocoa
- C) Verona, espresso blend
- D) Ferrara, espresso blend

Answer: A) Padua, espresso blend

Solution: Revenue for a cell is packs sold times price, in hundreds of euros: Padua espresso gives $15 \times 20 = 300$, Verona espresso $12 \times 20 = 240$, Ferrara espresso $9 \times 25 = 225$ and Ferrara cocoa $11 \times 20 = 220$. Padua espresso blend is therefore the largest at 30,000 euros; the Verona and Ferrara espresso entries trap candidates who compare only volumes or only prices.

Q33. [DILR-3 · mcsc · diff 0.5]

What was the combined revenue from cocoa across the three depots, in euros?

- A) 52,000
- B) 61,000
- C) 55,000 ✓
- D) 58,000

Answer: C) 55,000

Solution: Cocoa revenue in hundreds of euros is $5 \times 30 = 150$ at Verona, $9 \times 20 = 180$ at Padua and $11 \times 20 = 220$ at Ferrara, a total of 550 hundred euros, that is 55,000 euros. The 52,000 value comes from pricing Ferrara cocoa at 18 euros, 58,000 from swapping the Verona and Padua prices, and 61,000 from applying the 30 euro price to every depot.

Q34. [DILR-3 · mcsc · diff 0.5]

In how many of the nine depot and category combinations did revenue exceed 20,000 euros?

- A) 3
- B) 4 ✓
- C) 6
- D) 5

Answer: B) 4

Solution: The nine revenues in hundreds of euros are 240, 300, 225 for espresso blend, 80, 90, 100 for herbal tea and 150, 180, 220 for cocoa. Exceeding 20,000 euros means exceeding 200 hundred euros, which 240, 300, 225 and 220 do, giving 4 combinations. The count of 3 drops Ferrara cocoa at 220, while 5 and 6 wrongly admit the 180 and 150 cocoa figures.

Instructions (DILR-4): study the shared data below and answer its 3 question(s).

Four volunteers, Ines, Karim, Nadia and Omar, each work exactly one lab shift during a week, on Monday, Tuesday, Wednesday and Thursday, one volunteer per day. Each is also assigned a different single duty: filing, inventory, reception or training.

- Karim works on the day immediately after Nadia's shift. - Ines works neither on Monday nor on Thursday. - Omar's shift falls earlier in the week than Ines's shift. - The volunteer working on Wednesday is assigned inventory. - Ines is assigned reception, and Omar is assigned training.

Q35. [DILR-4 · mcsc · diff 0.5]

Which volunteer works on Monday?

- A) Omar ✓
- B) Karim
- C) Ines
- D) Nadia

Answer: A) Omar

Solution: Ines must work Tuesday or Wednesday, and Omar works before her. If Ines took Wednesday, the two days left for Nadia and Karim would not be consecutive, so Ines works Tuesday and Omar takes Monday, leaving Nadia on Wednesday and Karim on Thursday.

Q36. [DILR-4 · mcsc · diff 0.5]

Which volunteer and duty are correctly paired?

- A) Omar with reception
- B) Karim with inventory
- C) Nadia with filing
- D) Karim with filing ✓

Answer: D) Karim with filing

Solution: The schedule is Omar Monday, Ines Tuesday, Nadia Wednesday and Karim Thursday. Wednesday carries inventory, so that duty is Nadia's, while Ines has reception and Omar training; the only duty left for Karim is filing.

Q37. [DILR-4 · mcsc · diff 1]

Suppose Ines and Karim exchange shift days, every duty staying with the same person. Who then works on the day immediately before the inventory shift?

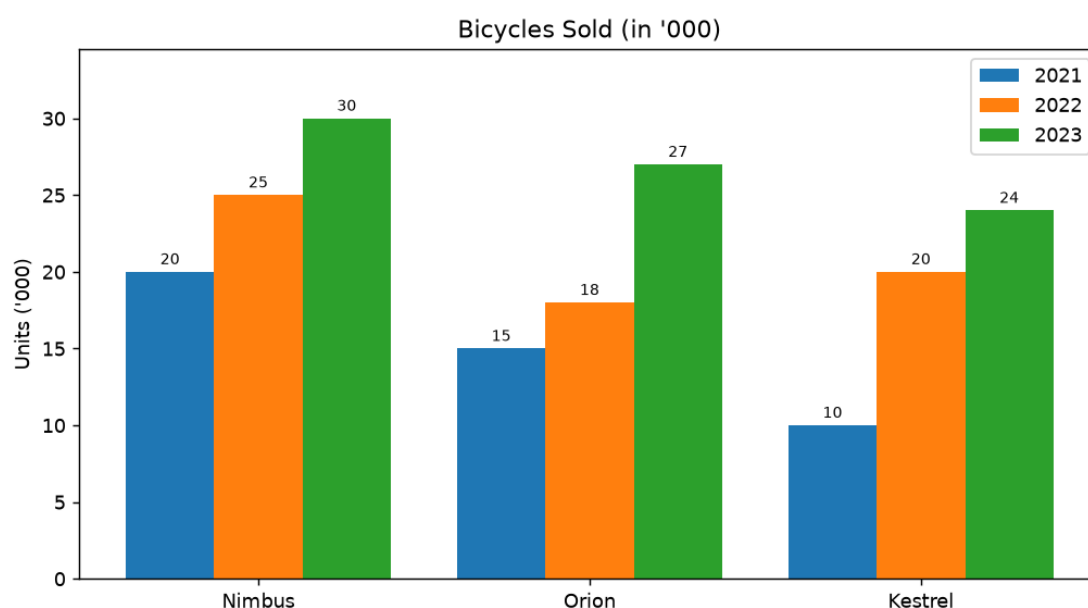
- A) Karim ✓
- B) Nadia
- C) Omar
- D) Ines

Answer: A) Karim

Solution: Originally Omar works Monday, Ines Tuesday, Nadia Wednesday and Karim Thursday, with inventory on Wednesday. After the exchange Karim works Tuesday and Ines Thursday, while Nadia keeps inventory on Wednesday, so Karim is the volunteer working the day before it. Ines has moved to Thursday, Omar stays on Monday, and Nadia holds the inventory shift itself.

Instructions (DILR-5): study the shared data below and answer its 5 question(s).

The bar chart shows the number of bicycles sold, in thousands, for three models of a manufacturer in each of the years 2021, 2022 and 2023.



Q38. [DILR-5 · mcsc · diff 0.5]

How many thousand bicycles were sold in total across the three models in 2022?

- A) 63 ✓
- B) 65
- C) 60
- D) 58

Answer: A) 63

Solution: For 2022 the bars read 25 thousand for Nimbus, 18 thousand for Orion and 20 thousand for Kestrel, so the total is $25 + 18 + 20 = 63$ thousand. The 58 figure keeps the 2021 Kestrel bar of 10 plus a misread, 60 rounds each bar to the nearest ten before adding, and 65 uses the 2021 Orion value of 15 with a doubled Kestrel reading.

Q39. [DILR-5 · mcsc · diff 0.5]

By what percentage did Kestrel sales grow from 2021 to 2023?

- A) 120%
- B) 140% ✓
- C) 58%
- D) 240%

Answer: B) 140%

Solution: Kestrel sales rise from 10 thousand in 2021 to 24 thousand in 2023, an increase of 14 thousand on a base of 10 thousand, which is 140%. The 240% value reports the 2023 figure as a percentage of the 2021 figure rather than the growth, 120% uses the 2022 value of 20 as the endpoint, and 58% divides the rise of 14 by the later year figure of 24 instead of the base.

Q40. [DILR-5 · mcsc · diff 0.5]

What is the ratio of total Orion sales over the three years to total Nimbus sales over the three years?

- A) 3 : 5
- B) 5 : 4
- C) 4 : 5 ✓
- D) 2 : 3

Answer: C) 4 : 5

Solution: Orion totals $15 + 18 + 27 = 60$ thousand and Nimbus totals $20 + 25 + 30 = 75$ thousand, so the ratio is 60 : 75, which reduces to 4 : 5. The 5 : 4 option reverses the two models, 2 : 3 comes from comparing only the 2021 bars of 15 and 20 wrongly reduced, and 3 : 5 arises from using the 2023 Orion bar against the Nimbus total.

Q41. [DILR-5 · mcsc · diff 0.5]

What were the average annual sales of Kestrel, in thousands, over the three years?

- A) 20
- B) 18 ✓
- C) 17
- D) 16

Answer: B) 18

Solution: Kestrel sells 10, 20 and 24 thousand units in the three years, a total of 54 thousand, so the average is 54 divided by 3, that is 18 thousand. The 16 value drops the 2023 bar to 18, 20 averages only the last two years, and 17 comes from taking the midpoint of the smallest and largest bars instead of the true mean.

Q42. [DILR-5 · mcsc · diff 0.5]

By how many thousand units did the three-model total in 2023 exceed the three-model total in 2021?

- A) 34
- B) 40
- C) 38
- D) 36 ✓

Answer: D) 36

Solution: The 2023 bars add to $30 + 27 + 24 = 81$ thousand and the 2021 bars add to $20 + 15 + 10 = 45$ thousand, so the excess is 81 minus 45, that is 36 thousand. The 34 value misreads the 2023 Orion bar as 25, 38 uses 2022 in place of 2021 for one model, and 40 comes from comparing 2023 against a rounded 2021 total of 41.

Section: QA (suggested 40 min)

Q43. [QA-MCQ-1 · mcsc · diff 0]

A school has 640 students, and 35% of them travel to school by bus. How many students travel by bus?

- A) 192
- B) 416
- C) 448
- D) 224 ✓

Answer: D) 224

Solution: 35% of 640 equals 640 multiplied by 35/100, which is 224 students, so 224 is correct. The value 416 counts the 65% who do not use the bus. The value 448 comes from taking 70% instead of 35%, and 192 comes from using 30% of 640.

Q44. [QA-MCQ-2 · mcsc · diff 0]

A trader buys a chair for 250 euros and sells it at a profit of 15% on the cost price. What is the selling price, in euros?

- A) 265.00
- B) 275.00
- C) 212.50
- D) 287.50 ✓

Answer: D) 287.50

Solution: The profit is 15% of 250, that is 37.50, so the selling price is 250 plus 37.50, giving 287.50 euros. The figure 212.50 applies a 15% loss instead of a profit, 265.00 adds a flat 15 euros rather than 15 percent, and 275.00 uses a 10% profit.

Q45. [QA-MCQ-3 · mcsc · diff 0]

A sum of 3300 euros is divided between Anna and Bruno in the ratio 4 : 7. How many euros does Bruno receive?

- A) 2100 ✓
- B) 1650
- C) 2200
- D) 1200

Answer: A) 2100

Solution: The 11 equal parts are worth $3300/11 = 300$ euros each, so Bruno's 7 parts come to 2100 euros. The value 1200 is Anna's 4 parts, 1650 splits the sum equally and ignores the ratio, and 2200 comes from using the fraction 2/3 instead of 7/11.

Q46. [QA-MCQ-4 · mcsc · diff 0]

Six readings taken during an experiment add up to 261. What is the average of the six readings?

- A) 37.3
- B) 52.2
- C) 44.5
- D) 43.5 ✓

Answer: D) 43.5

Solution: The average is the total divided by the count, so 261 divided by 6 gives 43.5. The value 52.2 divides by 5 readings, 37.3 divides by 7 readings, and 44.5 comes from rounding the total up to 267 before dividing.

Q47. [QA-MCQ-5 · mcsc · diff 0]

A circular flower bed has a diameter of 14 metres. What is the length of the fence needed to enclose it exactly once? (take $\pi = 22/7$)

- A) 22 m
- B) 154 m
- C) 88 m
- D) 44 m ✓

Answer: D) 44 m

Solution: The fence equals the circumference, which is π times the diameter, that is $(22/7)$ times 14 = 44 metres. The value 88 m treats 14 as the radius, 154 m is the area in square metres rather than a length, and 22 m halves the circumference.

Q48. [QA-MCQ-6 · mcsc · diff 0]

Two bells ring at intervals of 15 seconds and 18 seconds. If they ring together once, after how many seconds will they next ring together?

- A) 45
- B) 270
- C) 180
- D) 90 ✓

Answer: D) 90

Solution: They coincide after the LCM of 15 and 18, and since $15 = 3 \times 5$ and $18 = 2 \times 3 \times 3$, the LCM is $2 \times 3 \times 3 \times 5 = 90$ seconds. The value 270 is the plain product 15×18 without removing the common factor 3, 45 is only a multiple of 15, and 180 is twice the LCM.

Q49. [QA-MCQ-7 · mcsc · diff 0.5]

A sum of 12000 euros is lent at 8% per annum simple interest. What total amount, in euros, is repaid at the end of 2 years and 6 months?

- A) 14400 ✓
- B) 13920
- C) 14880
- D) 2400

Answer: A) 14400

Solution: The time is 2.5 years, so the interest is $12000 \times 8 \times 2.5 / 100 = 2400$ euros and the amount repaid is $12000 + 2400 = 14400$ euros. The value 13920 uses only 2 years, 14880 rounds the period up to 3 years, and 2400 is the interest alone without the principal.

Q50. [QA-MCQ-8 · mcsc · diff 0.5]

A cyclist rides from a village to a market 30 km away at 15 km/h and returns along the same road at 10 km/h. What is the average speed for the whole trip?

- A) 12.5 km/h
- B) 25 km/h
- C) 12 km/h ✓
- D) 11 km/h

Answer: C) 12 km/h

Solution: The trip takes $30/15 = 2$ hours out and $30/10 = 3$ hours back, so 60 km in 5 hours gives 12 km/h. The 12.5 km/h figure averages the two speeds directly, which is wrong because the times differ, 25 km/h adds the speeds, and 11 km/h comes from dividing 55 by 5.

Q51. [QA-MCQ-9 · mcsc · diff 0.5]

A water tank is $3/5$ full. After 24 litres are drawn out, the tank is $1/3$ full. What is the full capacity of the tank, in litres?

- A) 40
- B) 120
- C) 72
- D) 90 ✓

Answer: D) 90

Solution: The 24 litres correspond to $3/5$ minus $1/3$, that is $4/15$ of the capacity, so the capacity is 24 divided by $4/15 = 90$ litres. The value 40 comes from treating 24 litres as $3/5$ of the tank, 72 treats it as $1/3$ of the tank, and 120 comes from using a difference of $1/5$ instead of $4/15$.

Q52. [QA-MCQ-10 · mcsc · diff 0.5]

A sum of 12000 euros is lent for 2 years at 10% per annum. By how many euros does the compound interest, compounded annually, exceed the simple interest for the same period?

- A) 120 euros ✓
- B) 240 euros
- C) 252 euros
- D) 100 euros

Answer: A) 120 euros

Solution: Simple interest is $12000 \times 0.10 \times 2 = 2400$ euros. Compound interest is $12000 \times 1.1 \times 1.1 - 12000 = 2520$ euros, so the gap is 120 euros, which is simply the interest earned in year 2 on the first year's interest of 1200 euros. The 240 euro figure comes from doubling the gap by wrongly charging interest on interest twice, 252 euros is one tenth of the whole compound amount rather than the excess, and 100 euros ignores that the base for the extra interest is 1200 euros.

Q53. [QA-MCQ-11 · mcsc · diff 0.5]

Tap A alone can fill an empty tank in 15 minutes, while an outlet pipe B alone can empty a full tank in 20 minutes. If both are opened together when the tank is empty, how long will the tank take to fill completely?

- A) 60 minutes ✓
- B) 35 minutes
- C) 45 minutes
- D) 30 minutes

Answer: A) 60 minutes

Solution: In one minute tap A adds $1/15$ of the tank and pipe B removes $1/20$, so the net filling rate is $1/15 - 1/20 = 1/60$ of the tank per minute, giving 60 minutes. The 35 minute value adds the two times instead of combining the rates, 30 minutes doubles the smaller time by averaging the rates, and 45 minutes wrongly subtracts the times as $20 + 15 + 10$.

Q54. [QA-MCQ-12 · mcsc · diff 0.5]

A laboratory has 40 litres of a solution in which acid forms 15% of the volume. How many litres of pure water must be added so that acid forms exactly 10% of the resulting solution?

- A) 5 litres
- B) 10 litres
- C) 15 litres
- D) 20 litres ✓

Answer: D) 20 litres

Solution: The acid content stays fixed at $40 \times 0.15 = 6$ litres. For 6 litres to be 10% of the new volume the new volume must be 60 litres, so 20 litres of water is added. The 10 litre answer comes from treating the 5 percentage point drop as a 25% volume increase, 15 litres mixes up the concentration figure with a volume, and 5 litres assumes acid volume also changes.

Q55. [QA-MCQ-13 · mcsc · diff 0.5]

A box holds 5 red marbles and 3 green marbles. One marble is drawn at random, its colour is noted, and it is put back before a second marble is drawn. What is the probability that the two marbles drawn have the same colour?

- A) $17/32$ ✓
- B) $1/2$
- C) $13/28$
- D) $15/64$

Answer: A) $17/32$

Solution: With replacement each draw is independent, so $P(\text{both red}) = (5/8)(5/8) = 25/64$ and $P(\text{both green}) = (3/8)(3/8) = 9/64$. Adding gives $34/64 = 17/32$. The value $13/28$ is the answer for drawing without replacement, $15/64$ is the chance of one red and one green in a fixed order, and $1/2$ assumes the two colours are equally likely.

Q56. [QA-MCQ-14 · mcsc · diff 0.5]

A committee of 3 members is to be chosen from 5 men and 4 women. In how many ways can this be done if the committee must include at least one woman?

- A) 64
- B) 74 ✓
- C) 84
- D) 40

Answer: B) 74

Solution: Total selections of 3 from 9 people number 84, and those with no woman number 10, being the selections of 3 from the 5 men, so $84 - 10 = 74$ committees contain at least one woman. The 84 figure forgets the restriction, 64 subtracts the men themselves rather than the all-male triples, and 40 counts only one woman with two men.

Q57. [QA-MCQ-15 · mcsc · diff 0.5]

A closed solid cylinder has a base radius of 7 cm and a height of 10 cm. Taking $\pi = 22/7$, what is its total surface area?

- A) 594 square cm
- B) 440 square cm
- C) 902 square cm
- D) 748 square cm ✓

Answer: D) 748 square cm

Solution: Total surface area equals $2\pi r(h + r) = 2 \times (22/7) \times 7 \times 17 = 44 \times 17 = 748$ square cm. The 440 value is only the curved surface, 594 adds just one circular face instead of two, and 902 uses the volume-style factor by taking $h + 2r$ as the bracket.

Q58. [QA-MCQ-16 · mcsc · diff 1]

What is the smallest positive integer that leaves a remainder of 3 when divided by each of 6, 7 and 8, and is also exactly divisible by 11?

- A) 1683 ✓
- B) 1680
- C) 2523
- D) 171

Answer: A) 1683

Solution: Numbers leaving remainder 3 on division by 6, 7 and 8 have the form $168k + 3$, since the LCM of 6, 7 and 8 is 168. For divisibility by 11, $168k + 3$ leaves $3k + 3$ modulo 11, which is zero when $k = 10$, giving $1680 + 3 = 1683 = 11 \times 153$. The 1680 value drops the remainder, 171 is the first multiple of 11 in the sequence only if the modulus is mistaken as 84, and 2523 is the next valid number, not the smallest.

Q59. [QA-MCQ-17 · mcsc · diff 1]

Working in pairs, A and B finish a job in 12 days, B and C finish the same job in 15 days, and A and C finish it in 20 days. Working all three together, in how many days will they finish the job?

- A) 12 days
- B) 5 days
- C) 10 days ✓
- D) 16 days

Answer: C) 10 days

Solution: Adding the three pair rates gives $1/12 + 1/15 + 1/20 = 1/5$ of the job per day, but this counts every worker twice, so the combined rate of the three is $1/10$ per day and the time is 10 days. The 5 day answer forgets to halve the doubled rate, 12 days just repeats the fastest pair's time, and 16 days averages the three given times.

Q60. [QA-MCQ-18 · mcsc · diff 1]

In an arithmetic progression the sum of the first 15 terms is 600 and the 5th term is 22. What is the 20th term of this progression?

A) 112 ✓

B) 118

C) 100

D) 106

Answer: A) 112

Solution: The sum of 15 terms equals 15 times the 8th term, so the 8th term is 40. From the 5th term 22 to the 8th term 40 there are three common differences, so $d = 6$ and the first term is $22 - 24 = -2$. Hence the 20th term is $-2 + 19(6) = 112$. The 106 value stops at the 19th term, 118 goes one term too far, and 100 uses the 8th term as the first term.

Q61. [QA-TITA-1 · tita · diff 0]

In a geometric progression the first term is 5 and the common ratio is 3. What is the fourth term? Type your answer as an integer.

Answer (TITA): 135

Solution: Each term is three times the one before it, so the sequence runs 5, 15, 45, 135. Using the formula, the fourth term is $5 \times 3^3 = 5 \times 27 = 135$. Multiplying by the ratio four times instead of three gives 405, and adding 3 each time as if the progression were arithmetic gives 14.

Q62. [QA-TITA-2 · tita · diff 0]

What is the sum of all even numbers from 2 to 40, both included? Type your answer as an integer.

Answer (TITA): 420

Solution: The even numbers 2, 4, ..., 40 form 20 terms. Their sum is $2 \times (1 + 2 + \dots + 20) = 2 \times 210 = 420$. Equivalently, the average of the first term 2 and the last term 40 is 21, and 21×20 terms gives 420. Counting $40/2 = 20$ terms is the step most often missed; using 40 terms leads to the wrong total 840.

Q63. [QA-TITA-3 · tita · diff 0.5]

The population of a town is 12500. During the first year it rises by 20%, and during the second year it falls by 10%. What is the population at the end of the second year? Type your answer as an integer.

Answer (TITA): 13500

Solution: After the first year the population is $12500 \times 1.20 = 15000$. The 10% fall in the second year is applied to 15000, not to the original 12500, so the final figure is $15000 \times 0.90 = 13500$. Adding the two percentages into a single 10% net rise gives 13750, which is wrong because the second change uses the larger base 15000.

Q64. [QA-TITA-4 · tita · diff 0.5]

How many four-digit numbers can be formed using the digits 0, 1, 2, 3 and 4 if no digit is repeated in a number? Type your answer as an integer.

Answer (TITA): 96

Solution: The leading digit cannot be 0, so it has 4 choices from 1, 2, 3, 4. The remaining three places are filled from the four unused digits in $4 \times 3 \times 2 = 24$ ways. The count is $4 \times 24 = 96$. Ignoring the leading-zero restriction gives $5 \times 4 \times 3 \times 2 = 120$, and the 24 difference is exactly the strings beginning with 0.

Q65. [QA-TITA-5 · tita · diff 0.5]

A train 180 m long crosses a platform 240 m long in 21 seconds. What is the speed of the train in km/h? Type your answer as an integer.

Answer (TITA): 72

Solution: To clear the platform the train covers its own length plus the platform length, that is $180 + 240 = 420$ m, in 21 seconds. The speed is $420/21 = 20$ m/s, and multiplying by $18/5$ converts this to 72 km/h. Using only the 240 m platform gives $240/21$ m/s, about 41 km/h, and forgetting the $18/5$ conversion leaves the answer stuck at 20.

Q66. [QA-TITA-6 · tita · diff 1]

What is the smallest positive integer that leaves a remainder of 3 when divided by 5, a remainder of 4 when divided by 6, and a remainder of 5 when divided by 7? Type your answer as an integer.

Answer (TITA): 208

Solution: In each condition the remainder falls short of the divisor by exactly 2, since $5 - 3 = 6 - 4 = 7 - 5 = 2$. So the number plus 2 is divisible by 5, 6 and 7, whose LCM is 210. The smallest positive value is $210 - 2 = 208$, and a check confirms $208 = 41 \times 5 + 3$, $34 \times 6 + 4$, $29 \times 7 + 5$. Adding 2 instead of subtracting gives 212, and taking the LCM alone gives 210.

IIT Jodhpur Qualifier - MBA (Technology)

Time: 2 Hours | Maximum Marks: 204

Question Distribution

- **VARC:** 24 questions (suggested time 40 min)
- **DILR:** 22 questions (suggested time 40 min)
- **QA:** 22 questions (suggested time 40 min)
- Total: 68 questions | MCQ 63 | TITA (non-MCQ) 5

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Total duration is 2 Hours; the suggested per-section times are shown above.
- Sections may be attempted in any order; manage your time across all sections.

This is the answer-marked verification copy (correct option in green, with solution) and mirrors the content of the upload CSV.

Section: VARC (suggested 40 min)

Instructions (VARC-RC1): study the shared passage below and answer its 5 question(s).

Economists have long treated their instruments as mirrors: a price index reflects inflation, an unemployment rate reflects idleness, a credit score reflects the risk that a borrower will default. The metaphor is convenient and, within limits, harmless. It becomes misleading the moment a measure acquires an audience. Once a credit score governs access to a mortgage, borrowers begin to manage the score rather than the prudence it was built to detect; lenders, in turn, price loans off the managed score. What began as a description of behaviour becomes one of the pressures that shape it. The measure is not merely inaccurate; it is productive.

It is tempting to read this as an indictment of quantification, and some critics do. The inference is too quick. Nothing in the argument shows that unmeasured judgment is more faithful to the world; loan officers working on intuition were biased in ways no one audited, and their errors, being invisible, were also uncorrectable. What the argument establishes is narrower and more awkward: the accuracy of a social measure is not a fixed property of the measure but a function of how consequential it has been allowed to become. A statistic used only by researchers may describe reasonably well what it claims to describe. The same statistic, once written into a contract or a regulation, begins to erode its own validity, and does so at a rate that has little to do with the care taken in its construction.

This has an uncomfortable corollary for institutional design. If validity decays with use, then the standard remedy, refining the metric, addresses the wrong variable. Better specification may buy time; it cannot arrest the process, because the process is driven not by measurement error but by the incentives that consequence creates. The alternatives are unattractive in different ways: rotate metrics before they are gamed, and lose comparability across periods; keep several imperfect indicators in play, and abandon the crisp accountability a single number provides; or make the metric one input among several to a judgment that remains discretionary, and therefore hard to challenge. None of these is a solution. Each is a choice about which kind of failure a system prefers to live with.

Q1. [VARC-RC1 · mcsc · diff 0.5]

The passage is primarily concerned with establishing which of the following?

- A) Credit scoring shows how a poorly constructed statistic can end up distorting the very market that comes to depend on it.
- B) Because quantitative indicators are routinely gamed, consequential decisions are better left to experienced human judgment.
- C) Social measures fail chiefly because those who design them pay too little attention to how the underlying quantity is defined.
- D) A measure's validity falls once it is given consequences, so refining its design cannot fix the problem. ✓

Answer: D) A measure's validity falls once it is given consequences, so refining its design cannot fix the problem.

Solution: The passage argues that accuracy is a function of how consequential a measure has been allowed to become, and that refining the metric addresses the wrong variable. The appeal to human judgment is blocked by the paragraph on unaudited loan officers. Credit scoring is one illustration, not the thesis, and the passage explicitly says decay has little to do with care in construction.

Q2. [VARC-RC1 · mcsc · diff 0.5]

The reference to loan officers working on intuition serves mainly to:

- A) suggest that decisions taken without written records can never be usefully audited
- B) establish that discretionary human judgment was historically more faithful to the world
- C) illustrate the process by which lenders learn to price loans off a managed score
- D) block the conclusion that dispensing with measurement would improve accuracy ✓

Answer: D) block the conclusion that dispensing with measurement would improve accuracy

Solution: The sentence follows 'The inference is too quick' and exists to deny the anti-quantification reading: unmeasured judgment was biased and its errors were uncorrectable. Saying intuition was more faithful reverses the point; the pricing of managed scores belongs to the earlier example; and 'never be usefully audited' overstates a remark about invisibility.

Q3. [VARC-RC1 · mcsc · diff 1]

The author would agree with all of the following EXCEPT:

- A) A well specified metric retains its validity indefinitely if it is applied consistently. ✓
- B) A statistic may describe its object reasonably well while its use stays confined to research.
- C) Errors produced by unaudited discretion are harder to correct than errors of formal measurement.
- D) Relying on several imperfect indicators at once costs a system some of its clarity of accountability.

Answer: A) A well specified metric retains its validity indefinitely if it is applied consistently.

Solution: The passage says better specification may buy time but cannot arrest the decay, so indefinite validity from good specification is exactly what is denied. The other three restate claims made about research-only statistics, invisible and therefore uncorrectable intuition, and the loss of crisp accountability when several indicators are kept in play.

Q4. [VARC-RC1 · mcsc · diff 1]

It can be inferred that the author regards the erosion of a measure's validity as:

- A) incentive-driven, and hence beyond the reach of purely technical refinement ✓
- B) a consequence of borrowers and lenders misunderstanding what a score measures
- C) an unavoidable feature of every attempt to describe social behaviour in numbers
- D) slower in those cases where the measure has been constructed with greater care

Answer: A) incentive-driven, and hence beyond the reach of purely technical refinement

Solution: The third paragraph attributes the process to the incentives that consequence creates rather than to measurement error, which is why refinement addresses the wrong variable. Erosion is tied to consequential use, not to all quantification, so the universal claim overreaches; the passage denies the link with care of construction; and misunderstanding is never offered as the mechanism.

Q5. [VARC-RC1 · mcsc · diff 1]

Which of the following, if true, would most weaken the author's corollary for institutional design?

- A) Indicators published but never written into contracts decay at the same rate as those that are. ✓
- B) Regulators in practice revise their key indicators as soon as any clear evidence of widespread gaming reaches them.
- C) Firms devote considerable resources to learning exactly how scoring models weight the inputs they receive.
- D) Certain metrics embedded in regulation have gone on predicting the outcomes they were built to target for several decades.

Answer: A) Indicators published but never written into contracts decay at the same rate as those that are.

Solution: The corollary depends on consequence being the driver of decay; equal decay in indicators with no consequences attached removes that causal link and leaves refinement no worse a remedy than anything else. Decades of durability is the closest trap, but the author already concedes that specification can buy time. Revision matches the rotation option offered, and firms studying model weights is the gaming the author describes.

Instructions (VARC-RC2): study the shared passage below and answer its 4 question(s).

Reformers who export institutions tend to work from a plausible premise: if a commercial code, a bankruptcy statute or a securities regulator produced prosperity in one place, the same text should help elsewhere. The record is not encouraging. Codes copied faithfully have been ignored, applied selectively, or absorbed into arrangements their drafters never imagined. The usual explanation is cultural: rules that suit one society's habits will not suit another's. That explanation is loose enough to accommodate any outcome, which is a reason to distrust it rather than a reason to accept it.

A tighter account starts from what a rule is for. A rule does useful work only when parties can predict how others, including officials, will respond to its breach. That prediction rests on a stock of shared expectation built up through repeated enforcement, and such a stock cannot be imported, because it is not contained in the text. A transplanted rule therefore arrives without the very thing that made it effective at home. Whether it acquires one depends less on the fit between the rule and local values than on whether somebody with something to lose finds it worth invoking, repeatedly, before officials who face some cost for ignoring it.

This reframing has consequences. It predicts that transplants will take hold in narrow commercial domains, where a few well resourced parties litigate often, before they take hold in areas of diffuse benefit, where no single stake is large enough to sustain the litigation that generates expectation. It also predicts that the most durable transplants may be the least faithful: rules bent to serve interests that already existed get enforced, while pristine copies gather dust. Reformers who measure success by fidelity to the model will therefore misread their own results, treating as failure the drift that is the mechanism of adoption.

None of this restores the reformers' optimism; it relocates the difficulty. The obstacle is not that societies are incompatible with foreign rules, but that the asset which makes rules work is produced slowly, locally, and only as a by-product of use. Where no one uses a rule, no argument about its merits will make it operative.

Q6. [VARC-RC2 · mcsc · diff 0.5]

Which of the following comes closest to the claim the passage is defending?

- A) Reform programmes that export commercial codes to other jurisdictions should be discontinued as a matter of policy.
- B) Bankruptcy statutes and securities regulation are the two bodies of law least suited to transfer across borders.
- C) Transplants stall less from cultural mismatch than from missing shared expectation, which only local use builds. ✓
- D) Legal transplants take root only in those societies that already share the values held by the exporting country.

Answer: C) Transplants stall less from cultural mismatch than from missing shared expectation, which only local use builds.

Solution: The second and fourth paragraphs make the stock of shared expectation, which cannot travel in a text, the binding constraint, and expressly downgrade fit with local values. Requiring shared values reverses that; abandoning reform is never recommended, since the author only relocates the difficulty; and the named statutes are examples rather than a ranking.

Q7. [VARC-RC2 · mcsc · diff 0.5]

The author's principal objection to the cultural explanation is that it:

- A) draws its evidence almost entirely from transplants that are already known to have failed
- B) confuses the substance of a rule with the circumstances under which it was drafted
- C) can be reconciled with any observed result, and so accounts for nothing in particular ✓
- D) understates the extent to which commercial habits differ from one society to another

Answer: C) can be reconciled with any observed result, and so accounts for nothing in particular

Solution: The passage calls the cultural account loose enough to accommodate any outcome and treats that looseness as grounds for distrust, which is the standard charge of unfalsifiability. Nothing is said about selection of cases, the author does not claim habits differ more than culturalists allow, and no confusion of substance with drafting is alleged.

Q8. [VARC-RC2 · mcsc · diff 1]

On the author's account, a transplanted consumer protection rule would be expected to become operative more slowly than a transplanted secured-lending rule mainly because:

- A) consumer protection sits in sharper conflict with the commercial customs prevailing locally
- B) consumers as a group tend to be much less aware of the legal remedies formally available to them
- C) the benefit from enforcing it is spread too thinly for repeated litigation to be worthwhile ✓
- D) rules governing secured lending are drafted more simply and are easier for officials to apply

Answer: C) the benefit from enforcing it is spread too thinly for repeated litigation to be worthwhile

Solution: The mechanism given is that expectation is generated by frequent litigation, which requires a party whose stake is large enough to justify it; diffuse benefit therefore starves the process. Conflict with custom is the cultural story the author sets aside, drafting simplicity is never invoked, and low awareness is a plausible sounding substitute for the stake-size argument the passage actually makes.

Q9. [VARC-RC2 · mcsc · diff 1]

The author would agree with all of the following EXCEPT:

- A) Departures from the imported text can form part of the very process by which a rule is adopted.
- B) How a rule works depends on what parties predict officials will do when the rule is breached.
- C) Where local values are hostile to an imported rule, enforcement cannot make it effective. ✓
- D) Transplants settle sooner in those fields where a small number of parties go to court frequently.

Answer: C) Where local values are hostile to an imported rule, enforcement cannot make it effective.

Solution: The passage makes repeated invocation before officials who bear a cost for inaction the decisive factor and treats fit with local values as secondary, so a veto by hostile values contradicts it. The other three restate the prediction-of-response account, the claim that the least faithful transplants may be the most durable, and the prediction about narrow commercial domains.

Q10. [VARC-ODD-1 · mcsc · diff 0.5]

Five sentences are given below, labelled 1 to 5. Four of them can be arranged to form a coherent paragraph. Identify the sentence that does not belong.

- (1) Every map begins with an act of omission, since a representation that kept all the detail of a place would be as unusable as the place itself.
- (2) The cartographer must therefore decide in advance which features a reader will need and which can be silently discarded.
- (3) Recent surveys report that most commuters now consult digital route planners rather than printed atlases.
- (4) Those early decisions harden over time into conventions, and later users mistake the conventions for properties of the world.
- (5) What looks like a neutral record of terrain is thus the residue of a long series of judgements about relevance.

Which sentence is the odd one out?

- A) 1
- B) 2
- C) 3 ✓
- D) 4
- E) 5

Answer: C) 3

Solution: Sentences 1, 2, 4 and 5 build one argument: maps must omit, so a cartographer selects, the selections become conventions, and the finished map is therefore a record of judgements about relevance. Sentence 3 makes an empirical claim about which media commuters use today, which contributes nothing to the argument about selection and is the misfit. Sentence 1 is the natural opener, 2 follows from it through therefore, and 5 is the closing generalisation, so none of those can be removed without breaking the chain.

Q11. [VARC-ODD-2 · mcsc · diff 0.5]

Five sentences are given below, labelled 1 to 5. Four of them can be arranged to form a coherent paragraph. Identify the sentence that does not belong.

- (1) Experts seldom deliberate over the moves that most impress onlookers; they recognise a familiar pattern and act before reasons are available to them.
- (2) Such fluency is bought by long exposure to situations that return prompt and unambiguous feedback.
- (3) Where the feedback is delayed or noisy, however, long experience yields confidence without yielding accuracy.
- (4) Novices, by contrast, report that they enjoy the earliest stages of learning a craft far more than the later ones.
- (5) Intuition therefore deserves trust only after one has asked what the environment has actually been teaching.

Which sentence is the odd one out?

- A) 3
- B) 4 ✓
- C) 5
- D) 1
- E) 2

Answer: B) 4

Solution: Sentences 1, 2, 3 and 5 form a single line of reasoning about when expert intuition can be trusted: it looks effortless, it depends on good feedback, it fails where feedback is poor, and the practical rule follows. Sentence 4 reports how much novices enjoy learning, a claim about affect rather than about the reliability of intuition, so it is the outsider. Sentence 3 cannot be the answer because however marks exactly the contrast that sentence 5 then resolves.

Q12. [VARC-ODD-3 · mcsc · diff 1]

Five sentences are given below, labelled 1 to 5. Four of them can be arranged to form a coherent paragraph. Identify the sentence that does not belong.

- (1) A quantity chosen to stand for a goal starts to be pursued for its own sake as soon as rewards are attached to it.
- (2) Staff learn quickly which parts of their work are counted, and effort drains away from everything that is not.
- (3) The published figures then improve while the purpose they were meant to serve quietly deteriorates.
- (4) Organisations often adopt measurement schemes less to inform their own decisions than to demonstrate propriety to outside audiences.
- (5) Auditors, reading only the figures, record success and recommend that the scheme be extended to further departments.

Which sentence is the odd one out?

- A) 4 ✓
- B) 5
- C) 2
- D) 3
- E) 1

Answer: A) 4

Solution: Sentences 1, 2, 3 and 5 trace one causal sequence: attaching rewards to an indicator redirects effort, the indicator rises as the underlying purpose decays, and auditors who see only the indicator extend the scheme. Sentence 4 stays on the same subject but shifts the level of analysis, explaining why organisations take up measurement at all rather than what measurement does to behaviour, and it plays no part in the causal chain. Sentence 5 belongs because it supplies the consequence of the improved figures named in sentence 3.

Q13. [VARC-ODD-4 · mcsc · diff 1]

Five sentences are given below, labelled 1 to 5. Four of them can be arranged to form a coherent paragraph. Identify the sentence that does not belong.

- (1) No translator can carry every feature of a sentence across, so the work consists largely in deciding what to give up.
- (2) Readers tend to praise the translations that read most smoothly and to distrust those that preserve the strangeness of the original.
- (3) A pun that depends on sound may be replaced by one that depends on sense, on the ground that the joke matters more than its mechanism.
- (4) Rhythm surrendered in one line is sometimes repaid in another, the debt being settled across the page rather than within it.
- (5) Every such compensation is an argument about which layer of the original counts as the text itself.

Which sentence is the odd one out?

- A) 1
- B) 4
- C) 5
- D) 2 ✓
- E) 3

Answer: D) 2

Solution: Sentences 1, 3, 4 and 5 describe the translator's own practice: loss is unavoidable, so features are traded and debts are repaid elsewhere, and each trade amounts to a claim about what the text essentially is. Sentence 2 describes how readers receive finished translations, a claim about reception rather than about the decisions being made, and the paragraph never uses it. Sentence 4 is not the misfit because the repayment it describes is one instance of the compensation that sentence 5 generalises.

Q14. [VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) That demand is what makes boredom productive, since it pushes the mind to supply what the surroundings withhold.
- (2) Boredom is usually described as an absence, the gap left when nothing outside us claims our attention.
- (3) But the state has a content of its own, a restless demand that the world become more interesting than it is.
- (4) Children left with nothing to do therefore stay idle only briefly, building a game out of whatever lies nearest.

Which of the following is the correct order of the sentences?

- A) 2341
- B) 2314 ✓
- C) 3241
- D) 2134

Answer: B) 2314

Solution: Sentence 2 gives the standard description and must open. Sentence 3 begins with But and corrects that description by introducing the restless demand. Sentence 1 picks that demand up with That demand, so it can only follow. Sentence 4 closes with therefore and an illustration. The order beginning 2134 fails because That demand has no antecedent yet, 3241 puts the correction before what it corrects, and 2341 leaves sentence 1 stranded after the example it explains.

Q15. [VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) Once the date is fixed, the remaining hours acquire a rank, and the work is fitted to those that are left.
- (2) This is why arbitrary dates can serve as well as reasoned ones, provided they are actually believed.
- (3) A deadline works less by adding motivation than by removing alternatives.
- (4) Until a date is fixed, every hour is a candidate for the task and no hour is obliged to be.

Which of the following is the correct order of the sentences?

- A) 3421
- B) 3142
- C) 3412 ✓
- D) 4312

Answer: C) 3412

Solution: Sentence 3 states the thesis about removing alternatives and opens. Sentences 4 and 1 form the matched pair Until a date is fixed and Once the date is fixed, in that order, since the second echoes the first. Sentence 2 draws the inference with This is why and must end. The order 3142 breaks the Until and Once pair, 4312 buries the thesis, and 3421 places the inference before the mechanism it depends on.

Q16. [VARC-JUMBLE-3 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) The only reliable lever, then, is conduct that gives those custodians something else to carry.
- (2) Its custodians are third parties, who pass it on for reasons of their own and revise it without notice.
- (3) Reputation is information about us that we do not own and cannot directly edit.
- (4) This dependence explains why attempts at correction so often deepen the damage, for a correction is itself testimony and it travels.

Which of the following is the correct order of the sentences?

- A) 3421
- B) 2431
- C) 3241 ✓
- D) 2341

Answer: C) 3241

Solution: Only sentence 3 introduces reputation as a full noun, so the pronoun Its in sentence 2 must come after it. Sentence 4 opens with This dependence, which refers to the reliance on third parties just established, and sentence 1 closes with then and with those custodians. The order 2341 and 2431 both strand the pronoun Its ahead of its antecedent, and 3421 makes This dependence point at nothing but the definition.

Q17. [VARC-JUMBLE-4 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) A first reading of anything difficult is mostly navigation, since we are still learning where the parts are.
- (2) Readers who skip it are not saving time; they are trading comprehension for the feeling of having finished.
- (3) For this reason a second reading is not a repetition but a different task, undertaken with the map the first reading drew.
- (4) Only once the shape is known can attention be spent on the joints, which is where arguments usually fail.

Which of the following is the correct order of the sentences?

- A) 4132
- B) 1342
- C) 1432 ✓
- D) 1423

Answer: C) 1432

Solution: Sentence 1 must open, because Only once the shape is known in sentence 4 presupposes the navigating first reading. Sentence 3 then draws the conclusion with For this reason, and the pronoun it in sentence 2 refers to the second reading named there, so sentence 2 comes last. The order 1342 puts the conclusion before the reason, 4132 opens with a clause that has nothing to refer back to, and 1423 leaves it without its antecedent.

Q18. [VARC-JUMBLE-5 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) The reason travels less well than the exception, since the reason was particular and the exception is not.
- (2) What began as mercy is then read as entitlement, and the rule survives only as a memory.
- (3) Administrators who grasp this record their reasons loudly, hoping the record will do the limiting work the rule no longer can.
- (4) Every exception granted for a good reason becomes a fact that later applicants are free to cite.

Which of the following is the correct order of the sentences?

- A) 4213
- B) 1423
- C) 4132
- D) 4123 ✓

Answer: D) 4123

Solution: Sentence 4 must open, because the definite phrase The reason in sentence 1 refers back to the good reason mentioned there. Sentence 2 follows with then, marking the end point of the drift, and sentence 3 responds to the whole process with this. The order 1423 begins with a reference that has no antecedent, 4213 reverses the drift and its explanation, and 4132 makes this point at an explanation rather than at the completed process.

Q19. [VARC-SUMMARY-1 · mcsc · diff 0.5]

Infrastructure is noticed chiefly when it fails. A water main, a switching station, a clearing house: each is built to disappear into the routines it supports, and its success is measured by how little thought it demands. The consequence is political rather than technical. Systems nobody thinks about attract neither scrutiny nor investment, and their upkeep competes badly with projects that can be photographed. Failure then arrives as a surprise, and the surprise is used to argue that nobody could have foreseen it.

Which option best summarises the paragraph?

- A) Public money should be redirected from photogenic new construction towards the unglamorous upkeep of water mains, switching stations and clearing houses.
- B) Infrastructure fails because engineers deliberately design systems to disappear, and systems that are invisible are inherently more fragile than visible ones.
- C) Since infrastructure failures are always surprises, the political value of such systems lies in the attention that their breakdowns finally bring to them.
- D) Designed to go unnoticed, working infrastructure draws little scrutiny or funding, so its breakdowns look unforeseeable when they were merely unattended. ✓

Answer: D) Designed to go unnoticed, working infrastructure draws little scrutiny or funding, so its breakdowns look unforeseeable when they were merely unattended.

Solution: The paragraph links invisibility to neglect and neglect to apparently unforeseeable failure, which is exactly what the summary about unattended breakdowns captures. The claim that invisible systems are inherently more fragile substitutes a technical mechanism the paragraph explicitly rejects. The call to redirect public money is a recommendation the paragraph never makes. Treating breakdowns as valuable attention inverts the point that the surprise is used as an excuse.

Q20. [VARC-SUMMARY-2 · mcsc · diff 0.5]

Averaging many independent guesses often beats the best single guess, because errors pointing in opposite directions cancel. The condition doing the work is independence. Once the guessers can hear one another, their errors begin to align, and the average inherits a bias instead of shedding one. Committees that deliberate before voting therefore give up much of the advantage voting was supposed to secure, and the confidence produced by agreement rises just as the accuracy it is taken to signal falls.

Which option best summarises the paragraph?

- A) Confidence is a poor guide to accuracy, since agreement inside a group can show only that its members have all been exposed to the same limited information.
- B) Groups outperform their ablest members because pooling opinions cancels error, which is why committees ought always to vote instead of trusting one expert.
- C) Aggregation improves accuracy only while guesses stay independent; discussion correlates errors, so deliberating groups gain confidence and lose accuracy. ✓
- D) Deliberation harms committees chiefly because the members who happen to speak first impose their own private biases on all of those who speak later.

Answer: C) Aggregation improves accuracy only while guesses stay independent; discussion correlates errors, so deliberating groups gain confidence and lose accuracy.

Solution: The paragraph makes independence the load-bearing condition and ends on the divergence between rising confidence and falling accuracy, which the summary naming both captures. The version recommending that committees always vote keeps the cancelling of errors but drops the condition that makes it work. The claim about who speaks first invents a mechanism. The remark about confidence alone reports one consequence and omits independence entirely.

Q21. [VARC-SUMMARY-3 · mcsc · diff 0.5]

Options are usually treated as free, on the view that holding one costs nothing until it is exercised. In practice every open possibility levies a quiet charge: some attention reserved, some commitment withheld, some relationship kept provisional. Because the charge is never invoiced, options accumulate well past the point where their combined upkeep exceeds anything they could deliver. The person who cannot choose is not made richer by the alternatives; the alternatives are consuming the very capacity that choosing would have released.

Which option best summarises the paragraph?

- A) Open possibilities exact hidden upkeep in attention and commitment, so options collected beyond a point cost more than they could ever deliver. ✓
- B) Because an option costs nothing at all until it is exercised, a rational person should hold as many as possible and delay commitment for as long as possible.
- C) Indecisive people are poorer than decisive ones, because real wealth consists in commitments actually made rather than in alternatives that remain available.
- D) The chief trouble with unexercised options is that their true price is never invoiced, which makes any accounting of their value unreliable in practice.

Answer: A) Open possibilities exact hidden upkeep in attention and commitment, so options collected beyond a point cost more than they could ever deliver.

Solution: The paragraph identifies a hidden cost of holding options and locates a threshold past which the cost outweighs the benefit, which the summary about upkeep exceeding delivery states. The advice to hold as many options as possible repeats the view the paragraph is arguing against. The claim about indecisive people being poorer generalises into a thesis about wealth that the text does not assert. The uninvoiced price is only the reason the cost goes unnoticed, not the main point.

Q22. [VARC-SUMMARY-4 · mcsc · diff 1]

Forecasts about human affairs are not merely difficult; they are ingredients. A published prediction of shortage produces hoarding, and a published prediction of stability invites the risk-taking that ends it. This does not make forecasting useless, but it changes what accuracy could mean, since a forecast can be judged only against the world it helped to make, and a forecast that is widely ignored may be the only one ever fairly tested. Practitioners who forget this treat their failures as errors of measurement and answer them by refining instruments that were never the problem.

Which option best summarises the paragraph?

- A) Predictions about human affairs are self-fulfilling, so forecasters should publish nothing until their instruments are accurate enough to withstand contact with the public.
- B) Predictions about human affairs alter the conditions they describe, so accuracy can be judged only against an already altered world, and better instruments cannot help. ✓
- C) Social forecasts change the behaviour they describe, which is why only those forecasts that nobody happens to notice can ever be tested fairly against events.
- D) Failures of social forecasting come from measurement error, and sharper instruments will in time let practitioners predict hoarding and risk-taking reliably enough.

Answer: B) Predictions about human affairs alter the conditions they describe, so accuracy can be judged only against an already altered world, and better instruments cannot help.

Solution: The paragraph's main claim is that because forecasts feed back into events, accuracy must be redefined and better instruments cannot rescue the enterprise, which is the summary mentioning both the altered world and the instruments. The version about forecasts nobody notices repeats a subordinate aside and leaves out the conclusion about measurement. The advice to publish nothing adds a prescription and treats every forecast as self-fulfilling, which the shortage and stability cases contradict. Blaming measurement error states the mistake the paragraph diagnoses.

Q23. [VARC-SUMMARY-5 · mcsc · diff 1]

A standard is valuable in proportion to how many people already use it, so its merits and its market share are hard to tell apart. Once adoption passes a threshold, the standard is defended by everything built on top of it, and a technically superior rival must compensate not for a small gap in quality but for the whole accumulated weight of compatibility. Arguments about which standard is better therefore arrive too late to matter, and the winner's supporters read its survival as evidence that the argument was settled on the merits.

Which option best summarises the paragraph?

- A) Technically superior standards displace entrenched ones eventually, provided their supporters can show that the original choice was never really settled on the merits.
- B) A standard's worth grows with its user base, so entrenched standards defeat better rivals through compatibility, and survival is then misread as proof of merit. ✓
- C) Because compatibility counts for more than quality, users should simply adopt whichever standard is most widely used instead of assessing competing designs themselves.
- D) A standard's dominance owes far more to the number of people using it than to its quality, so late arguments about technical merit rarely change which one prevails.

Answer: B) A standard's worth grows with its user base, so entrenched standards defeat better rivals through compatibility, and survival is then misread as proof of merit.

Solution: The paragraph moves from network value to compatibility lock-out and ends with the mistaken inference from survival to superiority, all three of which the summary about misread survival preserves. The version about late arguments is accurate as far as it goes but stops before that inference, which is the paragraph's sting. The claim that superior standards eventually win reverses the argument. The advice to adopt the most popular standard converts a diagnosis into guidance the text does not offer.

Q24. [VARC-SUMMARY-6 · mcsc · diff 1]

An archive looks like a residue of the past, but it is a residue of past decisions about what would be worth keeping. Records survive because someone was obliged to file them, and the obligations fell unevenly, heavily on property and litigation and hardly at all on ordinary agreement. Historians who read the surviving mass as a sample of what happened therefore reconstruct a world of disputes and transactions and miss the settled life around them. The remedy is not more documents but a habit of asking, of every silence, whether it records an absence or only an exemption from paperwork.

Which option best summarises the paragraph?

- A) Because filing obligations bore most heavily on property and litigation, surviving archives over-represent disputes and transactions at the expense of ordinary agreement.
- B) Every silence in an archive records a genuine absence, which is why the past appears to consist mainly of litigation, transactions and disputes rather than agreement.
- C) Historians ought to seek out further documents about settled everyday life, since the archives that do survive were shaped by the interests of owners and courts.
- D) Archives keep whatever filing duties happened to capture, so reading them as a fair sample distorts the past unless each silence is itself interrogated. ✓

Answer: D) Archives keep whatever filing duties happened to capture, so reading them as a fair sample distorts the past unless each silence is itself interrogated.

Solution: The paragraph diagnoses a bias created by uneven filing duties and then prescribes a way of reading silences, and only the summary combining distortion with interrogating silences carries both halves. The statement about property and litigation is true but stops at the diagnosis and omits the remedy. The suggestion to seek more documents is precisely the remedy the text rules out. The assertion that every silence records a real absence states the error the paragraph warns against.

Section: DILR (suggested 40 min)

Instructions (DILR-1): study the shared data below and answer its 4 question(s).

Five inspectors (Lena, Marco, Nadia, Omar and Priya) each audited exactly one of five sites (Alpha, Birch, Cedar, Dune and Elm) during a single working week. Every inspector worked on a different weekday from Monday to Friday, and no site was visited twice. Each audit received a distinct compliance score from the list 12, 15, 18, 21 and 24.

- Lena's audit took place before Marco's, and Marco's took place before Omar's. - Cedar was audited on Wednesday. - Nadia audited Alpha, on the day immediately after Priya's audit. - Elm was audited earlier in the week than Birch, and Dune was audited on Friday. - Lena did not audit Cedar. - The Elm audit scored 18, and Marco's score was exactly 9 more than Nadia's. - Priya's score was lower than both Nadia's score and Omar's score.

Q25. [DILR-1 · mcsc · diff 1]

Which inspector audited Birch?

- A) Priya
- B) Nadia
- C) Lena
- D) Marco ✓

Answer: D) Marco

Solution: Dune is Friday and Cedar is Wednesday, so the Nadia-immediately-after-Priya pair must sit inside Monday to Thursday. With Lena before Marco before Omar, Omar is pushed to Friday (Dune) and the only consistent fit is Lena Monday, Marco Tuesday, Priya Wednesday, Nadia Thursday. Since Nadia has Alpha, Priya has Cedar and Elm must precede Birch, Lena takes Elm on Monday and Marco takes Birch on Tuesday. Naming Priya or Nadia ignores that their sites are already fixed as Cedar and Alpha, and naming Lena ignores the Elm-before-Birch order.

Q26. [DILR-1 · mcsc · diff 1]

What is the combined compliance score of the Monday audit and the Friday audit?

- A) 42
- B) 33
- C) 30
- D) 39 ✓

Answer: D) 39

Solution: The schedule is Lena Monday Elm, Marco Tuesday Birch, Priya Wednesday Cedar, Nadia Thursday Alpha, Omar Friday Dune. Elm scored 18, so Lena has 18. Marco being 9 above Nadia leaves the pairs 24 with 15 or 21 with 12; the second pair forces Nadia to 12, which contradicts Priya being lower than Nadia, so Marco has 24 and Nadia 15. Priya then takes 12 and Omar 21, giving 18 plus 21 equals 39. The value 42 comes from pairing 18 with 24, 33 from using Nadia's 15 and 30 from using Priya's 12.

Q27. [DILR-1 · mcsc · diff 1]

Which of the following statements is true?

- A) Priya recorded the lowest of the five scores and worked on Wednesday. ✓
- B) Lena audited Elm and scored higher than Marco did on the same week.
- C) Omar recorded the highest of the five scores, and he worked on Friday.
- D) Nadia audited Alpha on the day immediately before Cedar was audited.

Answer: A) Priya recorded the lowest of the five scores and worked on Wednesday.

Solution: The unique solution is Lena Monday Elm 18, Marco Tuesday Birch 24, Priya Wednesday Cedar 12, Nadia Thursday Alpha 15, Omar Friday Dune 21. Priya's 12 is indeed the lowest score and her day is Wednesday. Omar does work on Friday but 21 is not the highest score, since Marco has 24. Nadia's Thursday comes after, not before, the Wednesday Cedar audit. Lena did audit Elm but her 18 is below Marco's 24.

Q28. [DILR-1 · mcsc · diff 1]

Suppose the information that Dune was audited on Friday is replaced by the information that Dune was audited on Thursday, with every other statement unchanged. Which site is then audited on Friday?

- A) Cedar
- B) Birch
- C) Dune
- D) Alpha ✓

Answer: D) Alpha

Solution: With Dune on Thursday and Cedar on Wednesday, the consecutive Priya then Nadia pair can only be Thursday then Friday, so Nadia takes Friday with Alpha. Lena before Marco before Omar then fills Monday, Tuesday and Wednesday in that order, and Elm before Birch forces Lena on Elm and Marco on Birch, leaving Omar with Cedar on Wednesday and Priya with Dune on Thursday. Cedar and Dune are ruled out by their stated days, and Birch is locked to Tuesday by the ordering.

Instructions (DILR-2): study the shared data below and answer its 3 question(s).

Six crates labelled K, L, M, N, O and R are stacked one above another in a single tower. Position 1 is the bottom of the tower and position 6 is the top, and each position holds exactly one crate.

- Crate M rests immediately above crate K. - Exactly two crates lie between L and M. - R is neither at the bottom nor at the top of the tower. - O is somewhere above L. - Exactly one crate lies between N and R.

Q29. [DILR-2 · mcsc · diff 0.5]

Which crate is at the bottom of the tower?

- A) K
- B) L
- C) N ✓
- D) R

Answer: C) N

Solution: The K then M block plus two crates between L and M leaves only a few layouts, and requiring exactly one crate between N and R while R avoids both ends fixes the tower as N, L, R, K, M, O from bottom to top. So N is at the bottom. L sits second, K fourth and R is barred from position 1 outright by the end restriction.

Q30. [DILR-2 · mcsc · diff 0.5]

How many crates rest above R?

- A) 3 ✓
- B) 4
- C) 2
- D) 1

Answer: A) 3

Solution: The tower is N, L, R, K, M, O from bottom to top, so R occupies position 3 and the crates K, M and O lie above it, giving 3. Answering 2 counts only the K and M block, 4 comes from placing R at position 2 and 1 comes from reading the stack top down instead of bottom up.

Q31. [DILR-2 · mcsc · diff 1]

If the condition that O is somewhere above L is dropped and all the other conditions still hold, how many different stacking orders are possible?

- A) 3
- B) 4 ✓
- C) 2
- D) 5

Answer: B) 4

Solution: Without the O above L condition, the K then M block can sit at positions 1 and 2, 4 and 5 or 5 and 6 while keeping two crates between L and M, R off both ends and exactly one crate between N and R. Checking each case gives K M O R L N, N L R K M O, O N L R K M and O R L N K M, so 4 orders survive. Answering 3 misses the case with the block at the very bottom, while 5 and 2 come from mishandling the N and R gap.

Instructions (DILR-3): study the shared data below and answer its 4 question(s).

An audit of four solar module assemblers covers three panel grades, G1, G2 and G3. Table 1 gives the number of modules installed and Table 2 gives the share of those modules found defective in the audit.

Table 1: Modules installed (in hundreds)

Firm	G1	G2	G3
Arkin	24	36	20
Bhima	30	18	42
Cygna	45	25	15
Dhruv	16	40	34

Table 2: Defective modules as a percentage of modules installed

Firm	G1	G2	G3
Arkin	2	5	4
Bhima	6	3	2
Cygna	4	2	8
Dhruv	5	4	3

Q32. [DILR-3 · mcsc · diff 1]

In grade G3, how many more defective modules did Cygna and Dhruv together have than Arkin and Bhima together?

- A) 46
- B) 64
- C) 58 ✓
- D) 38

Answer: C) 58

Solution: Defective modules equal installed hundreds times the rate, so each product is (hundreds x rate). G3 gives Arkin $20 \times 4 = 80$, Bhima $42 \times 2 = 84$, Cygna $15 \times 8 = 120$, Dhruv $34 \times 3 = 102$. So 222 against 164, a gap of 58. The 46 trap swaps two rate entries, 64 comes from reading G2 rates, and 38 subtracts firm totals rather than G3 alone.

Q33. [DILR-3 · mcsc · diff 1]

Taking all four firms and all three grades together, defective modules form approximately what percentage of all modules installed?

- A) 3.82% ✓
- B) 3.90%
- C) 4.05%
- D) 3.68%

Answer: A) 3.82%

Solution: Total installed is $(80+90+85+90)$ hundreds = 34500 modules. Total defective is $308+318+350+342 = 1318$, so the overall rate is $1318/34500 = 3.82$ percent. The 3.90 and 4.05 values come from averaging the twelve percentages instead of weighting them by volume, and 3.68 drops one grade column from the defective total.

Q34. [DILR-3 · mcsc · diff 1]

Which firm has the lowest defect rate across its entire installed base?

- A) Dhruv
- B) Bhima ✓
- C) Cygna
- D) Arkin

Answer: B) Bhima

Solution: Overall rates must be volume weighted: Arkin $308/8000 = 3.85$ percent, Bhima $318/9000 = 3.53$ percent, Cygna $350/8500 = 4.12$ percent, Dhruv $342/9000 = 3.80$ percent. Bhima is lowest despite carrying the single worst cell rate of 6 percent, because that rate sits on its smallest grade mix. Dhruv and Arkin look lower if the three grade rates are simply averaged.

Q35. [DILR-3 · mcsc · diff 1]

By how many modules does the firm with the largest number of defective modules exceed the firm with the smallest?

- A) 24
- B) 42 ✓
- C) 34
- D) 56

Answer: B) 42

Solution: Firm defective totals are Arkin 308, Bhima 318, Cygna 350 and Dhruv 342. The largest is Cygna at 350 and the smallest is Arkin at 308, a difference of 42. The 34 value compares Bhima with Arkin, 24 compares Dhruv with Bhima, and 56 comes from using installed volumes rather than defective counts.

Instructions (DILR-4): study the shared data below and answer its 4 question(s).

A depot dispatched exactly 48 crates of packaged tea to four towns, Alwar, Betul, Chirala and Dhanbad, on a single morning. The crates for each town went on that town's own truck, and the four trucks left the depot one after another, first to fourth.

- Each town received a distinct number of crates, and every one of those numbers was a multiple of 4 lying between 4 and 24, both inclusive. - Chirala received more crates than Alwar. - The town whose truck left first received exactly twice as many crates as the town whose truck left fourth. - Dhanbad's truck left immediately after Betul's truck. - Alwar's truck did not leave first, and Alwar received fewer crates than Betul. - The town whose truck left third received exactly 12 crates. - Chirala's truck left before Betul's truck.

Q36. [DILR-4 · mcsc · diff 0.5]

How many crates did Chirala receive?

- A) 12 crates
- B) 24 crates
- C) 4 crates
- D) 8 crates ✓

Answer: D) 8 crates

Solution: Chirala leaves before Betul and Alwar is not first, while Dhanbad follows Betul immediately, so the order can only be Chirala, Betul, Dhanbad, Alwar or Chirala, Alwar, Betul, Dhanbad. The third truck carries 12 crates and the first carries twice the fourth. Testing distinct multiples of 4 from 4 to 24 that total 48 leaves only Chirala 8, Betul 24, Dhanbad 12, Alwar 4 in the order Chirala, Betul, Dhanbad, Alwar. Hence Chirala received 8 crates. The 4 figure is Alwar's share, 12 is Dhanbad's (the third truck), and 24 is Betul's, the largest allocation.

Q37. [DILR-4 · mcsc · diff 1]

Which town's truck left second, and how many crates did that town receive?

- A) Chirala, with 8 crates
- B) Betul, with 24 crates ✓
- C) Alwar, with 20 crates
- D) Dhanbad, with 12 crates

Answer: B) Betul, with 24 crates

Solution: The only allocation consistent with every clue is Chirala 8, Betul 24, Dhanbad 12, Alwar 4, dispatched in the order Chirala, Betul, Dhanbad, Alwar. So the second truck was Betul's, carrying 24 crates. Dhanbad with 12 crates describes the third truck, Chirala with 8 the first, and Alwar never received 20 crates in any consistent arrangement since 24 is fixed for Betul.

Q38. [DILR-4 · mcsc · diff 1]

If the clue that the third truck's town received exactly 12 crates is withdrawn and all the other clues still hold, how many complete arrangements of dispatch order and crate counts are possible?

- A) 7 arrangements ✓
- B) 6 arrangements
- C) 9 arrangements
- D) 5 arrangements

Answer: A) 7 arrangements

Solution: Without the 12-crate clue the order is still either Chirala, Betul, Dhanbad, Alwar or Chirala, Alwar, Betul, Dhanbad. For the first order the possibilities are Alwar 4 with Betul and Dhanbad taking 12 and 24, 16 and 20, 20 and 16 or 24 and 12 alongside Chirala 8, plus Alwar 8 with Chirala 16, Betul 20, Dhanbad 4, which is five cases; the second order adds two more, giving 7 in all. The 5 count stops at the first order only, 6 misses one of the extra cases, and 9 double counts allocations that repeat a crate figure.

Q39. [DILR-4 · mcsc · diff 1]

Suppose the first truck's town received three times, rather than twice, as many crates as the fourth truck's town, every other clue remaining as stated. How many crates would Dhanbad then receive?

- A) 8 crates ✓
- B) 24 crates
- C) 4 crates
- D) 12 crates

Answer: A) 8 crates

Solution: Under the revised ratio the arrangement becomes Chirala 24, Alwar 4, Betul 12, Dhanbad 8, dispatched in that order: the third truck is Betul's with 12 crates, the first is Chirala's with 24, which is three times Alwar's 4 on the fourth truck, and the four counts still total 48. So Dhanbad receives 8 crates. The 4 belongs to Alwar, 12 to the third truck Betul, and 24 to Chirala, whose figure is often read off as Dhanbad's by mistake.

Instructions (DILR-5): study the shared data below and answer its 3 question(s).

A training coordinator must schedule exactly four of seven half-day workshops, namely Auditing, Branding, Coding, Design, Ethics, Finance and Grammar, for the coming quarter.

- Coding is scheduled only if Design is also scheduled. - Branding and Ethics are run by the same trainer, so they cannot both be scheduled. - Auditing and Finance need the same laboratory, so they cannot both be scheduled. - Grammar is scheduled if and only if Auditing is scheduled. - If Design is scheduled, then Branding is not scheduled.

Q40. [DILR-5 · mcsc · diff 0.5]

How many different groups of four workshops can the coordinator schedule?

- A) 3 groups ✓
- B) 4 groups
- C) 5 groups
- D) 2 groups

Answer: A) 3 groups

Solution: Checking all 35 groups of four against the conditions leaves only Auditing, Coding, Design and Grammar; Auditing, Design, Ethics and Grammar; and Coding, Design, Ethics and Finance. That is 3 groups. The 2 count comes from overlooking the group that omits Ethics, while 4 and 5 come from admitting groups that break either the Grammar and Auditing pairing or the ban on Design with Branding.

Q41. [DILR-5 · mcsc · diff 0.5]

Which workshop cannot appear in any acceptable group of four?

- A) Grammar
- B) Branding ✓
- C) Auditing
- D) Coding

Answer: B) Branding

Solution: If Branding is scheduled then Ethics and Design are both out, so the four must be drawn from Auditing, Coding, Finance and Grammar; but Coding needs Design, Auditing and Finance clash, and Grammar needs Auditing, so no such group exists. Branding therefore never appears. Auditing and Grammar appear together in two acceptable groups, and Coding appears in two acceptable groups as well.

Q42. [DILR-5 · mcsc · diff 1]

If Finance is scheduled, which workshops make up the rest of the group?

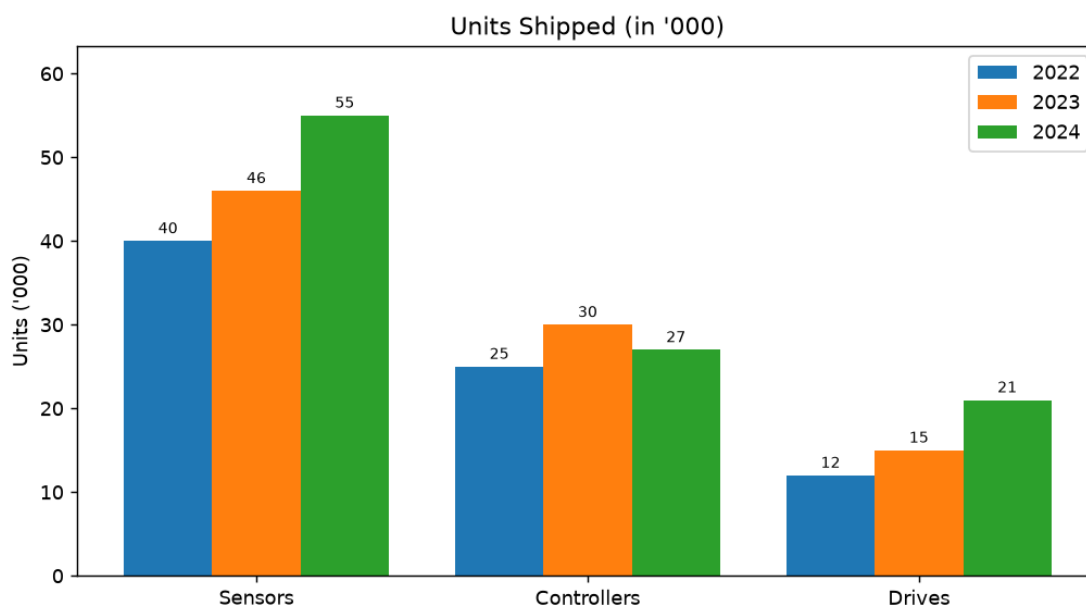
- A) Coding, Design and Grammar
- B) Auditing, Design and Grammar
- C) Design, Ethics and Grammar
- D) Coding, Design and Ethics ✓

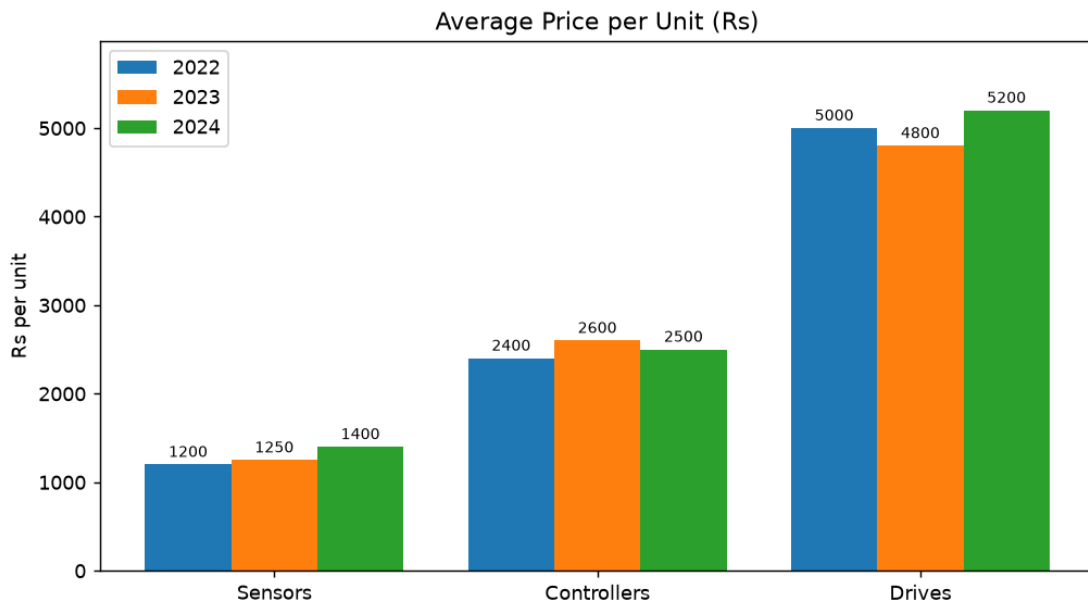
Answer: D) Coding, Design and Ethics

Solution: Finance rules out Auditing, and with Auditing out Grammar is out too, so the remaining three come from Branding, Coding, Design and Ethics. Branding cannot join Design, and dropping Design would force Coding out as well, leaving too few workshops, so Design stays, Branding goes, and Coding and Ethics complete the group: Coding, Design and Ethics. Any option containing Grammar fails because Grammar travels only with Auditing, which Finance excludes.

Instructions (DILR-6): study the shared data below and answer its 4 question(s).

An industrial electronics maker reports units shipped and the average price realised per unit for its three product lines over three years. Revenue for a line equals units shipped multiplied by average price.





Q43. [DILR-6 · mcsc · diff 1]

What was the total revenue of the three lines together in 2024, in Rs lakh?

- A) 2612
- B) 2537 ✓
- C) 2350
- D) 2465

Answer: B) 2537

Solution: Revenue in Rs lakh is units in thousands times price divided by 100. For 2024 that gives Sensors $55 \times 1400 / 100 = 770$, Controllers $27 \times 2500 / 100 = 675$ and Drives $21 \times 5200 / 100 = 1092$, summing to 2537. The 2465 and 2350 values carry over a 2023 price or volume, and 2612 uses the 2023 Drives price of 4800 against the wrong volume.

Q44. [DILR-6 · mcsc · diff 1]

The line with the highest percentage growth in revenue from 2022 to 2024 grew by approximately what percentage?

- A) 92%
- B) 60%
- C) 82% ✓
- D) 71%

Answer: C) 82%

Solution: Revenues in Rs lakh move from Sensors 480 to 770, Controllers 600 to 675 and Drives 600 to 1092. The growth rates are about 60 percent, 12.5 percent and 82 percent, so Drives leads at 82 percent. The 60 percent figure belongs to Sensors, 71 percent comes from growing only the volume of Drives from 12 to 21 read against price, and 92 percent overstates the base as 570.

Q45. [DILR-6 · mcsc · diff 1]

In 2023, what was the average price per unit realised across all units shipped, in Rs?

- A) 2280 ✓
- B) 2124
- C) 2217
- D) 2350

Answer: A) 2280

Solution: Total 2023 revenue is $575 + 780 + 720 = 2075$ Rs lakh on $46 + 30 + 15 = 91$ thousand units, so the blended price is $207500000 / 91000$, about Rs 2280. The 2217 value is the plain average of 1250, 2600 and 4800 scaled wrongly, 2350 mixes in 2024 prices, and 2124 uses 2022 volumes as weights instead of 2023 volumes.

Q46. [DILR-6 · mcsc · diff 1]

By approximately what percentage did the combined revenue of the three lines change from 2023 to 2024?

- A) 18.2%
- B) 22.3% ✓
- C) 24.9%
- D) 20.5%

Answer: B) 22.3%

Solution: Combined revenue is $575+780+720 = 2075$ Rs lakh in 2023 and $770+675+1092 = 2537$ Rs lakh in 2024, an increase of 462 on 2075, that is 22.3 percent. The 18.2 percent value divides the rise by the 2024 total, 24.9 percent averages the three line growth rates, and 20.5 percent tracks units alone while ignoring the price changes.

Section: QA (suggested 40 min)

Q47. [QA-MCQ-1 · mcsc · diff 0.5]

A dealer marks the price of a ceiling fan 60% above its cost price and then allows two successive discounts of 20% and 10% on the marked price. What is his profit percentage on this sale?

- A) 15.2% ✓
- B) 12.0%
- C) 16.0%
- D) 28.0%

Answer: A) 15.2%

Solution: Take cost as 100, so marked price is 160. After 20% off the price is 128, and after a further 10% off it is 115.2, giving a profit of 15.2 on 100, that is 15.2%. The 12.0% figure comes from treating the two discounts as one flat 30% cut, since $160 \times 0.70 = 112$. The 28.0% figure comes from applying only the 20% discount and stopping. The 16.0% figure comes from reading 10% of 160 as the whole profit.

Q48. [QA-MCQ-2 · mcsc · diff 0.5]

Pranav invests Rs 42,000 for 8 months, Qadir invests Rs 56,000 for 5 months and Rohit invests Rs 35,000 for 12 months in a partnership firm. At the end of the year the firm earns a profit of Rs 74,000, shared in proportion to capital multiplied by time. What is Qadir's share?

- A) Rs 24,000
- B) Rs 22,000
- C) Rs 20,000 ✓
- D) Rs 30,000

Answer: C) Rs 20,000

Solution: The capital-months are $42000 \times 8 = 336000$, $56000 \times 5 = 280000$ and $35000 \times 12 = 420000$, which reduce to 12 : 10 : 15, a total of 37 parts. One part is $74000 / 37 = 2000$, so Qadir's 10 parts give Rs 20,000. The Rs 24,000 value is Pranav's 12 parts and the Rs 30,000 value is Rohit's 15 parts, both read off the wrong partner. The Rs 22,000 value comes from mis-simplifying 280 into 11 parts instead of 10.

Q49. [QA-MCQ-3 · mcsc · diff 0.5]

A vessel contains 60 litres of a mixture of milk and water in which milk and water are in the ratio 7 : 3. How many litres of water must be added so that the ratio of milk to water in the new mixture becomes 7 : 5?

- A) 15 litres
- B) 12 litres ✓
- C) 8 litres
- D) 10 litres

Answer: B) 12 litres

Solution: The vessel holds 42 litres of milk and 18 litres of water. Milk does not change, so for the ratio 7 : 5 the water must become $42 \times 5 / 7 = 30$ litres, needing $30 - 18 = 12$ litres more. The 10 litres value comes from wrongly making total volume 70. The 15 litres value comes from taking the new water as one quarter of a 60 litre base. The 8 litres value comes from using the ratio difference $5 - 3 = 2$ on 4 litre parts.

Q50. [QA-MCQ-4 · mcsc · diff 0.5]

A motorboat whose speed in still water is 12 km/h goes 45 km downstream along a river and returns to the starting point, taking 8 hours in all. What is the speed of the current?

- A) 3 km/h ✓
- B) 2.5 km/h
- C) 4 km/h
- D) 6 km/h

Answer: A) 3 km/h

Solution: If the current is s , then $45 / (12 + s) + 45 / (12 - s) = 8$, so $45 \times 24 = 8 \times (144 - s^2)$, giving $144 - s^2 = 135$ and $s = 3$ km/h. The 4 km/h value comes from averaging the two legs at $45 / 4$ hours each. The 2.5 km/h value comes from solving with a total of 7.5 hours. The 6 km/h value comes from equating the round trip to a single 90 km run at $12 - s$.

Q51. [QA-MCQ-5 · mcsc · diff 0.5]

Anil alone can paint a hall in 20 days and Bimal alone can paint it in 30 days. They begin painting together, but Anil stops working 5 days before the hall is finished, and Bimal completes the remaining part alone. In how many days is the hall painted?

- A) 14 days
- B) 12 days
- C) 18 days
- D) 15 days ✓

Answer: D) 15 days

Solution: If the job takes T days, Bimal works all T days and Anil works $T - 5$ days, so $(T - 5) / 20 + T / 30 = 1$. Multiplying by 60 gives $3T - 15 + 2T = 60$, so $T = 15$ days. The 12 days value is the time if both worked throughout. The 14 days value comes from letting Bimal, not Anil, be the one who leaves early. The 18 days value comes from adding 5 idle days to a 13 day estimate.

Q52. [QA-MCQ-6 · mcsc · diff 0.5]

A sum of Rs 16,000 is lent for one year at 10% per annum. By how much does the interest earned when the rate is compounded half-yearly exceed the interest earned when it is compounded annually?

- A) Rs 20
- B) Rs 80
- C) Rs 1,640
- D) Rs 40 ✓

Answer: D) Rs 40

Solution: Half-yearly, the amount is $16000 \times 1.05 \times 1.05 = 17640$, so interest is 1640, while annual compounding for one year gives just 1600. The excess is Rs 40. The Rs 80 value comes from using 10% per half year instead of 5%. The Rs 20 value comes from halving the correct gap. The Rs 1,640 value is the half-yearly interest itself, not the excess over annual compounding.

Q53. [QA-MCQ-7 · mcsc · diff 0.5]

Four boys and three girls are to be seated in a row of seven chairs. In how many ways can they be seated if no two girls sit next to each other?

- A) 5040
- B) 720
- C) 2880
- D) 1440 ✓

Answer: D) 1440

Solution: Seat the four boys first in $4! = 24$ ways, creating five gaps including the two ends. Choose and fill three of those gaps with the three girls in $5 \times 4 \times 3 = 60$ ordered ways, giving $24 \times 60 = 1440$. The 720 value drops the ordering of the girls by dividing by $3!$. The 2880 value comes from allowing six gaps instead of five. The 5040 value is $7!$, the count with no restriction at all.

Q54. [QA-MCQ-8 · mcsc · diff 1]

Bag A contains 4 red and 6 green balls, while bag B contains 7 red and 3 green balls. One of the two bags is chosen at random and two balls are drawn from it together. Both drawn balls turn out to be red. What is the probability that bag A was the bag chosen?

- A) $\frac{1}{3}$
- B) $\frac{2}{9}$ ✓
- C) $\frac{7}{9}$
- D) $\frac{2}{15}$

Answer: B) $\frac{2}{9}$

Solution: Each bag has 10 balls, so two reds have chance $\frac{6}{45}$ from bag A and $\frac{21}{45}$ from bag B. Since the bags are equally likely, the required probability is $\frac{6}{6 + 21} = \frac{2}{9}$. The $\frac{2}{15}$ value is the unconditional chance of two reds from bag A alone. The $\frac{7}{9}$ value is the probability for bag B, the complement. The $\frac{1}{3}$ value comes from using $\frac{4}{10}$ and $\frac{7}{10}$ as single-draw odds and rounding the ratio.

Q55. [QA-MCQ-9 · mcsc · diff 1]

A solid metal cone of base radius 6 cm and height 24 cm is melted down, and 10% of the metal is lost in the process. The remaining metal is recast into solid spheres of radius 2 cm each. What is the largest number of complete spheres that can be made?

- A) 8
- B) 25
- C) 24 ✓
- D) 27

Answer: C) 24

Solution: The cone has volume 288π cubic cm, and 90% of it is 259.2π . Each sphere takes $\frac{32\pi}{3}$ cubic cm, so $259.2 \times \frac{3}{32} = 24.3$, and only 24 complete spheres are possible. The 27 value ignores the 10% metal loss. The 25 value comes from rounding 24.3 upward instead of down. The 8 value comes from omitting the factor $\frac{1}{3}$ in the volume of a sphere.

Q56. [QA-MCQ-10 · mcsc · diff 1]

Let N be the number whose prime factorisation is $2^a \times 3^b \times 5^c$. How many positive divisors of N are perfect squares and are also multiples of 12?

- A) 12
- B) 8 ✓
- C) 4
- D) 6

Answer: B) 8

Solution: A divisor $2^a \times 3^b \times 5^c$ is a perfect square only when a, b and c are all even, and it is a multiple of 12 only when a is at least 2 and b is at least 1. So a is 2 or 4, b is 2 or 4, and c is 0 or 2, giving $2 \times 2 \times 2 = 8$ divisors. The 12 value allows b = 0 as well, dropping the multiple of 3 condition. The 6 value allows b = 0 but also forces c = 0. The 4 value forgets that c can be 2.

Q57. [QA-MCQ-11 · mcsc · diff 1]

Three positive numbers are in arithmetic progression. Their sum is 27 and the sum of their squares is 293. What is the product of the three numbers?

- A) 405
- B) 585
- C) 504 ✓
- D) 729

Answer: C) 504

Solution: Write the numbers as $9 - d$, 9 and $9 + d$, since the sum 27 forces the middle term to be 9. Then $243 + 2d^2 = 293$, so $d^2 = 25$ and the numbers are 4, 9 and 14, whose product is 504. The 729 value comes from taking all three numbers equal to 9, which fits the sum but not the square sum. The 585 value uses $d = 4$ and the 405 value uses $d = 6$, neither of which gives 293.

Q58. [QA-MCQ-12 · mcsc · diff 1]

A non-zero real number x satisfies $x + 1/x = 3$. What is the value of $x^5 + 1/x^5$?

- A) 141
- B) 243
- C) 123 ✓
- D) 126

Answer: C) 123

Solution: Write $s(n) = x^n + 1/x^n$. Multiplying $s(n)$ by $(x + 1/x)$ gives $s(n+1) + s(n-1)$, so $s(n+1) = 3 s(n) - s(n-1)$ with $s(0) = 2$ and $s(1) = 3$. This gives $s(2) = 7$, $s(3) = 18$, $s(4) = 47$ and $s(5) = 123$. The value 243 comes from assuming the answer is 3^5 , that is from treating $x + 1/x$ as if it were x itself. The value 126 is the product 7 times 18, an invented shortcut $s(2)$ times $s(3)$. The value 141 follows from the slip $s(4) = 3$ times $18 - 3 = 51$ and then 3 times $51 - 18$.

Q59. [QA-MCQ-13 · mcsc · diff 1]

If a and b are positive real numbers with $a + b = 10$, what is the maximum possible value of $a^2 b^3$?

- A) 3087
- B) 4000
- C) 3125
- D) 3456 ✓

Answer: D) 3456

Solution: Split the sum as $a/2 + a/2 + b/3 + b/3 + b/3$, five positive terms adding to 10. By AM-GM the product $(a^2 b^3)/108$ is largest when all five terms are equal, so $a/2 = b/3 = 2$, giving $a = 4$ and $b = 6$. Then $a^2 b^3 = 16$ times $216 = 3456$. The value 3125 comes from setting $a = b = 5$, which maximises only symmetric products. The value 4000 comes from the guess $a = 2$, $b = 8$ followed by 4 times 1000. The value 3087 corresponds to $a = 3$ and $b = 7$, a nearby but smaller choice.

Q60. [QA-MCQ-14 · mcsc · diff 1]

The equation $(\log_3 x)^2 + 3 = \log_3 (x^4)$ holds for exactly two positive values of x . What is the sum of these two values?

- A) 30 ✓
- B) 84
- C) 12
- D) 28

Answer: A) 30

Solution: Put $t = \log_3 x$. Since $\log_3 (x^4) = 4t$, the equation becomes $t^2 - 4t + 3 = 0$, so $t = 1$ or $t = 3$, giving $x = 3$ and $x = 27$, whose sum is 30. The value 84 comes from misreading the roots as $t = 1$ and $t = 4$, which gives $3 + 81$. The value 12 comes from adding the roots 1 and 3 after cubing only one of them, that is $9 + 3$. The value 28 comes from dropping the constant 3 and using $t = 0$ and $t = 3$, so $1 + 27$.

Q61. [QA-MCQ-15 · mcsc · diff 1]

In the coordinate plane, $A = (0, 2)$ and $B = (12, 3)$. A point P moves along the x -axis. What is the smallest possible value of $PA + PB$?

- A) 13 ✓
- B) 17
- C) 15
- D) 12

Answer: A) 13

Solution: Reflect A in the x -axis to get $A' = (0, -2)$. For any P on the axis $PA = PA'$, so $PA + PB$ equals $PA' + PB$, which is least when P lies on the segment $A'B$. The minimum is the length $A'B = \sqrt{12^2 + 5^2} = 13$, reached at $P = (4.8, 0)$. The value 17 comes from adding the two heights and the horizontal gap, $2 + 3 + 12$. The value 12 is only the horizontal gap between A and B . The value 15 comes from subtracting the heights instead of adding them and then rounding $\sqrt{12^2 + 1^2}$ upward.

Q62. [QA-MCQ-16 · mcsc · diff 1]

Among 200 students, 120 play cricket, 90 play football and 70 play tennis. Exactly 40 play both cricket and football, 30 play both football and tennis, 35 play both cricket and tennis, and 15 play all three games. How many students play exactly one of the three games?

- A) 145
- B) 115 ✓
- C) 130
- D) 100

Answer: B) 115

Solution: By inclusion and exclusion the number playing at least one game is $120 + 90 + 70 - 40 - 30 - 35 + 15 = 190$. The pairwise counts include the 15 triple players, so the numbers playing exactly two games are $40 - 15 = 25$, $30 - 15 = 15$ and $35 - 15 = 20$, a total of 60. Subtracting the 60 and the 15 from 190 leaves 115 who play exactly one game. The value 130 comes from subtracting only 60 and forgetting the triple group. The value 100 comes from subtracting 2 times 15 twice over. The value 145 treats the pairwise figures as exactly-two counts already.

Q63. [QA-MCQ-17 · mcsc · diff 1]

What is the value of the sum $2/(1 \cdot 3) + 2/(3 \cdot 5) + 2/(5 \cdot 7) + \dots + 2/(47 \cdot 49)$?

- A) 24/25
- B) 48/49 ✓
- C) 47/49
- D) 24/49

Answer: B) 48/49

Solution: Each term splits as $2/((2k-1)(2k+1)) = 1/(2k-1) - 1/(2k+1)$, so consecutive parts cancel. With the last denominator pair being 47 and 49, k runs from 1 to 24 and the sum collapses to $1 - 1/49 = 48/49$. The value 24/49 comes from halving the telescoped result, that is from keeping the factor 1/2 that the numerator 2 has already cancelled. The value 47/49 comes from stopping the cancellation one term early at $1 - 2/49$. The value 24/25 comes from miscounting the terms and telescoping to $1 - 1/25$.

Q64. [QA-TITA-1 · tita · diff 0.5]

What is the sum of all the odd positive factors of 1800? Type your answer as an integer.

Answer (TITA): 403

Solution: 1800 factorises as $2^3 \times 3^2 \times 5^2$. An odd factor uses no power of 2, so the odd factors are exactly the factors of $3^2 \times 5^2 = 225$. Their sum is $(1 + 3 + 9)(1 + 5 + 25) = 13 \times 31 = 403$. Using the full factor sum $(1 + 2 + 4 + 8)(13)(31) = 6045$ wrongly keeps the even factors, and 13×31 miscounted as $12 \times 30 = 360$ comes from dropping the term 1 in each bracket.

Q65. [QA-TITA-2 · tita · diff 0.5]

In a colony of 120 households, every household subscribes to at least one of three newspapers A, B and C. Exactly 70 subscribe to A, 60 subscribe to B and 50 subscribe to C, while 12 households subscribe to all three. How many households subscribe to exactly two of the three newspapers? Type your answer as an integer.

Answer (TITA): 36

Solution: By inclusion and exclusion, $120 = 70 + 60 + 50$ minus (sum of the three pairwise overlaps) plus 12, so the pairwise overlap total is $180 + 12 - 120 = 72$. Each pairwise overlap counts the 12 all-three households once, so the exactly-two count is $72 - 3 \times 12 = 36$. Answering 72 stops at the raw pairwise total, and 48 comes from subtracting 12 only once instead of three times.

Q66. [QA-TITA-3 · tita · diff 1]

What is the remainder when $3^{2024} + 4^{2024}$ is divided by 100? Type your answer as an integer.

Answer (TITA): 37

Solution: The last two digits of powers of 3 repeat with period 20, and 2024 leaves remainder 4 on division by 20, so 3^{2024} ends in the same two digits as $3^4 = 81$. The last two digits of powers of 4 repeat with period 10 from the first power onward, and 2024 leaves remainder 4 on division by 10, so 4^{2024} ends like $4^4 = 256$, that is 56. Then $81 + 56 = 137$, whose remainder on division by 100 is 37. Answering 137 forgets the final reduction, and 97 comes from using a period of 4 for powers of 4 and getting 16.

Q67. [QA-TITA-4 · tita · diff 1]

The positive integers x and y satisfy $x + y = 30$. What is the largest possible value of $x^2 y$? Type your answer as an integer.

Answer (TITA): 4000

Solution: Write the product as $(x/2)(x/2)y$, three quantities whose sum is $x + y = 30$ and hence fixed. The product of such quantities is largest when they are equal, giving $x/2 = y$, so $x = 20$ and $y = 10$, and the value is $400 \times 10 = 4000$. Checking the integer neighbours confirms it, since $19^2 \times 11 = 3971$ and $21^2 \times 9 = 3969$. Answering 3375 comes from wrongly setting $x = y = 15$, and 8100 comes from maximising x^2 alone at $x = 27$.

Q68. [QA-TITA-5 · tita · diff 1]

In how many distinct arrangements of all eleven letters of the word MISSISSIPPI do no two letters S stand next to each other? Type your answer as an integer.

Answer (TITA): 7350

Solution: Set the four letters S aside and arrange the remaining seven letters M, I, I, I, I, P, P, which can be done in $7!/(4! \times 2!) = 105$ ways. Those seven letters create eight gaps, and choosing four of them for the four identical letters S keeps every pair of S separated, giving $C(8,4) = 70$ choices. The count is $105 \times 70 = 7350$. Answering 34650 is the total number of arrangements with no separation condition, and 8400 comes from using $C(8,4)$ with 120 instead of 105 for the non-S block.

IIT Jodhpur Qualifier - BS

Time: 2 Hours | Maximum Marks: 198

Question Distribution

- **VARC:** 24 questions (suggested time 40 min)
- **DILR:** 18 questions (suggested time 40 min)
- **QA:** 24 questions (suggested time 40 min)
- Total: 66 questions | MCQ 60 | TITA (non-MCQ) 6

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Total duration is 2 Hours; the suggested per-section times are shown above.
- Sections may be attempted in any order; manage your time across all sections.

This is the answer-marked verification copy (correct option in green, with solution) and mirrors the content of the upload CSV.

Section: VARC (suggested 40 min)

Instructions (VARC-RC1): study the shared passage below and answer its 4 question(s).

For centuries, cartographers left blank the interiors of continents they had not surveyed, and some filled the emptiness with drawings of imagined beasts. These flourishes were not mere decoration. A map is a claim about what is worth recording, and the choice of what to include always reflects the purposes of those who commission it. A trade map emphasises ports, rivers, and the shortest routes between markets; a military map foregrounds ridges, passes, and the width of fords. Neither is a neutral mirror of the land, because no map can show everything at once. To be useful, a map must leave things out.

This selective quality is precisely what makes maps powerful and, at times, misleading. A traveller who trusts a road map assumes that the roads shown are the ones that matter, and rarely wonders about the villages the mapmaker chose to omit. When an authority draws a boundary as a firm line, the line can come to feel as solid as a wall, even where the ground itself offers no such division. In this way a map does not simply describe a territory; it quietly shapes how people imagine and argue about that territory.

Modern digital maps intensify the effect rather than dissolving it. They appear complete and objective, yet every one rests on decisions about which streets to label, which businesses to display, and which neighbourhoods to render in detail. The reader sees a clean surface and forgets the editing beneath it. Understanding a map, then, requires more than reading its symbols. It requires asking who made it, for whom, and what they had reason to leave in the blank spaces.

Q1. [VARC-RC1 · mcsc · diff 0.5]

Which of the following best expresses the main idea of the passage?

- A) Trade maps and military maps differ chiefly because rivers matter more to merchants than mountain passes ever will.
- B) Early cartographers decorated unknown regions with imaginary creatures to disguise the limits of their knowledge from buyers.
- C) Digital maps have finally removed the distortions that troubled the printed maps of earlier centuries and travellers.
- D) Maps are selective representations whose contents reflect the aims of their makers rather than a neutral picture of the land. ✓

Answer: D) Maps are selective representations whose contents reflect the aims of their makers rather than a neutral picture of the land.

Solution: The passage repeatedly argues that maps leave things out according to the maker's purpose and thus are not neutral mirrors, which the selective-representation option captures. The creatures point is a single opening example, not the thesis. The claim that digital maps remove distortion is contradicted, since the text says they intensify the effect. The trade-versus-military contrast is one illustration, not the overall point.

Q2. [VARC-RC1 · mcsc · diff 0.5]

According to the passage, all of the following are true of maps EXCEPT:

- A) The features a map emphasises depend on the purpose for which it was commissioned by its makers.
- B) A digital map presents a complete and objective record precisely because it labels every street and business. ✓
- C) A boundary drawn as a firm line can come to feel solid even where the land shows no real division.
- D) A map cannot display everything at once, so being useful depends on deliberately leaving certain things out.

Answer: B) A digital map presents a complete and objective record precisely because it labels every street and business.

Solution: The passage says digital maps only appear complete and objective while resting on hidden editing choices, so treating them as genuinely complete is the claim the text denies. The other statements are directly supported: maps must omit to be useful, a firm boundary line can feel like a wall, and emphasis follows the commissioning purpose.

Q3. [VARC-RC1 · mcsc · diff 0.5]

It can be inferred from the passage that a reader who wants to judge a map fairly should:

- A) assume that the roads a road map displays are always the ones that truly matter
- B) prefer digital maps to printed ones because their surfaces are cleaner and easier
- C) consider the intentions behind the map and notice what its maker chose not to show ✓
- D) accept that any boundary drawn as a firm line marks a genuine division in the land

Answer: C) consider the intentions behind the map and notice what its maker chose not to show

Solution: The closing lines say understanding a map requires asking who made it, for whom, and what was left in the blank spaces, so a fair reader weighs purpose and omission. Preferring digital maps ignores the warning that clean surfaces hide editing. Trusting firm boundaries and displayed roads repeats the very errors the passage cautions against.

Q4. [VARC-RC1 · mcsc · diff 0.5]

Why does the author mention trade maps and military maps in the first paragraph?

- A) To suggest that the two kinds of map were usually produced by the same workshops
- B) To prove that military needs have always mattered far more than trade did to early states everywhere
- C) To show that different aims lead makers to highlight different features of the same land ✓
- D) To argue that rivers are harder to chart accurately than ridges and mountain passes

Answer: C) To show that different aims lead makers to highlight different features of the same land

Solution: The two examples illustrate the point that a map's emphasis follows its purpose, one stressing routes to markets and the other stressing terrain for movement. The passage makes no ranking of military over trade, offers no claim about charting difficulty, and says nothing about shared workshops.

Instructions (VARC-RC2): study the shared passage below and answer its 4 question(s).

When a forest burns, the loss looks total. Yet many woodlands have evolved not merely to survive fire but to depend on it. The cones of certain pines are sealed shut with resin and open only when the heat of a blaze melts that seal, releasing seeds onto ground newly cleared of competitors and enriched with ash. For such trees, a fire is not a disaster but the signal to reproduce. Ecologists describe these systems as fire-adapted, and they are far more common than the popular image of fire as pure destruction suggests.

The difficulty is that not all fires are alike. A frequent, low-intensity fire clears undergrowth while leaving mature trees standing, and many species are tuned to exactly this rhythm. When people suppress every small fire in order to protect property, dead wood and dry brush accumulate year after year. The next fire, when it finally comes, burns hotter and higher than anything the forest is built to withstand, killing even the trees that would ordinarily endure. In trying to prevent fire altogether, a policy can make the eventual blaze far more severe.

This paradox has forced a rethinking of how forests are managed. In several regions, foresters now set controlled burns on purpose, choosing cool, damp days when a fire can be kept small and contained. The aim is not to eliminate fire but to restore its natural frequency, so that fuel never builds to dangerous levels. The approach demands patience and public trust, since a deliberate fire is an unsettling sight. But it reflects a hard lesson: a landscape shaped by fire cannot be kept safe by pretending fire away.

Q5. [VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the central argument of the passage?

- A) Foresters in several regions have recently begun to distrust the public's reaction to any deliberate fire.
- B) Because many forests are adapted to regular fire, suppressing all fires can make later blazes more destructive. ✓
- C) The resin-sealed cones of certain pines will open only when a passing blaze first melts the seal and then frees the seeds.
- D) Controlled burns should be set on damp days so that the flames stay small, contained, and easy to watch.

Answer: B) Because many forests are adapted to regular fire, suppressing all fires can make later blazes more destructive.

Solution: The passage builds from fire-adapted forests to the paradox that blanket suppression lets fuel accumulate and produces worse fires, which the correct option summarises. The damp-day detail and the pine cones are supporting examples, and public distrust is a practical hurdle mentioned in passing, not the core claim.

Q6. [VARC-RC2 · mcsc · diff 0.5]

According to the passage, why does suppressing every small fire eventually lead to a more severe blaze?

- A) The resin seals on pine cones grow thicker each season and demand a hotter fire to open.
- B) Dead wood and dry brush pile up over the years, giving a later fire far more fuel to burn. ✓
- C) Mature trees lose their natural resistance to heat when they are protected for too long.
- D) Damp weather becomes rarer over time, so foresters can no longer schedule any cool burns.

Answer: B) Dead wood and dry brush pile up over the years, giving a later fire far more fuel to burn.

Solution: The second paragraph states that when small fires are suppressed, dead wood and dry brush accumulate year after year, so the next fire burns hotter and higher. The other options invent mechanisms the passage never gives: thickening seals, lost heat resistance, or vanishing damp weather.

Q7. [VARC-RC2 · mcsc · diff 0.5]

It can be inferred that the author would most likely agree with which of the following statements?

- A) Popular opinion about fire is usually a reliable guide to how forests should actually be managed.
- B) The safest forest is one in which every fire, however small, is put out as quickly as it can be found.
- C) Protecting the property of nearby residents should never once be counted among the goals of any modern forest-management plan.
- D) A management policy works better when it cooperates with a landscape's natural patterns than when it fights them. ✓

Answer: D) A management policy works better when it cooperates with a landscape's natural patterns than when it fights them.

Solution: The passage praises restoring fire's natural frequency and warns against pretending fire away, so the author favours working with natural patterns. Dismissing property protection goes further than the text, which treats it as a real concern. Extinguishing every fire is the very policy criticised, and the passage says the popular image of fire is misleading.

Q8. [VARC-RC2 · mcsc · diff 0.5]

Which word best describes the author's overall tone in the passage?

- A) Measured and explanatory, presenting a corrective view of a common belief ✓
- B) Alarmed and urgent, warning that forests everywhere face imminent collapse
- C) Playful and ironic, mocking the foresters who still set deliberate burns
- D) Nostalgic and mournful, lamenting woodlands that can no longer be saved

Answer: A) Measured and explanatory, presenting a corrective view of a common belief

Solution: The author calmly explains fire adaptation and the suppression paradox to correct the popular image of fire as pure destruction, a measured and explanatory stance. The tone is not alarmed about imminent collapse, not mournful about lost forests, and not mocking of foresters, whose controlled-burn practice the passage endorses.

Instructions (VARC-RC3): study the shared passage below and answer its 4 question(s).

The public library began as a simple idea: that useful books should be available to anyone, not only to those who could afford to buy them. Early lending libraries charged a small fee, which kept out the very readers who might have gained the most. The breakthrough came when towns agreed to pay for libraries out of shared taxes, so that borrowing became free at the point of use. A reader needed only to live in the town and to return the books on time. For many families, the library was the first place where knowledge cost nothing.

Over time, libraries grew into something larger than shelves of books. They became quiet places to study, warm rooms to sit in, and centres where people could ask a trained librarian for help finding what they needed. Children attended reading hours, job seekers used the desks, and newcomers to a town found notices about local life. The building offered not just texts but a shared space open to all, regardless of wealth.

Today, when a great deal of information sits a click away, some people ask whether libraries are still needed. Yet the answer many communities give is yes. Not everyone owns a computer or a quiet place to work, and not every question is easily answered by a search engine. The library remains what it has long been: a place where help and knowledge are offered freely, to whoever walks through the door.

Q9. [VARC-RC3 · mcsc · diff 0]

Which of the following best captures the main point of the passage?

- A) Librarians spend more of their time helping job seekers than assisting young children.
- B) The public library has long offered free access to knowledge and remains valuable today. ✓
- C) Early lending libraries failed mainly because the fees they charged were set far too high.
- D) Search engines have replaced most of the services that public libraries once provided.

Answer: B) The public library has long offered free access to knowledge and remains valuable today.

Solution: The passage traces the library from free tax-funded lending to a broad community space and argues it is still needed, which the correct option captures. The fee point is background, the search-engine claim is the view the passage rejects, and the comparison of job seekers to children is never made.

Q10. [VARC-RC3 · mcsc · diff 0]

According to the passage, what change made borrowing free at the point of use?

- A) Librarians volunteered their time so that no salaries had to be paid at all.
- B) Towns agreed to fund libraries through shared taxes instead of charging each reader. ✓
- C) Publishers lowered the price of books until borrowing them cost almost nothing.
- D) Wealthy donors bought up large private collections of books and then handed them over to the towns.

Answer: B) Towns agreed to fund libraries through shared taxes instead of charging each reader.

Solution: The first paragraph says the breakthrough came when towns agreed to pay for libraries out of shared taxes, making borrowing free at the point of use. Donors, volunteer librarians, and cheaper book prices are not mentioned as the cause.

Q11. [VARC-RC3 · mcsc · diff 0.5]

It can be inferred from the passage that the author regards the claim that libraries are no longer needed as:

- A) reasonable, since a search engine now answers nearly every possible question
- B) unimportant, because libraries were never truly meant to serve the whole public
- C) mistaken, because many people still lack the resources a library freely provides ✓
- D) proven, given that most families already own computers of their own at home

Answer: C) mistaken, because many people still lack the resources a library freely provides

Solution: The final paragraph answers the doubt with a yes, noting that not everyone owns a computer or a quiet workspace and that search engines cannot answer every question, so the author treats the claim as mistaken. The other options either accept the doubt or contradict the passage's insistence that the library serves everyone.

Q12. [VARC-RC3 · mcsc · diff 0.5]

What is the main purpose of the second paragraph?

- A) To compare the reading hours for children with the desks used by job seekers
- B) To show that libraries came to serve as shared community spaces, not only stores of books ✓
- C) To describe how librarians are trained before they begin helping the public
- D) To explain the exact set of rules that a reader had to carefully follow in order to borrow any book

Answer: B) To show that libraries came to serve as shared community spaces, not only stores of books

Solution: The second paragraph lists study space, warmth, librarian help, reading hours, and notices to show the library grew into a shared community space beyond its shelves. It does not lay out borrowing rules, describe librarian training, or set up a comparison between children and job seekers.

Q13. [VARC-ODD-1 · mcsc · diff 0.5]

Five sentences are given below. Four of them can be joined to form a coherent paragraph on a single idea, while one does not fit.

- (1) Honeybees communicate the location of food through a series of movements known as the waggle dance.
- (2) The angle of the dance relative to vertical tells other bees the direction of the flowers.
- (3) The duration of each waggle run signals roughly how far away the food source lies.
- (4) A large hive can hold tens of thousands of individual worker bees during summer.
- (5) Observers can even estimate distance by timing how long each run lasts.

Which sentence is the odd one out?

- A) Sentence 1
- B) Sentence 4 ✓
- C) Sentence 3
- D) Sentence 5
- E) Sentence 2

Answer: B) Sentence 4

Solution: Sentences 1, 2, 3 and 5 all describe how the waggle dance encodes the direction and distance of food, forming one connected idea. Sentence 4 shifts to hive population, an unrelated statistic that breaks the theme, so it is the misfit. The other options each belong to the dance-communication chain.

Q14. [VARC-ODD-2 · mcsc · diff 0.5]

Five sentences are given below. Four of them can be joined to form a coherent paragraph on a single idea, while one does not fit.

- (1) The invention of movable type allowed identical pages to be printed quickly and cheaply.
- (2) Books that once took months to copy by hand could suddenly be produced in days.
- (3) Wider access to printed texts helped standardise spelling across many regions.
- (4) Lower prices meant that ideas spread far beyond wealthy monasteries and courts.
- (5) Many early manuscripts were decorated with gold leaf and elaborate hand-painted borders.

Which sentence is the odd one out?

- A) Sentence 2
- B) Sentence 3
- C) Sentence 5 ✓
- D) Sentence 1
- E) Sentence 4

Answer: C) Sentence 5

Solution: Sentences 1, 2, 3 and 4 all trace the effects of printing, faster output, standard spelling and cheaper wider access, so they cohere. Sentence 5 describes the decoration of hand-made manuscripts, which runs against the theme of mass printing and is the odd one out. The remaining options each extend the printing-impact idea.

Q15. [VARC-ODD-3 · mcsc · diff 0.5]

Five sentences are given below. Four of them can be joined to form a coherent paragraph on a single idea, while one does not fit.

- (1) Coral reefs support a quarter of all marine species despite covering a tiny fraction of the ocean.
- (2) The corals depend on algae living inside their tissues to provide most of their food.
- (3) When water grows too warm, the corals expel these algae and turn a pale white.
- (4) A prolonged loss of algae can leave the corals starving and eventually dead.
- (5) Deep-sea trenches remain among the least explored regions of the entire planet.

Which sentence is the odd one out?

- A) Sentence 5 ✓
- B) Sentence 3
- C) Sentence 1
- D) Sentence 4
- E) Sentence 2

Answer: A) Sentence 5

Solution: Sentences 1 to 4 build one picture of coral reefs and their reliance on algae, including the bleaching that follows warming. Sentence 5 jumps to deep-sea trenches, a separate ocean topic that does not connect, making it the misfit. Each other option contributes to the coral-and-algae narrative.

Q16. [VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, form a paragraph when placed in the correct order. Rearrange them into the most logical sequence.

1. A city planted thousands of trees along its hottest streets to cool them down.
2. Within a few summers, shaded pavements were measurably cooler than bare ones.
3. Residents also reported walking more often once the routes felt pleasant.
4. The idea began as a simple response to rising temperatures in the centre.

Select the correct order.

- A) 4-1-2-3 ✓
- B) 1-2-3-4
- C) 4-2-1-3
- D) 1-4-3-2

Answer: A) 4-1-2-3

Solution: Sentence 4 opens with the origin of the idea, sentence 1 states the action taken, sentence 2 gives the first measured result, and sentence 3 adds a further outcome. The order 4-1-2-3 moves from motive to action to results. Sequences starting with 1 skip the framing motive, and 4-2-1-3 places a result before the action that caused it.

Q17. [VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, form a paragraph when placed in the correct order. Rearrange them into the most logical sequence.

1. This layer traps heat and keeps the surface warm enough for liquid water.
2. The atmosphere of a planet is the blanket of gases held by its gravity.
3. Without it, the same surface would swing between scorching days and freezing nights.
4. On our world that blanket is mostly nitrogen with a smaller share of oxygen.

Select the correct order.

- A) 2-4-1-3 ✓
- B) 2-1-4-3
- C) 4-2-1-3
- D) 2-3-1-4

Answer: A) 2-4-1-3

Solution: Sentence 2 defines the atmosphere, sentence 4 names its main gases, sentence 1 explains the warming effect of that layer, and sentence 3 states the consequence of its absence. The order 2-4-1-3 flows from definition to composition to function to contrast. Options that put 1 before 4 mention the layer before it is described, and 4 cannot open since it refers back to a blanket not yet named.

Q18. [VARC-JUMBLE-3 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, form a paragraph when placed in the correct order. Rearrange them into the most logical sequence.

1. She began by sorting the beans by size before the roasting could start.
2. A small cafe decided to roast its own coffee for the first time.
3. Only then did the owner adjust the heat to bring out the flavour.
4. Uneven beans, she knew, would scorch and cook at different rates.

Select the correct order.

A) 2-1-4-3 ✓

B) 2-3-1-4

C) 1-4-2-3

D) 2-4-1-3

Answer: A) 2-1-4-3

Solution: Sentence 2 sets the scene, sentence 1 describes the first step of sorting, sentence 4 explains why that step matters, and sentence 3 uses 'only then' to mark the next stage. The order 2-1-4-3 keeps action before its justification and reserves 3 for last. Placing 4 before 1 explains a step not yet mentioned, and 3 cannot precede the sorting it depends on.

Q19. [VARC-JUMBLE-4 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, form a paragraph when placed in the correct order. Rearrange them into the most logical sequence.

1. Early maps of the region were drawn mostly from travellers' rough sketches.
2. As a result, coastlines were often placed dozens of miles from their true position.
3. Later surveys with proper instruments corrected many of these old errors.
4. Today satellites fix the same coastlines to within a few metres.

Select the correct order.

A) 3-1-2-4

B) 1-3-2-4

C) 1-2-4-3

D) 1-2-3-4 ✓

Answer: D) 1-2-3-4

Solution: The paragraph is chronological: sentence 1 describes early sketch-based maps, sentence 2 gives their result, sentence 3 introduces later instrument surveys, and sentence 4 ends with modern satellites. The order 1-2-3-4 follows time and cause. Swapping 2 and 3 separates the error from its cause, and ending on 3 rather than 4 reverses the march toward greater accuracy.

Q20. [VARC-SUMMARY-1 · mcsc · diff 0]

Libraries once measured their worth by how many books they owned, but many now judge themselves by how well they connect people to information wherever it lives. A modern library lends digital files, offers work and meeting space, and guides visitors through vast online collections it does not physically hold. The building still matters, yet its main job has shifted from storing objects to giving access.

Which option best summarises the paragraph?

A) Libraries now judge their worth by how well they connect people to information rather than by how many books they own. ✓

B) Modern libraries lend out digital files and provide quiet meeting rooms as well as long shelves of printed books for their many visitors.

C) Libraries no longer need any physical building at all, since nearly every part of their collection is now stored and read online.

D) The sheer number of books that a library owns remains the clearest and most reliable measure of how useful it truly is.

Answer: A) Libraries now judge their worth by how well they connect people to information rather than by how many books they own.

Solution: The passage contrasts an older ownership measure with a new access measure, which the correct option captures fully. The digital-and-meeting-room option is too narrow, listing services without the central shift. Claiming buildings are unneeded overreaches beyond the text, which says the building still matters, and the counting-books option restates the old view the passage rejects.

Q21. [VARC-SUMMARY-2 · mcsc · diff 0.5]

Sleep is often treated as lost time, yet the brain stays busy during it. While a person rests, the brain sorts through the day's experiences, strengthens useful memories and clears away waste that builds up during waking hours. Far from being idle, sleep is active maintenance that keeps the mind working well the next day.

Which option best summarises the paragraph?

- A) Sleep feels like wasted time to most people because nothing worth measuring ever happens inside the resting brain.
- B) People who sleep for longer will always think faster and remember far more facts than those who choose to sleep much less.
- C) During each period of sleep the brain quietly strengthens the useful memories a person has formed across the whole of the previous day.
- D) Sleep is not lost time but active brain upkeep that sorts memories and clears waste to keep the mind sharp.

✓

Answer: D) Sleep is not lost time but active brain upkeep that sorts memories and clears waste to keep the mind sharp.

Solution: The main point is that sleep is productive upkeep involving memory sorting and waste clearing, which the correct option states in full. The memory-strengthening option is too narrow, covering only one function. The 'always faster' option overreaches into a claim the passage never makes, and the 'nothing happens' option distorts the text, which argues the opposite.

Q22. [VARC-SUMMARY-3 · mcsc · diff 0.5]

Fixing a bicycle taught the student more than any lecture had. By taking the gears apart and putting them back together, she saw how each part depended on the others in a way diagrams had never made clear. The lesson was not really about bicycles; it was that handling something directly can reveal what mere description leaves hidden.

Which option best summarises the paragraph?

- A) Bicycles are surprisingly complicated machines whose many gears turn out to depend on one another in unexpected ways.
- B) The student slowly learned how to take the gears of a bicycle apart and then reassemble them correctly during her afternoon repair.
- C) Handling a real object directly can reveal how its parts relate in ways that descriptions and diagrams cannot.
- D) Lectures and diagrams are quite useless for teaching almost anything and should be replaced entirely by hands-on practice sessions.

✓

Answer: C) Handling a real object directly can reveal how its parts relate in ways that descriptions and diagrams cannot.

Solution: The paragraph generalises from one repair to a claim about hands-on learning, which the correct option captures. The gear-reassembly option is too narrow, sticking to the literal task. The 'lectures are useless' option overreaches, since the text only says direct handling reveals more, and the bicycle-complexity option treats the example as the point rather than the lesson about learning.

Q23. [VARC-SUMMARY-4 · mcsc · diff 0.5]

A town worried about litter tried fines for a year with little effect. It then placed more bins and painted cheerful footpaths leading to them, and litter fell sharply. The change suggested that making the right action easy and inviting often works better than simply punishing the wrong one.

Which option best summarises the paragraph?

- A) Fines can never change the way the public behaves and ought to be abolished by every town that seriously hopes to stay clean.
- B) Making a desired action easy and inviting can cut a problem more than punishing the unwanted behaviour does. ✓
- C) The town first tried out fines for a full year before it finally decided to add many more bins around its busy streets.
- D) The town carefully painted a set of cheerful footpaths that were meant to guide residents toward its newly installed litter bins.

Answer: B) Making a desired action easy and inviting can cut a problem more than punishing the unwanted behaviour does.

Solution: The passage draws a general lesson that easing the right action beats punishing the wrong one, which the correct option states. The painted-footpaths option is too narrow, naming one tactic only. The 'fines never work' option overreaches beyond the single case, and the fines-first-for-a-year option reports a detail without the main conclusion.

Q24. [VARC-SUMMARY-5 · mcsc · diff 0.5]

Maps always leave things out, and that is their strength. A useful subway map ignores true distances and street layouts so that riders can see the lines and stops at a glance. By dropping detail that would only clutter the task at hand, a good map makes the information that matters easy to read.

Which option best summarises the paragraph?

- A) Maps that leave out any information are basically misleading and cannot really be trusted for any kind of serious purpose.
- B) Subway riders simply need to see the coloured lines and the stops so that they can plan their daily journeys across the city.
- C) Subway maps very deliberately ignore both the true distances between stations and the real layout of the streets above them.
- D) A good map is useful because it leaves out detail that would clutter the task, showing only what matters. ✓

Answer: D) A good map is useful because it leaves out detail that would clutter the task, showing only what matters.

Solution: The central claim is that leaving out detail is what makes a map useful, which the correct option expresses. The subway-distances option is too narrow, giving only the example. The 'maps are misleading' option distorts the passage, which praises omission, and the riders-plan-journeys option states a side detail rather than the main idea.

Section: DILR (suggested 40 min)

Instructions (DILR-1): study the shared data below and answer its 3 question(s).

Five runners, Neha, Omar, Priya, Rahul and Sana, finished a road race in positions one (first) through five (last), with no ties.

- Omar finished last. - Priya finished immediately before Rahul. - Neha finished ahead of both Priya and Sana. - Sana did not finish fourth.

Q25. [DILR-1 · mcsc · diff 0.5]

Who finished in second place?

- A) Sana ✓
- B) Priya
- C) Neha
- D) Rahul

Answer: A) Sana

Solution: Omar is fifth. The Priya-Rahul block sits at positions three and four, so Neha and Sana take positions one and two. Since Neha finishes ahead of Sana, Neha is first and Sana is second. Neha is the wrong pick from reading the block, while Priya and Rahul are fixed at third and fourth.

Q26. [DILR-1 · mcsc · diff 0.5]

Who finished in fourth place?

- A) Priya
- B) Sana
- C) Rahul ✓
- D) Omar

Answer: C) Rahul

Solution: The Priya-Rahul block occupies positions three and four, placing Rahul fourth. Priya is one place earlier at third, Sana is second, and Omar is fixed last rather than fourth.

Q27. [DILR-1 · mcsc · diff 0.5]

How many runners finished between Sana and Omar?

- A) Three
- B) Zero
- C) One
- D) Two ✓

Answer: D) Two

Solution: Sana is second and Omar is fifth, so Priya (third) and Rahul (fourth) sit between them, which is two runners. Counting only one or three misreads the gap, and zero ignores the block entirely.

Instructions (DILR-2): study the shared data below and answer its 3 question(s).

Four teams, Falcons, Hawks, Owls and Ravens, took part in a quiz. Their final scores were the four distinct values 30, 40, 50 and 60 points, in some order.

- The Hawks scored exactly 20 points more than the Owls. - The Ravens outscored the Falcons. - The Falcons did not finish with the lowest score.

Q28. [DILR-2 · mcsc · diff 0.5]

Which team won the quiz with the highest score?

- A) Hawks
- B) Owls
- C) Ravens ✓
- D) Falcons

Answer: C) Ravens

Solution: The Owls are not lowest only if Falcons take 30, but Falcons are not last, so the Owls score 30 and the Hawks score 50. That leaves 40 and 60 for Falcons and Ravens, and since Ravens outscore Falcons, Ravens take 60 and win. Hawks at 50 and Falcons at 40 are runners-up, and Owls are last.

Q29. [DILR-2 · mcsc · diff 0.5]

How many points did the Falcons score?

- A) 40 ✓
- B) 30
- C) 60
- D) 50

Answer: A) 40

Solution: With Owls on 30 and Hawks on 50, the remaining 40 and 60 go to Falcons and Ravens. Ravens outscore Falcons, so Falcons take 40. Reading 30 mistakes them for last, while 50 and 60 belong to Hawks and Ravens.

Q30. [DILR-2 · mcsc · diff 0.5]

What is the difference between the highest and the lowest team scores?

- A) 30 ✓
- B) 10
- C) 20
- D) 60

Answer: A) 30

Solution: The top score is the Ravens on 60 and the bottom is the Owls on 30, a gap of 30. The 20 trap echoes the Hawks-Owls margin, 60 mistakes the top score for the gap, and 10 halves the difference.

Instructions (DILR-3): study the shared data below and answer its 4 question(s).

A regional juice company sells three flavours, Mango, Guava and Litchi, at four riverside kiosks named P, Q, R and S. Table 1 gives the number of bottles sold and Table 2 gives the selling price of each bottle. All bottle counts are in thousands of bottles; all prices are in rupees per bottle; revenue = bottles sold x price per bottle, so each revenue figure comes out in thousands of rupees.

Table 1: Bottles sold (in thousands)

Flavour	P	Q	R	S
Mango	12	8	10	6
Guava	5	9	7	4
Litchi	8	6	5	9

Table 2: Price per bottle (in rupees)

Flavour	P	Q	R	S
Mango	25	25	30	30
Guava	20	20	20	20
Litchi	15	15	10	10

Q31. [DILR-3 · mcsc · diff 0.5]

How many bottles in total (in thousands) were sold at kiosk Q across all three flavours?

- A) 22 thousand
- B) 23 thousand ✓
- C) 21 thousand
- D) 24 thousand

Answer: B) 23 thousand

Solution: Kiosk Q is the second data column: Mango 8, Guava 9 and Litchi 6 bottles, all in thousands. Adding the column gives $8 + 9 + 6 = 23$ thousand bottles. The 21 trap drops the 8, the 22 trap misreads Guava as 8, and the 24 trap adds an extra unit.

Q32. [DILR-3 · mcsc · diff 0.5]

What was the total revenue (in thousands of rupees) generated by kiosk P?

- A) 540 thousand
- B) 470 thousand
- C) 520 thousand ✓
- D) 500 thousand

Answer: C) 520 thousand

Solution: At kiosk P: Mango $12 \times 25 = 300$, Guava $5 \times 20 = 100$ and Litchi $8 \times 15 = 120$, all in thousands of rupees. The total is $300 + 100 + 120 = 520$ thousand. The 500 and 540 traps mis-price the Litchi bottles and 470 is the revenue of kiosk Q instead of P.

Q33. [DILR-3 · mcsc · diff 0.5]

What was the combined revenue of the Mango flavour (in thousands of rupees) across all four kiosks?

- A) 880 thousand
- B) 1000 thousand
- C) 900 thousand
- D) 980 thousand ✓

Answer: D) 980 thousand

Solution: Mango revenue by kiosk: $12 \times 25 = 300$, $8 \times 25 = 200$, $10 \times 30 = 300$ and $6 \times 30 = 180$, all in thousands of rupees. The row total is $300 + 200 + 300 + 180 = 980$ thousand. The 900 and 880 traps apply a flat price of 25 or 30 to every kiosk, and 1000 rounds carelessly.

Q34. [DILR-3 · mcsc · diff 0.5]

In how many of the twelve flavour-kiosk cells did the revenue exceed 150 thousand rupees?

- A) 3 cells
- B) 4 cells
- C) 5 cells ✓
- D) 6 cells

Answer: C) 5 cells

Solution: Cell revenues in thousands are Mango 300, 200, 300, 180; Guava 100, 180, 140, 80; Litchi 120, 90, 50, 90. Those above 150 are the four Mango cells plus Guava at kiosk Q (180), giving 5 cells. The 4 trap forgets the Guava cell, while 6 and 3 miscount the Mango row.

Instructions (DILR-4): study the shared data below and answer its 3 question(s).

Four employees, Arjun, Bela, Carlos and Diana, each attended a one-day training session. The sessions were held on four different days: Monday, Tuesday, Wednesday and Thursday, one employee per day.

- Arjun trained on an earlier day than Bela. - Carlos trained exactly two days after Diana. - Diana did not train on Monday.

Q35. [DILR-4 · mcsc · diff 0.5]

On which day did Arjun train?

- A) Thursday
- B) Monday ✓
- C) Tuesday
- D) Wednesday

Answer: B) Monday

Solution: Carlos is two days after Diana, and Diana avoids Monday, so Diana is Tuesday and Carlos is Thursday. Monday and Wednesday remain for Arjun and Bela, and since Arjun is earlier, Arjun takes Monday. Tuesday and Thursday are taken by Diana and Carlos, and Wednesday belongs to Bela.

Q36. [DILR-4 · mcsc · diff 0.5]

Who trained on Wednesday?

- A) Diana
- B) Carlos
- C) Arjun
- D) Bela ✓

Answer: D) Bela

Solution: Diana is Tuesday and Carlos is Thursday, leaving Monday and Wednesday for Arjun and Bela. Arjun is earlier, so Arjun is Monday and Bela is Wednesday. Diana and Carlos are fixed on other days, and Arjun sits on Monday.

Q37. [DILR-4 · mcsc · diff 1]

Who trained on the day immediately before Carlos?

- A) Arjun
- B) No one, Carlos trained on the first day
- C) Bela ✓
- D) Diana

Answer: C) Bela

Solution: Carlos trains on Thursday, so the day immediately before is Wednesday, which is Bela. Diana trains two days earlier on Tuesday rather than the day before, Arjun is on Monday, and Carlos is certainly not on the first day.

Instructions (DILR-5): study the shared data below and answer its 5 question(s).

The grouped bar chart below shows the number of units sold by three solar lamp models, Solaris, Lumen and Vireo, in the years 2022, 2023 and 2024. All figures are in thousands of units, so a bar of height 40 means 40 thousand units were sold.

Q38. [DILR-5 · mcsc · diff 0.5]

How many units in total (in thousands) were sold across all three models in 2023?

- A) 115 thousand
- B) 130 thousand
- C) 110 thousand
- D) 120 thousand ✓

Answer: D) 120 thousand

Solution: The 2023 bars read 50 for Solaris, 40 for Lumen and 30 for Vireo, all in thousands of units. Their sum is $50 + 40 + 30 = 120$ thousand. The 110 trap drops a Vireo unit, 130 borrows a 2024 bar, and 115 comes from misreading Solaris as 45.

Q39. [DILR-5 · mcsc · diff 0.5]

By what percentage did the units sold of Solaris grow from 2022 to 2024?

- A) 50 percent ✓
- B) 33 percent
- C) 20 percent
- D) 40 percent

Answer: A) 50 percent

Solution: Solaris rose from 40 thousand units in 2022 to 60 thousand in 2024, an increase of 20 thousand. As a percentage of the 2022 base, $20 / 40 = 50$ percent. The 33 trap wrongly uses 60 as the base, 20 is the raw increase read as a percentage, and 40 mixes up the models.

Q40. [DILR-5 · mcsc · diff 0.5]

In which year were the total units sold across all three models the highest?

- A) Equal in 2023 and 2024
- B) 2022
- C) 2024 ✓
- D) 2023

Answer: C) 2024

Solution: Year totals in thousands are $40 + 30 + 20 = 90$ for 2022, $50 + 40 + 30 = 120$ for 2023 and $60 + 50 + 40 = 150$ for 2024. The largest total, 150 thousand, belongs to 2024. Because 150 clearly beats 120, the years are not tied and neither earlier year leads.

Q41. [DILR-5 · mcsc · diff 0.5]

What is the ratio of Lumen units sold in 2024 to Vireo units sold in 2022?

- A) 5 : 2 ✓
- B) 5 : 3
- C) 3 : 1
- D) 2 : 1

Answer: A) 5 : 2

Solution: Lumen sold 50 thousand units in 2024 and Vireo sold 20 thousand units in 2022. The ratio $50 : 20$ divides by 10 to give $5 : 2$. The $2 : 1$ and $5 : 3$ traps come from misreading one of the bars, and $3 : 1$ wrongly pairs Lumen with a 2023 Vireo value.

Q42. [DILR-5 · mcsc · diff 0.5]

What was the average number of units (in thousands) sold per model in 2024?

- A) 45 thousand
- B) 48 thousand
- C) 50 thousand ✓
- D) 55 thousand

Answer: C) 50 thousand

Solution: The 2024 bars are 60 for Solaris, 50 for Lumen and 40 for Vireo, all in thousands of units, summing to 150 thousand. Dividing by the three models gives $150 / 3 = 50$ thousand. The 45 trap averages the 2023 figures, and 55 and 48 come from arithmetic slips on the total.

Section: QA (suggested 40 min)

Q43. [QA-MCQ-1 · mcsc · diff 0]

In a class of 250 students, 36% opted for the science club. How many students opted for the science club?

- A) 90 ✓
- B) 214
- C) 160
- D) 36

Answer: A) 90

Solution: 36% of 250 equals $0.36 \times 250 = 90$ students, so 90 is correct. The value 36 wrongly reads the percentage as the count. The value 160 counts the remaining 64% who did not opt in, and 214 subtracts 36 from 250 by treating the percent as an absolute figure.

Q44. [QA-MCQ-2 · mcsc · diff 0]

A shopkeeper buys a bag for 400 rupees and sells it for 460 rupees. What is the profit percentage?

- A) 13%
- B) 15% ✓
- C) 12%
- D) 60%

Answer: B) 15%

Solution: Profit is $460 - 400 = 60$, and profit percent is $60/400 \times 100 = 15\%$, so 15% is correct. The 13% figure wrongly uses the selling price 460 as the base. The 60% reads the absolute profit of 60 rupees as a percentage, and 12% divides 60 by 500.

Q45. [QA-MCQ-3 · mcsc · diff 0]

Two numbers are in the ratio 3 : 5, and their sum is 96. What is the larger number?

- A) 48
- B) 36
- C) 60 ✓
- D) 58

Answer: C) 60

Solution: The total of parts is $3 + 5 = 8$, so each part is $96/8 = 12$ and the larger number is $5 \times 12 = 60$. The 36 value gives the smaller number 3×12 . The 48 splits the sum evenly in half, and 58 comes from an incorrect near-half split.

Q46. [QA-MCQ-4 · mcsc · diff 0]

The average of five numbers is 24. What is their total sum?

- A) 24
- B) 120 ✓
- C) 100
- D) 29

Answer: B) 120

Solution: The sum equals the average times the count, so $24 \times 5 = 120$, making 120 correct. The 29 value wrongly adds the count 5 to the average. The 24 mistakes the average itself for the sum, and 100 uses an incorrect count of about 4.

Q47. [QA-MCQ-5 · mcsc · diff 0]

A sum of 2000 rupees is invested at 6% simple interest per year. What is the interest earned in one year?

- A) 120 ✓
- B) 2120
- C) 60
- D) 112

Answer: A) 120

Solution: Simple interest for one year is $2000 \times 6/100 = 120$ rupees, so 120 is correct. The 2120 value adds the principal to the interest, giving the maturity amount. The 60 halves the correct interest, and 112 uses a rate near 5.6%.

Q48. [QA-MCQ-6 · mcsc · diff 0]

A rectangular garden is 14 metres long and 9 metres wide. What is its perimeter?

- A) 52
- B) 126
- C) 23
- D) 46 ✓

Answer: D) 46

Solution: Perimeter of a rectangle is $2 \times (\text{length} + \text{width}) = 2 \times (14 + 9) = 46$ metres, so 46 is correct. The 126 value computes the area 14×9 instead. The 23 gives just length plus width without doubling, and 52 uses an incorrect width of 12.

Q49. [QA-MCQ-7 · mcsc · diff 0.5]

A car travels 165 kilometres in 3 hours, then 120 kilometres in 2 hours. What is its average speed for the whole journey?

- A) 55
- B) 57 ✓
- C) 60
- D) 56

Answer: B) 57

Solution: Total distance is $165 + 120 = 285$ km over $3 + 2 = 5$ hours, so average speed is $285/5 = 57$ km/h, making 57 correct. The 60 value averages the two segment speeds of 55 and 60 wrongly weighted. The 55 uses only the first segment speed, and 56 averages the two speeds arithmetically.

Q50. [QA-MCQ-8 · mcsc · diff 0.5]

The smallest number that is exactly divisible by 8, 12, and 18 is:

- A) 144
- B) 36
- C) 24
- D) 72 ✓

Answer: D) 72

Solution: The LCM of 8, 12, and 18 is found from prime factors 2^3 , $2^2 \times 3$, and 2×3^2 , giving $2^3 \times 3^2 = 72$, so 72 is correct. The 36 value is only the LCM of 12 and 18 and misses the factor of 8. The 144 doubles the answer unnecessarily, and 24 is only the LCM of 8 and 12 while ignoring 18.

Q51. [QA-MCQ-9 · mcsc · diff 0.5]

A shirt is marked at 800 rupees. After a discount of 15%, the shopkeeper still makes a profit of 20% on the cost price. What was the cost price?

- A) 567 ✓
- B) 600
- C) 680
- D) 544

Answer: A) 567

Solution: The selling price after 15% discount is $800 \times 0.85 = 680$, and this equals 1.2 times the cost, so cost is $680/1.2 = 567$ rupees approximately, making 567 correct. The 544 wrongly takes 20% off 680. The 680 mistakes the selling price for the cost, and 600 uses a $800/1.333$ division.

Q52. [QA-MCQ-10 · mcsc · diff 0.5]

A sum of Rs 8000 is invested at 10% per annum compound interest, compounded annually. What is the compound interest earned at the end of 2 years?

- A) Rs 1600
- B) Rs 968
- C) Rs 1728
- D) Rs 1680 ✓

Answer: D) Rs 1680

Solution: Amount after 2 years is $8000 \times 1.1 \times 1.1 = 9680$, so the compound interest is $9680 - 8000 = 1680$. The Rs 1600 trap uses simple interest (800 per year). Rs 1728 wrongly compounds for 3 years, and Rs 968 takes only the second-year interest.

Q53. [QA-MCQ-11 · mcsc · diff 0.5]

Pipe A can fill a tank in 12 hours and pipe B can fill the same tank in 18 hours. If both pipes are opened together, how long will they take to fill the tank?

- A) 7.5 hours
- B) 15 hours
- C) 6 hours
- D) 7.2 hours ✓

Answer: D) 7.2 hours

Solution: Combined rate is $1/12 + 1/18 = 3/36 + 2/36 = 5/36$ tank per hour, so the time is $36/5 = 7.2$ hours. The 15-hour value wrongly adds the two times, 6 hours halves the smaller time, and 7.5 comes from averaging 12 and 18 incorrectly.

Q54. [QA-MCQ-12 · mcsc · diff 0.5]

A 40-litre mixture contains milk and water in the ratio 3:1. How many litres of water must be added so that the ratio of milk to water becomes 3:2?

- A) 5 litres
- B) 10 litres ✓
- C) 20 litres
- D) 8 litres

Answer: B) 10 litres

Solution: The mixture has 30 litres of milk and 10 litres of water. Milk stays fixed, so $30/(10 + x) = 3/2$ gives $10 + x = 20$, hence $x = 10$ litres. The 5-litre answer targets a 3:1.5 split, 8 litres misreads the target ratio, and 20 litres doubles the water to reach 1:1.

Q55. [QA-MCQ-13 · mcsc · diff 0.5]

Two fair six-sided dice are rolled together. What is the probability that the sum of the two numbers shown is exactly 8?

- A) 1/9
- B) 5/36 ✓
- C) 1/6
- D) 7/36

Answer: B) 5/36

Solution: The favourable outcomes are (2,6), (3,5), (4,4), (5,3) and (6,2), which is 5 of the 36 equally likely pairs, so the probability is 5/36. The 7/36 trap over-counts the pairs, 1/6 assumes 6 outcomes, and 1/9 counts only 4.

Q56. [QA-MCQ-14 · mcsc · diff 0.5]

In how many distinct ways can all the letters of the word BANANA be arranged?

- A) 720
- B) 120
- C) 360
- D) 60 ✓

Answer: D) 60

Solution: BANANA has 6 letters with three A's and two N's repeated, so the count is $6!/(3! \times 2!) = 720/12 = 60$. The 720 value forgets all repeats, 120 divides by 3! only, and 360 divides by 2! only.

Q57. [QA-MCQ-15 · mcsc · diff 0.5]

A solid sphere has a radius of 3 cm. What is its volume?

- A) $36\pi \text{ cm}^3$ ✓
- B) $27\pi \text{ cm}^3$
- C) $12\pi \text{ cm}^3$
- D) $9\pi \text{ cm}^3$

Answer: A) $36\pi \text{ cm}^3$

Solution: Volume of a sphere is $(4/3)\pi r^3 = (4/3)\pi(27) = 36\pi$ cubic cm. The 27π value drops the 4/3 factor, 12π uses r^2 instead of r^3 , and 9π confuses volume with a surface-related figure.

Q58. [QA-MCQ-16 · mcsc · diff 1]

A certain sum invested at compound interest, compounded annually, amounts to Rs 4840 in 2 years and to Rs 5324 in 3 years. What was the original sum invested?

- A) Rs 3993
- B) Rs 4400
- C) Rs 4000 ✓
- D) Rs 4200

Answer: C) Rs 4000

Solution: The third-year interest is $5324 - 4840 = 484$ on 4840, giving a rate of $484/4840 = 10\%$. Then the principal satisfies $P \times 1.1 \times 1.1 = 4840$, so $P = 4840/1.21 = 4000$. The Rs 4400 value is the amount after 1 year, Rs 4200 subtracts interest linearly, and Rs 3993 mis-applies a 3-year discount.

Q59. [QA-MCQ-17 · mcsc · diff 1]

In an arithmetic progression the third term is 12 and the seventh term is 24. What is the sum of the first 10 terms of this progression?

- A) 195 ✓
- B) 165
- C) 210
- D) 180

Answer: A) 195

Solution: From $a + 2d = 12$ and $a + 6d = 24$, subtracting gives $4d = 12$ so $d = 3$ and $a = 6$. The sum of 10 terms is $(10/2)(2a + 9d) = 5(12 + 27) = 5 \times 39 = 195$. The 210 value uses $d = 3$ but $a = 9$, 180 forgets the $9d$ part, and 165 uses only $8d$ in the bracket.

Q60. [QA-MCQ-18 · mcsc · diff 1]

In a group of 120 students, 70 like tea, 80 like coffee, and 20 like neither tea nor coffee. How many students like both tea and coffee?

- A) 40
- B) 30
- C) 50 ✓
- D) 60

Answer: C) 50

Solution: The number liking at least one drink is $120 - 20 = 100$. By inclusion-exclusion, both = $70 + 80 - 100 = 50$. The 30 trap forgets to remove the 20 who like neither, 40 subtracts 20 twice, and 60 mis-adds the totals.

Q61. [QA-TITA-1 · tita · diff 0]

Find the value of the expression $(2^5 \times 3^2)$ divided by $(2^3 \times 3)$.

Answer (TITA): 12

Solution: Subtract exponents of like bases: $2^{(5-3)} \times 3^{(2-1)}$ equals $2^2 \times 3^1$, which is 4×3 equals 12. A reader who forgets to reduce the powers of 3 and writes 4 lands short, while cancelling only the twos gives 4×9 equals 36.

Q62. [QA-TITA-2 · tita · diff 0]

A train travels at a constant speed of 72 kilometres per hour. Express this speed in metres per second.

Answer (TITA): 20

Solution: To convert kilometres per hour to metres per second, multiply by 5 and divide by 18: $72 \times 5 / 18$ equals $360 / 18$ equals 20 metres per second. Multiplying by $18/5$ instead of $5/18$ gives the wrong figure of about 259, a common inversion slip.

Q63. [QA-TITA-3 · tita · diff 0.5]

In an election between two candidates, the winner secured 60 percent of the total valid votes and won by a margin of 4800 votes. Find the total number of valid votes polled.

Answer (TITA): 24000

Solution: The winner got 60 percent and the loser 40 percent, so the margin is 20 percent of the total. If 20 percent equals 4800, the total is $4800 \times 100 / 20$ equals 24000. Treating 4800 as the winner's 60 percent share instead of the 20 percent margin gives a wrong total of 8000.

Q64. [QA-TITA-4 · tita · diff 0.5]

How many positive integer factors does 180 have?

Answer (TITA): 18

Solution: 180 factors as $2^2 \times 3^2 \times 5^1$, so the number of positive factors is $(2+1)(2+1)(1+1) = 3 \times 3 \times 2 = 18$. Counting only 1, 180 and the primes, or forgetting a prime, undercounts; squaring an exponent overcounts.

Q65. [QA-TITA-5 · tita · diff 0.5]

In a group of students, 60 like tea, 50 like coffee, and 20 like both tea and coffee. How many students like exactly one of the two drinks?

Answer (TITA): 70

Solution: Students liking only tea number 60 minus 20 equals 40, and those liking only coffee number 50 minus 20 equals 30, so exactly one is 40 plus 30 equals 70. Adding 60 and 50 to get 110 counts the 20 who like both twice and answers the wrong question.

Q66. [QA-TITA-6 · tita · diff 1]

Find the remainder when 7 raised to the power 103 is divided by 100.

Answer (TITA): 43

Solution: The last two digits of powers of 7 cycle with period 4 as 07, 49, 43, 01 and then repeat. Since 103 leaves remainder 3 when divided by 4, the value matches the third term of the cycle, which is 43. Taking 103 modulo 4 as 0 instead of 3 would wrongly point to the ending 01.

IIT Jodhpur Qualifier - MBA (Technology)

Time: 2 Hours | Maximum Marks: 204

Question Distribution

- **VARC:** 24 questions (suggested time 40 min)
- **DILR:** 22 questions (suggested time 40 min)
- **QA:** 22 questions (suggested time 40 min)
- Total: 68 questions | MCQ 63 | TITA (non-MCQ) 5

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Total duration is 2 Hours; the suggested per-section times are shown above.
- Sections may be attempted in any order; manage your time across all sections.

This is the answer-marked verification copy (correct option in green, with solution) and mirrors the content of the upload CSV.

Section: VARC (suggested 40 min)

Instructions (VARC-RC1): study the shared passage below and answer its 5 question(s).

When we ask what a poem means, we are tempted to reach for what its author intended it to mean, as though the work were a sealed letter and criticism the act of opening it. The temptation is understandable, since intentions do govern most of what we make and say. Yet a strand of twentieth-century criticism insisted that, once a work is finished and released, the author's private aims cease to be the arbiter of its meaning. On this view the poem is a public object made of language, and language carries implications its user may neither have foreseen nor endorsed. A reader who recovers a documented intention has learned something about the poet's biography, not necessarily something decisive about the poem.

The argument is easy to caricature as a licence for arbitrary reading, but its careful proponents claimed no such thing. They distinguished the meaning of a work from the occasion of its composition, and held that evidence internal to the text, its diction, structure, and the conventions it invokes, constrains interpretation more tightly than any appeal to what the author confided in a letter. Where internal and external evidence agree, the appeal to intention is redundant; where they conflict, it is the text, not the confession, that the critic is professionally bound to honour. Intention, in short, is neither available nor desirable as a final standard of literary judgement.

Critics of this position have replied that meaning without a meaner is incoherent, and that to strip intention from interpretation is to reduce reading to the free play of association. The reply has force against the caricature but less against the original claim, which never denied that someone meant something, only that the someone's report settles the question. A more searching objection notes that the very conventions the text invokes are themselves the sediment of countless past intentions, so that intention returns, collectivised, at the level of the language itself. The debate, then, is less about whether intention matters than about whose intention, and at what remove, a competent reading is obliged to reconstruct.

Q1. [VARC-RC1 · mcsc · diff 0.5]

The passage is primarily concerned with:

- A) defending the claim that a reader is free to assign a poem almost any meaning that the language will loosely permit
- B) tracing how twentieth-century literary criticism gradually abandoned nearly all close attention to the words on the page

C) presenting a view that denies the author's intention fixes a poem's meaning, and weighing the replies it draws ✓

- D) arguing that an author's stated intention is wholly irrelevant to every form of literary and biographical inquiry

Answer: C) presenting a view that denies the author's intention fixes a poem's meaning, and weighing the replies it draws

Solution: The passage lays out the anti-intention position (paragraphs 1 to 2) and then the objections to it (paragraph 3), ending on what the dispute is really about. It does not claim intention is irrelevant everywhere; it grants intention is biographically informative and even returns collectivised, so the 'wholly irrelevant' option overstates. The 'any meaning' option is the caricature the passage explicitly rejects. The historical-abandonment option is a scope shift the text never makes.

Q2. [VARC-RC1 · mcsc · diff 0.5]

The careful proponents of the anti-intention view would agree with all of the following EXCEPT:

- A) The diction and structure of a text limit its readings more tightly than a private confession by its author
- B) Where internal and external evidence point the same way, invoking the author's intention adds nothing new
- C) A finished poem has no determinate meaning until its author supplies an authoritative gloss on it ✓
- D) Recovering an author's documented intention can still teach us something about that author's biography

Answer: C) A finished poem has no determinate meaning until its author supplies an authoritative gloss on it

Solution: The proponents hold that the text, not the author's gloss, fixes meaning, so the claim that a poem lacks meaning until the author supplies an authoritative gloss inverts their position and is the EXCEPT. The other three restate points made directly: intention is biographically informative, it is redundant when the evidence agrees, and internal evidence constrains reading more than a confession.

Q3. [VARC-RC1 · mcsc · diff 1]

It can be inferred that the 'more searching objection' differs from the simpler reply that precedes it because it:

- A) abandons the appeal to textual evidence and defends interpretation as the reader's free play of association
- B) denies outright that any individual author ever meant anything determinate by the words chosen for a poem
- C) rests on the premise that a work can carry no meaning at all unless a single mind stands directly behind it
- D) grants the priority given to the text while relocating intention to the conventions the language carries ✓

Answer: D) grants the priority given to the text while relocating intention to the conventions the language carries

Solution: The searching objection accepts that the text constrains reading yet points out that the conventions are themselves 'the sediment of countless past intentions', so intention returns collectivised rather than being reasserted for the lone author. The 'meaning without a meaner' premise belongs to the simpler reply, not this one. The other options attribute denials or a free-play stance that the objection does not make.

Q4. [VARC-RC1 · mcsc · diff 1]

The author calls the appeal to intention 'redundant' when evidence agrees and to be overruled when evidence conflicts chiefly in order to:

- A) show that on either alignment of the evidence the author's intention does no decisive work ✓
- B) establish the separate point that authors are habitually unreliable witnesses to their own real aims
- C) suggest that external evidence about a work is nearly always misleading and is best set entirely aside
- D) concede that in a narrow range of disputes an author's own stated intention does settle the reading

Answer: A) show that on either alignment of the evidence the author's intention does no decisive work

Solution: The two-case structure is a dilemma: if internal and external evidence agree, intention is superfluous; if they conflict, the text wins anyway. Either way intention is not the deciding factor, which is the rhetorical work the sentence performs. It does not concede that intention sometimes settles the reading, nor argue authors are unreliable witnesses, nor dismiss external evidence wholesale.

Q5. [VARC-RC1 · mcsc · diff 1]

Which of the following claims about the debate is fully supported by the passage?

- A) The passage shows that intention of every kind can be dispensed with once a text is complete
- B) The live question is which intentions, and how far mediated, a competent reading must reconstruct ✓
- C) The critics have demonstrated that the anti-intention view collapses into arbitrary reading
- D) The passage concludes that an individual author's intention proves indispensable to interpretation after all

Answer: B) The live question is which intentions, and how far mediated, a competent reading must reconstruct

Solution: The final sentence recasts the dispute as one about whose intention and at what remove a reading must reconstruct, which the correct option states exactly. The 'indispensable after all' option overreads the collectivised return of intention as a rescue of the individual author, which the text does not grant. The 'dispensed with entirely' option ignores that collectivised return. The 'collapses into arbitrary reading' option is the caricature the passage says has force only against a straw version.

Instructions (VARC-RC2): study the shared passage below and answer its 4 question(s).

For a generation, the standard verdict on shared resources was pessimistic: where no one owns a pasture, a fishery, or an aquifer, each user has an incentive to take as much as possible before a rival does, and the resource is ruined by the sum of individually rational choices. The remedy seemed to follow with the force of a theorem. Either the state must regulate the resource from above, or it must be carved into private parcels whose owners will husband what they hold. Between the regulator and the auctioneer there appeared to be no third path.

Field research complicated the theorem. In mountain villages, coastal fisheries, and irrigation systems that had endured for centuries, users had neither nationalised nor privatised their commons, yet had not destroyed them either. Instead they had crafted their own rules: who could harvest, when, how much, and what would befall a cheat. These arrangements were neither spontaneous nor costless. They depended on the users being able to monitor one another, to communicate, to revise the rules as conditions changed, and to impose graduated penalties that punished a first offence lightly and a repeated one severely. Where such conditions were absent, the pessimistic prediction often held.

The lesson was not that the original model was false, but that it described one case rather than the general one. The model assumed users who could not talk, could not watch, and could not bind themselves, and for such users the grim conclusion does follow. Real communities are frequently able to do all three, and the capacity to make and enforce local agreements is itself a resource, one that erodes if outsiders, however well intentioned, override it. A government that replaces a working local arrangement with a uniform national rule may extinguish the very monitoring and trust on which restraint depended, and so produce the tragedy it meant to prevent.

Q6. [VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the central argument of the passage?

- A) Communities can sometimes govern shared resources by self-made rules, so the pessimistic model fits a special case ✓
- B) State regulation from above is shown to be the single most dependable remedy for the overuse of any common resource
- C) Traditional rural communities, unlike modern states, essentially never deplete the natural resources placed in their care
- D) Shared resources are inevitably destroyed unless they are divided up into privately owned parcels by their own users

Answer: A) Communities can sometimes govern shared resources by self-made rules, so the pessimistic model fits a special case

Solution: The passage argues the grim model holds only for users who cannot monitor, communicate, or bind themselves, while many real communities can and so govern their commons without nationalising or privatising them. The privatisation-inevitable and regulation-most-dependable options each pick one of the two paths the passage relativises. The 'never deplete' option overstates: the text says the prediction 'often held' where the enabling conditions were absent.

Q7. [VARC-RC2 · mcsc · diff 0.5]

The passage credits the durable commons arrangements with all of the following features EXCEPT:

- A) provisions allowing the governing rules to be revised as surrounding conditions change
- B) penalties that fall lightly on a first offence and grow heavier when the offence is repeated
- C) the ability of users to keep watch on one another and to communicate about the resource
- D) the assignment of a separately owned private parcel of the resource to each individual user ✓

Answer: D) the assignment of a separately owned private parcel of the resource to each individual user

Solution: The enduring commons were explicitly neither privatised nor nationalised, so assigning each user a private parcel is the feature the passage denies and is the EXCEPT. Graduated penalties, mutual monitoring and communication, and revisable rules are each named directly as pillars of the successful arrangements.

Q8. [VARC-RC2 · mcsc · diff 1]

The author observes that a well-meaning government can 'produce the tragedy it meant to prevent' mainly in order to:

- A) warn that overriding a working local arrangement can destroy the conditions that sustained restraint ✓
- B) argue that governments are, almost as a rule, indifferent to whether shared resources survive or collapse
- C) claim that locally made rules are plainly superior to state rules in virtually every respect and setting
- D) show that dividing a resource into private holdings will reliably outperform any centrally imposed public rule

Answer: A) warn that overriding a working local arrangement can destroy the conditions that sustained restraint

Solution: The point is that a uniform national rule can extinguish the monitoring and trust on which local restraint depended, so intervention can backfire. It does not impute indifference to governments, and it does not claim local rules win everywhere: the text stresses local success is conditional on monitoring and communication. Privatisation is not endorsed here at all.

Q9. [VARC-RC2 · mcsc · diff 1]

The author's stance toward the original pessimistic model is best described as:

- A) full endorsement of it as a fixed law that governs shared resources in every setting
- B) studied indifference, on the ground that the model's assumptions are said to be untestable
- C) outright rejection of it as an account that field evidence has now proved plainly false
- D) qualified acceptance, treating it as accurate for a restricted range of conditions ✓

Answer: D) qualified acceptance, treating it as accurate for a restricted range of conditions

Solution: The author says explicitly the lesson is 'not that the original model was false' but that it fits one case, and that for users who cannot talk, watch, or bind themselves 'the grim conclusion does follow'. That is qualified acceptance. Outright rejection contradicts the disclaimer; full endorsement ignores the restriction to a special case; indifference is wrong because the model is engaged with closely and its assumptions are tested against field cases.

Q10. [VARC-ODD-1 · mcsc · diff 0.5]

Five sentences are given below. Four of them combine to form a single coherent paragraph; one does not belong. Read the sentences and identify the misfit.

- (1) A tidal wetland acts as a natural buffer, absorbing the energy of incoming storm surges before they reach inland settlements.
- (2) Its dense mat of roots binds loose sediment, so the shoreline erodes far more slowly than an unvegetated coast would.
- (3) The same roots filter runoff, trapping pollutants that would otherwise flow straight into open water.
- (4) Because of these functions, planners increasingly treat wetlands as infrastructure rather than as idle land.
- (5) Modern desalination plants can convert seawater into drinking water at a steadily falling cost per litre.

Which sentence is the odd one out?

- A) (3)
- B) (4)
- C) (5) ✓
- D) (2)
- E) (1)

Answer: C) (5)

Solution: Sentences 1, 2, 3 and 4 build one argument: wetlands buffer surges, hold sediment, filter runoff, and so count as infrastructure. The desalination-plant sentence introduces an unrelated technology for producing drinking water and shares none of that thread, so it is the intruder. Sentence 4 begins with 'these functions', which points back to the wetland benefits, confirming the other four cohere.

Q11. [VARC-ODD-2 · mcsc · diff 0.5]

Five sentences are given below. Four of them combine to form a single coherent paragraph; one does not belong. Read the sentences and identify the misfit.

- (1) When a folk tale passes from one teller to the next, small changes creep in with every retelling.
- (2) A teller may drop a name, sharpen a joke, or shift the setting to a place the listeners know.
- (3) Over generations these tiny edits accumulate into strikingly different versions of one story.
- (4) Scholars can sometimes trace those versions back to a shared ancestor by comparing their fixed motifs.
- (5) Printing presses lowered the price of books so sharply that private libraries became common.

Which sentence is the odd one out?

- A) (2)
- B) (1)
- C) (3)
- D) (5) ✓
- E) (4)

Answer: D) (5)

Solution: The four connected sentences describe how oral folk tales mutate as they are retold and how scholars reconstruct their common origin. The sentence about printing presses and the price of books concerns written distribution, a different subject that breaks the chain about oral transmission, making it the misfit. The word 'those versions' in sentence 4 links back to the accumulated edits, tying the other four together.

Q12. [VARC-ODD-3 · mcsc · diff 1]

Five sentences are given below. Four of them combine to form a single coherent paragraph; one does not belong. Read the sentences and identify the misfit.

- (1) A scientific model is deliberately simpler than the reality it represents, keeping only the features that matter for a given question.
- (2) By stripping away detail, it lets researchers reason about a system that would otherwise be too tangled to follow.
- (3) The value of a model therefore lies not in its completeness but in the useful predictions it yields.
- (4) A model that reproduced every detail of the world would be as hard to understand as the world itself.
- (5) A well-designed model can be tested against fresh observations and discarded the moment it fails to predict them.

Which sentence is the odd one out?

- A) (5) ✓
- B) (2)
- C) (1)
- D) (3)
- E) (4)

Answer: A) (5)

Solution: Sentences 1, 2, 3 and 4 develop one idea, that a model earns its worth through simplification rather than through completeness. Sentence 5 is subtly off: it stays on the topic of models but shifts to a different claim, about falsification and testing against new data, which the other four never raise. Because 5 introduces a separate virtue instead of extending the completeness-versus-usefulness argument, it is the odd one out.

Q13. [VARC-ODD-4 · mcsc · diff 1]

Five sentences are given below. Four of them combine to form a single coherent paragraph; one does not belong. Read the sentences and identify the misfit.

- (1) A currency holds its value only because the people using it expect others to accept it tomorrow.
- (2) That expectation is a shared belief, sustained by habit and by trust in the issuing authority.
- (3) When the belief weakens, holders rush to spend, and prices climb faster than the notes can be printed.
- (4) A sudden loss of confidence can thus unravel a currency long before any change in the goods it can buy.
- (5) Central banks set interest rates to steer how much borrowing and lending take place across the economy.

Which sentence is the odd one out?

- A) (4)
- B) (5) ✓
- C) (2)
- D) (1)
- E) (3)

Answer: B) (5)

Solution: The cohering four trace one causal story: a currency's value rests on shared expectation, that expectation is belief, weakened belief triggers a spending rush, and confidence can collapse before real goods change. Sentence 5 also concerns central banking but makes a different claim, about interest-rate policy and credit, rather than the psychology of confidence. It sits in the same field yet breaks the single argument, so it is the intruder.

Q14. [VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. Every autumn, millions of songbirds leave the cooling north for warmer southern feeding grounds.
2. To survive the journey, each bird nearly doubles its body weight by gorging on insects and berries beforehand.
3. This stored fat becomes the fuel that powers days of almost continuous flight.
4. Without it, a small migrant could not cross the wide deserts and seas that lie along the route.

Select the correct order.

- A) 2134
- B) 1324
- C) 1243
- D) 1234 ✓

Answer: D) 1234

Solution: Sentence 1 sets the scene with the autumn departure and is the natural opener. Sentence 2 explains the preparation, doubling body weight, which sentence 3 then labels 'this stored fat' and calls the fuel. Sentence 4 closes with 'Without it', referring to that fat, and names the obstacles. The chain 1-2-3-4 keeps each pronoun next to its referent, so the order is 1234.

Q15. [VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. A public library began, in most towns, as a single donated room lined with second-hand books.
2. As demand grew, the room acquired a paid keeper who lent volumes and tracked their return.
3. That keeper slowly became a trained profession, with rules for cataloguing and preserving the collection.
4. Today those same rules underpin vast digital archives that no single room could ever hold.

Select the correct order.

- A) 1324
- B) 1243
- C) 2134
- D) 1234 ✓

Answer: D) 1234

Solution: The paragraph moves through time. Sentence 1 gives the humble origin, a donated room. Sentence 2 adds the paid keeper as demand grows, and sentence 3 turns 'that keeper' into a profession with rules. Sentence 4 begins 'Today those same rules', extending the cataloguing rules to digital archives. Each step depends on the one before, so the sequence is 1234.

Q16. [VARC-JUMBLE-3 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. Critics once dismissed the invention as a toy with no serious use.
2. Its defenders replied that the same had been said of the telephone in its first decade.
3. They pointed out that such judgements track familiarity, not the real limits of a device.
4. History, they argued, rarely rewards those who mistake a rough prototype for a finished tool.

Select the correct order.

- A) 1324
- B) 1234 ✓
- C) 2134
- D) 1243

Answer: B) 1234

Solution: Sentence 1 states the dismissal and must open. Sentence 2 begins 'Its defenders replied', a direct rebuttal that has to follow the criticism. Sentence 3 opens with 'They', the defenders, generalising their reply, and sentence 4 continues 'they argued' with the closing lesson. Reordering breaks the pronoun 'They', which needs 'defenders' named first, so only 1234 keeps the references intact.

Q17. [VARC-JUMBLE-4 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. A map must leave things out, for a chart that showed every detail would be useless.
2. Yet what it leaves out is never neutral, since each omission quietly reflects the mapmaker's priorities.
3. Such choices, repeated across thousands of maps, shape how a whole society pictures its own territory.
4. For this reason a map is best read not only as a guide but as an argument about what deserves notice.

Select the correct order.

- A) 2134
- B) 1324
- C) 1234 ✓
- D) 1243

Answer: C) 1234

Solution: Sentence 1 introduces the necessary omission and opens the paragraph. Sentence 2 begins 'Yet what it leaves out', a contrast that must follow the claim about leaving things out. Sentence 3 refers to 'Such choices', the priority-driven omissions of sentence 2, and sentence 4 draws the conclusion with 'For this reason'. The connectives Yet, Such choices, and For this reason force the order 1234.

Q18. [VARC-JUMBLE-5 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. Silence in a conversation is often read as a gap, an awkward failure to keep talking.
2. In many cultures, however, it carries meaning as rich as any spoken reply.
3. There a considered pause signals respect, telling the other that the question deserved real thought.
4. To fill that pause too quickly can therefore seem not friendly but careless.

Select the correct order.

- A) 1243
- B) 1234 ✓
- C) 2134
- D) 1324

Answer: B) 1234

Solution: Sentence 1 sets up the common view of silence as an awkward gap. Sentence 2 answers with 'however', reframing silence as meaningful, and sentence 3 begins 'There', pointing to those cultures and explaining what the pause signals. Sentence 4 concludes with 'that pause' and 'therefore', drawing the consequence. The linking words however, There, and therefore fix the reading order as 1234.

Q19. [VARC-SUMMARY-1 · mcsc · diff 0.5]

Many species that appear wasteful are in fact insuring themselves against bad years. An oak may drop thousands of acorns although only a handful can ever grow into trees, because the surplus swamps the animals that eat them and guarantees that a few seeds survive in almost any season. What looks like extravagance is a hedge: by producing far more than the average conditions require, the tree buys protection against the years when conditions are worst.

Which option best summarises the paragraph?

A) Apparent overproduction in nature is often a safeguard, letting a species survive poor years by making far more offspring than good conditions alone would need. ✓

B) Animals that feed on seeds are usually overwhelmed by the sheer number a plant produces and so fail to eat the entire crop.

C) Oak trees drop many thousands of acorns every single year even though almost none of those scattered acorns will ever manage to grow into a mature standing tree.

D) Nature tends to be wasteful, and much of what living things produce serves no real purpose beyond simple excess.

Answer: A) Apparent overproduction in nature is often a safeguard, letting a species survive poor years by making far more offspring than good conditions alone would need.

Solution: The passage argues that seeming waste is really insurance against bad years, using the oak only as an example. The correct summary captures that general point about overproduction as a safeguard. The acorn statement and the seed-eater statement are single supporting details, too narrow to be the summary, and the 'nature is wasteful' option flatly contradicts the passage, which says the surplus has a clear purpose.

Q20. [VARC-SUMMARY-2 · mcsc · diff 0.5]

A new road built to relieve congestion often ends up just as crowded as before. The extra capacity lowers the effective cost of driving, so people who once avoided the trip, took the train, or travelled at odd hours now join the traffic. Within a few years the freed-up space fills again, and the promised relief quietly disappears even though the road itself was built exactly as planned.

Which option best summarises the paragraph?

A) Roads are frequently built exactly to their original plans, yet the traffic engineers still misjudge how drivers will respond.

B) Adding road capacity tends to attract new trips that fill the space, so the expected easing of congestion often fails to last. ✓

C) Traffic congestion is one of the most stubborn and expensive problems that fast-growing modern cities are forced to confront.

D) People who used to take the train will switch to driving as soon as a wider and more convenient road becomes available to them.

Answer: B) Adding road capacity tends to attract new trips that fill the space, so the expected easing of congestion often fails to last.

Solution: The paragraph's core claim is that new capacity induces extra demand that refills the road, defeating the intended relief. The correct option states exactly that cause and effect. The train-switching option is only one instance of the induced demand, the 'built to plan' option seizes a side remark, and the 'congestion is stubborn' option is a general theme the passage never actually argues.

Q21. [VARC-SUMMARY-3 · mcsc · diff 0.5]

Learning a second language as an adult rarely produces a flawless accent, yet the effort is far from wasted. Even partial fluency reshapes how the learner hears speech, sharpens attention to grammar in the first language, and opens doors to texts and people that were closed before. The goal need not be to pass as a native; the practical and mental gains arrive long before that distant point.

Which option best summarises the paragraph?

- A) Studying grammar in a new language happens to sharpen a learner's awareness of the grammar of their own first language.
- B) Reaching the fluent level of a genuine native speaker is really the one true and proper measure of success for anyone setting out to learn a new foreign language.
- C) Adult language learning brings real practical and mental benefits well before fluency, so a perfect accent need not be the aim. ✓
- D) Adults who study a second language almost never manage to acquire an accent that a native speaker would accept as flawless.

Answer: C) Adult language learning brings real practical and mental benefits well before fluency, so a perfect accent need not be the aim.

Solution: The passage says adult learners seldom sound native but gain much sooner, so a perfect accent is not the point. The correct summary states both halves, benefits arrive early and native-level need not be the goal. The accent option and the grammar option are isolated details, and the 'native speaker is the true measure' option reverses the passage's actual conclusion.

Q22. [VARC-SUMMARY-4 · mcsc · diff 1]

Rating systems promise to guide us toward quality, but a high average can mislead as easily as it informs. A product reviewed by only a dozen buyers may score higher than a rival reviewed by thousands, not because it is better but because a small sample swings more wildly. Worse, the people who bother to rate are rarely typical: the delighted and the furious speak up while the merely satisfied stay silent. A score, then, summarises not the product alone but the peculiar crowd that chose to judge it, and reading it wisely means asking who did the rating before trusting what they said.

Which option best summarises the paragraph?

- A) A product judged by only a handful of reviewers can post a higher average than a rival that thousands of buyers have carefully rated.
- B) Rating systems were originally designed to steer ordinary shoppers toward higher quality goods, and for the most part they seem to carry out that useful task quite reliably.
- C) People who feel delighted or furious about a purchase are much more likely to leave a review than those who feel merely satisfied.
- D) Average ratings can mislead because small samples fluctuate and reviewers are unrepresentative, so a score reflects who rated as much as the product. ✓

Answer: D) Average ratings can mislead because small samples fluctuate and reviewers are unrepresentative, so a score reflects who rated as much as the product.

Solution: The passage makes two linked points, that small samples swing and that reviewers are a skewed crowd, concluding a score reflects its raters. Only the correct option carries that full scope. The small-sample option and the delighted-or-furious option are each one of the two reasons, true but partial, and the reliability option contradicts the passage by calling ratings reliable, which is the very claim being questioned.

Q23. [VARC-SUMMARY-5 · mcsc · diff 1]

We tend to praise the lone inventor, but most breakthroughs are less a single leap than a last step. By the time a discovery is announced, dozens of others have usually reached the same threshold, held back only by a missing tool or an unasked question. The celebrated name is often simply the one who arrived days ahead, or who framed the result in words the age was ready to hear. To credit a single mind, then, is to mistake the visible tip of a long collective effort for the whole of it.

Which option best summarises the paragraph?

A) The lone individual inventor still fully deserves the praise that we give, because framing a raw result in exactly the right words is itself the single hardest part of any discovery.

B) Great discoveries are usually the final step of a wide shared effort, so crediting one inventor mistakes a visible moment for the whole. ✓

C) A breakthrough is frequently delayed by nothing more than a missing instrument or a question that no one has yet thought to ask.

D) Whoever announces a discovery first has often arrived only a few days ahead of several rivals working on the very same problem.

Answer: B) Great discoveries are usually the final step of a wide shared effort, so crediting one inventor mistakes a visible moment for the whole.

Solution: The passage argues that breakthroughs cap a broad collective effort and that honouring one mind mistakes the tip for the whole. The correct option states both the shared-effort claim and its consequence. The 'arrived days ahead' and 'missing instrument' options are supporting details, and the final option endorses the lone-inventor view the passage sets out to overturn.

Q24. [VARC-SUMMARY-6 · mcsc · diff 1]

A law that everyone quietly ignores can be more corrosive than no law at all. When a rule is kept on the books but never enforced, it does not simply lie dormant; it teaches citizens that the written word and the real world need not match. Each unpunished breach makes the next one easier to excuse, and over time the habit of selective obedience spreads to rules that were meant to bind. The damage lies less in the single ignored statute than in the lesson it spreads about how seriously any law should be taken.

Which option best summarises the paragraph?

A) Once one violation of a statute goes unpunished, excusing the next similar violation becomes noticeably easier to justify.

B) An unenforced law erodes respect for law in general, because tolerating its breach teaches that rules need not be obeyed. ✓

C) Lawmakers should take care to repeal statutes they no longer intend to enforce rather than leaving them on the books.

D) A rule that stays on the books without ever being enforced does not vanish but quietly remains available for later use.

Answer: B) An unenforced law erodes respect for law in general, because tolerating its breach teaches that rules need not be obeyed.

Solution: The passage's thesis is that an ignored law spreads a corrosive lesson, weakening obedience to law as a whole. The correct option captures that broad erosion and its cause. The 'remains available' option distorts a minor phrase, the 'next violation easier' option is one step in the argument rather than the whole, and the repeal recommendation is advice the passage never offers.

Section: DILR (suggested 40 min)

Instructions (DILR-1): study the shared data below and answer its 4 question(s).

Five employees, Mira, Omar, Priya, Rehan and Sana, sit at a row of five desks numbered 1 to 5 from left to right. Each works in exactly one of five departments (Finance, Legal, Sales, Tech, HR), and each has a distinct number of years of experience drawn from 2, 4, 6, 8 and 10.

- The Finance employee sits immediately to the left of the Legal employee.
- Priya has more experience than exactly three of her colleagues.
- The Tech employee has the least experience.
- Omar sits at desk 3 and works in Finance.
- Sana has exactly 4 years more experience than the Sales employee, and Sana does not work in Sales.
- The employee at desk 1 has 6 years of experience.
- Mira sits somewhere to the right of Rehan.
- The HR employee sits at desk 5.
- The Legal employee has 8 years of experience.

Q25. [DILR-1 · mcsc · diff 1]

Which employee sits at desk 4?

- A) Sana
- B) Mira
- C) Priya ✓
- D) Omar

Answer: C) Priya

Solution: Priya has more experience than exactly three others, so she holds 8 years, and the Legal person also holds 8, making Priya the Legal employee. Omar is Finance at desk 3, so Legal sits immediately to his right at desk 4. Hence Priya is at desk 4. Mira, Sana and Omar occupy other desks, so they are error traps from misplacing the adjacency clue.

Q26. [DILR-1 · mcsc · diff 1]

The employee with 10 years of experience works in which department?

- A) HR ✓
- B) Finance
- C) Sales
- D) Legal

Answer: A) HR

Solution: Sales sits at desk 1 with 6 years, so Sana has $6+4=10$ years and is not in Sales. Since Legal holds 8 and Finance is Omar with 4, and the Tech employee holds the least at 2, the only department left for the 10-year employee is HR at desk 5, which is Sana. Legal and Finance are traps from ignoring the experience gap.

Q27. [DILR-1 · mcsc · diff 1]

Suppose Omar and Priya exchange only their departments, keeping their desks and their years of experience unchanged. Which statement would then become true?

- A) The Tech employee would end up with the most experience overall.
- B) The Finance and Legal employees would no longer sit adjacently. ✓
- C) The employee at desk 1 would then work in the Legal department.
- D) Exactly two of the five employees would then share one department.

Answer: B) The Finance and Legal employees would no longer sit adjacently.

Solution: Originally Omar is Finance at desk 3 and Priya is Legal at desk 4, satisfying the adjacency rule. After the swap Priya becomes Finance at desk 4 and Omar becomes Legal at desk 3, so Finance now sits to the right of Legal and the two are no longer adjacent in the required order. Desk 1 stays Sales, Tech keeps its 2 years, and each department is still held by exactly one person, so those statements remain false.

Q28. [DILR-1 · mcsc · diff 1]

What is the combined experience, in years, of the Finance employee and the Legal employee?

- A) 12 ✓
- B) 16
- C) 10
- D) 14

Answer: A) 12

Solution: Omar is the Finance employee with 4 years, and Priya is the Legal employee with 8 years, so together they hold $4+8=12$ years. The value 14 comes from pairing Legal with Sales, 10 from using Tech instead of Legal, and 16 from adding the two most experienced staff rather than the two named departments.

Instructions (DILR-2): study the shared data below and answer its 3 question(s).

Four flights, operated by the airlines Aster, Birch, Cedar and Delta, depart one after another from a regional airport, no two at the same time. Each flight goes to a different city among Lisbon, Cairo, Oslo and Tokyo.

- Aster departs earlier than Birch. - The flight to Tokyo departs first. - Cedar flies to Cairo. - The flight to Oslo departs immediately after the flight to Lisbon. - Delta departs third. - Cedar departs earlier than Delta.

Q29. [DILR-2 · mcsc · diff 0.5]

Which airline operates the last flight to depart?

- A) Aster
- B) Delta
- C) Cedar
- D) Birch ✓

Answer: D) Birch

Solution: Tokyo departs first and Aster is earlier than Birch, so Aster takes the first slot with Tokyo. Cedar flies to Cairo and departs before Delta, who is third, so Cedar is second. That leaves Birch for the fourth and final slot. Delta is third and Aster is first, so both are traps for readers who stop after placing Delta.

Q30. [DILR-2 · mcsc · diff 0.5]

To which city does Delta fly?

- A) Lisbon ✓
- B) Cairo
- C) Tokyo
- D) Oslo

Answer: A) Lisbon

Solution: Aster (first) goes to Tokyo and Cedar (second) goes to Cairo. Oslo must follow Lisbon immediately, and with Delta third and Birch fourth the only consecutive pair left is Delta then Birch, so Delta flies to Lisbon and Birch to Oslo. Cairo and Tokyo are already assigned, so they are traps for readers who ignore the Lisbon-then-Oslo order.

Q31. [DILR-2 · mcsc · diff 1]

Suppose the airport swaps the departure slots of the two flights currently in the second and fourth positions, leaving every destination with its own aircraft unchanged. After the swap, the flight to which city departs immediately before the flight to Oslo?

- A) Cairo
- B) Oslo
- C) Tokyo ✓
- D) Lisbon

Answer: C) Tokyo

Solution: In the original order the slots are Tokyo, Cairo, Lisbon, Oslo. Swapping the second and fourth flights gives Tokyo, Oslo, Lisbon, Cairo. Oslo now sits in the second slot, so the flight departing immediately before it is Tokyo in the first slot. Lisbon and Cairo have moved away from that position, and Oslo cannot precede itself, so they are traps.

Instructions (DILR-3): study the shared data below and answer its 4 question(s).

A tour operator sells three package types across four cities. Table 1 gives the number of packages sold (in thousands) and Table 2 gives the price per package (in thousand rupees). Revenue for any cell = packages sold (in '000) x price per package (in '000 rupees), so all revenue figures are in millions of rupees.

****Packages sold (in thousands)****

Package	Jaipur	Kochi	Shimla	Goa
Heritage	4	10	8	6
Coastal	30	8	5	4
Mountain	2	6	10	20

****Price per package (in thousand rupees)****

Package	Jaipur	Kochi	Shimla	Goa
Heritage	20	22	25	18
Coastal	15	16	14	12
Mountain	30	28	26	24

Q32. [DILR-3 · mcsc · diff 1]

What is the total revenue earned from Coastal packages across all four cities?

- A) 664 million rupees
- B) 720 million rupees
- C) 696 million rupees ✓
- D) 648 million rupees

Answer: C) 696 million rupees

Solution: Coastal revenue per city is packages sold times price: Jaipur $30 \times 15 = 450$, Kochi $8 \times 16 = 128$, Shimla $5 \times 14 = 70$, Goa $4 \times 12 = 48$, giving 696 million rupees in total. Dropping Goa by mistake leaves 648, which is the tempting near miss.

Q33. [DILR-3 · mcsc · diff 1]

The ratio of total Mountain revenue to total Heritage revenue (across all cities) is closest to which of the following?

- A) 7 : 4
- B) 3 : 2
- C) 8 : 5 ✓
- D) 5 : 3

Answer: C) 8 : 5

Solution: Mountain revenue is $60 + 168 + 260 + 480 = 968$ million rupees and Heritage revenue is $80 + 220 + 200 + 108 = 608$ million rupees. The ratio $968:608$ works out to about 1.59, which is nearest to 8:5 (1.60); 3:2 undershoots at 1.50.

Q34. [DILR-3 · mcsc · diff 1]

Which city has the lowest overall average price per package (total revenue divided by total packages sold)?

- A) Goa
- B) Shimla
- C) Kochi
- D) Jaipur ✓

Answer: D) Jaipur

Solution: Overall average price is total revenue over total packages. Jaipur is $590/36 = 16.4$, Goa $636/30 = 21.2$, Kochi $516/24 = 21.5$, Shimla $530/23 = 23.0$ (thousand rupees). Jaipur is lowest because its huge cheap Coastal volume pulls the weighted price down; taking a plain average of the three listed prices wrongly flags Goa (mean 18).

Q35. [DILR-3 · mcsc · diff 1]

By how much does Goa's total revenue exceed Kochi's total revenue?

- A) 96 million rupees
- B) 120 million rupees ✓
- C) 132 million rupees
- D) 108 million rupees

Answer: B) 120 million rupees

Solution: Goa's total revenue is $108 + 48 + 480 = 636$ million rupees and Kochi's is $220 + 128 + 168 = 516$ million rupees, so Goa leads by $636 - 516 = 120$ million rupees. The distractors come from small slips in the Mountain or Coastal products.

Instructions (DILR-4): study the shared data below and answer its 4 question(s).

A cultural trust runs four community centres, named Aster, Birch, Cedar and Dune. Over one season the four centres together host a total of exactly 26 workshops. Each centre hosts a different whole number of workshops, and each hosts between 3 and 10 workshops (both inclusive). Separately, the four centres open for the season in four different months, one each in March, April, May and June.

The following facts are known about the workshop counts:

- Aster hosts more workshops than every other centre, and Dune hosts fewer than every other centre. - Birch hosts more workshops than Cedar. - Cedar hosts exactly 6 workshops. - Aster hosts an odd number of workshops. - Aster hosts exactly two more workshops than Birch.

The following facts are known about the opening months:

- Cedar opens in the earliest of the four months, and Birch opens in the latest of the four months. - Aster opens in an earlier month than Dune.

Q36. [DILR-4 · mcsc · diff 0.5]

How many workshops does Dune host?

- A) 4 ✓
- B) 9
- C) 7
- D) 6

Answer: A) 4

Solution: The distinct counts sum to 26 with Cedar fixed at 6. Aster is the largest and odd, Dune the smallest, Birch above Cedar, and Aster is Birch plus 2. The only fit is Aster 9, Birch 7, Cedar 6, Dune 4, so Dune hosts 4. The 6 trap copies Cedar, 7 copies Birch, and 9 copies Aster.

Q37. [DILR-4 · mcsc · diff 1]

Which centre opens in the third of the four months?

- A) Cedar
- B) Birch
- C) Aster
- D) Dune ✓

Answer: D) Dune

Solution: Cedar opens earliest and Birch latest, so those take the first and fourth slots. Aster opens before Dune, which forces Aster into the second slot and Dune into the third. So Dune opens third. Aster is the second-month trap, Cedar is fixed first, and Birch is fixed last.

Q38. [DILR-4 · mcsc · diff 1]

Suppose only these workshop facts are kept: the counts are distinct whole numbers from 3 to 10 summing to 26, Aster hosts the most, Dune the fewest, and Birch hosts more than Cedar. The facts fixing Cedar at 6, making Aster odd and setting Aster two above Birch are dropped. How many different count-assignments to the four centres are then possible?

- A) 10
- B) 6
- C) 8 ✓
- D) 12

Answer: C) 8

Solution: With Aster the strict maximum, Dune the strict minimum, Birch above Cedar, distinct values 3 to 10, and a total of 26, a direct enumeration gives exactly 8 valid assignments. The 6 and 10 traps come from stopping the count early or double-counting a mirrored case, and 12 overcounts the middle pair.

Q39. [DILR-4 · mcsc · diff 1]

What is the combined number of workshops hosted by the centre that opens first and the centre that opens last?

- A) 16
- B) 11
- C) 15
- D) 13 ✓

Answer: D) 13

Solution: Cedar opens first with 6 workshops and Birch opens last with 7, so the combined total is 13. Using Aster instead of Cedar for the first slot gives the 15 trap, pairing Dune with Birch gives 11, and pairing Aster with Birch gives 16.

Instructions (DILR-5): study the shared data below and answer its 3 question(s).

A field consultant visits five clients, named Vernon, Wesley, Xavier, Yara and Zane, over a single working week. She visits exactly one client on each of the five days from Monday to Friday, and no client is visited more than once.

The following conditions all hold:

- Wesley is visited on Monday.
- Zane is visited on the day immediately after the day on which Xavier is visited.
- Xavier is visited on an earlier day than Yara.
- Vernon is visited on an earlier day than Xavier.

Q40. [DILR-5 · mcsc · diff 0.5]

Which client does the consultant visit on Friday?

- A) Yara ✓
- B) Zane
- C) Vernon
- D) Xavier

Answer: A) Yara

Solution: Wesley takes Monday. Vernon comes before Xavier, and Zane sits immediately after Xavier, so Vernon, Xavier and Zane fill Tuesday, Wednesday and Thursday in that order. That leaves Yara for Friday, matching the requirement that Xavier precedes Yara. Zane and Xavier fall on Thursday and Wednesday, and Vernon on Tuesday.

Q41. [DILR-5 · mcsc · diff 0.5]

On which day is Xavier visited?

- A) Tuesday
- B) Wednesday ✓
- C) Monday
- D) Thursday

Answer: B) Wednesday

Solution: With Wesley on Monday and Vernon before Xavier, Vernon must take Tuesday. Xavier then takes Wednesday, Zane the immediately following Thursday, and Yara Friday. So Xavier is visited on Wednesday. Tuesday belongs to Vernon, Thursday to Zane, and Monday to Wesley.

Q42. [DILR-5 · mcsc · diff 1]

Suppose the condition that Vernon is visited before Xavier is dropped, while the other three conditions still hold. How many different visiting orders for the week are then possible?

- A) 4
- B) 5
- C) 3 ✓
- D) 2

Answer: C) 3

Solution: Keeping Wesley on Monday, Zane immediately after Xavier, and Xavier before Yara, the Xavier-Zane block and the remaining two clients can be placed in exactly three valid ways. The 2 trap forgets one placement of Vernon, while 4 and 5 over-count arrangements that break the immediate-after or the Xavier-before-Yara rule.

Instructions (DILR-6): study the shared data below and answer its 4 question(s).

A ceramic tile maker ships four tile designs. The first chart shows boxes shipped (in thousands of boxes) and the second shows the price per box (in rupees) for three financial years FY22, FY23 and FY24. Revenue for any design in a year = boxes shipped (in '000) x price per box (in rupees), so all revenue figures are in thousands of rupees.

Q43. [DILR-6 · mcsc · diff 1]

What is the total revenue from all four designs in FY24?

- A) 44,300 thousand rupees
- B) 46,200 thousand rupees
- C) 45,710 thousand rupees ✓
- D) 43,950 thousand rupees

Answer: C) 45,710 thousand rupees

Solution: FY24 revenue per design is boxes ('000) times price: Slate $60 \times 220 = 13200$, Marble $45 \times 350 = 15750$, Terra $24 \times 240 = 5760$, Onyx $20 \times 550 = 11000$, summing to 45,710 thousand rupees. The distractors drop or mis-multiply one design.

Q44. [DILR-6 · mcsc · diff 1]

Which design recorded the highest percentage growth in revenue from FY22 to FY24?

- A) Terra
- B) Onyx ✓
- C) Marble
- D) Slate

Answer: B) Onyx

Solution: FY22 revenue is Slate 8000, Marble 9000, Terra 5000, Onyx 5000; FY24 is Slate 13200, Marble 15750, Terra 5760, Onyx 11000 (thousand rupees). Growth is Onyx +120%, Marble +75%, Slate +65%, Terra +15%, so Onyx leads. Judging by box counts alone still favours Onyx but hides the size of the gap.

Q45. [DILR-6 · mcsc · diff 1]

Over the three years, what is Slate's average price per box weighted by the number of boxes shipped?

- A) 215.00 rupees
- B) 210.00 rupees
- C) 211.33 rupees ✓
- D) 213.33 rupees

Answer: C) 211.33 rupees

Solution: Weighting Slate prices by boxes shipped gives $(40 \times 200 + 50 \times 210 + 60 \times 220) / (40 + 50 + 60) = 31700 / 150 = 211.33$ rupees. A plain average of 200, 210 and 220 ignores volumes and wrongly yields 210.

Q46. [DILR-6 · mcsc · diff 1]

By how much did the combined total revenue of all four designs change from FY22 to FY23?

- A) increased by 9,780 thousand rupees
- B) increased by 8,900 thousand rupees
- C) increased by 8,320 thousand rupees
- D) increased by 9,320 thousand rupees ✓

Answer: D) increased by 9,320 thousand rupees

Solution: FY22 combined revenue is $8000 + 9000 + 5000 + 5000 = 27000$; FY23 is $10500 + 11520 + 6500 + 7800 = 36320$ (thousand rupees). The rise is $36320 - 27000 = 9320$ thousand rupees. The distractors mishandle one design's price or volume.

Section: QA (suggested 40 min)

Q47. [QA-MCQ-1 · mcsc · diff 0.5]

A trader marks his goods 60% above cost price and then allows two successive discounts of 25% and 20% on the marked price. What is his overall profit or loss percentage?

- A) 4% loss ✓
- B) 6% loss
- C) 4% profit
- D) 2% loss

Answer: A) 4% loss

Solution: Take cost 100, so marked price is 160. After 25% off it is 120, and after a further 20% off it is 96. Selling at 96 against cost 100 is a 4% loss. The 4% profit trap forgets one discount, and the 6% figure comes from misapplying the second discount to cost.

Q48. [QA-MCQ-2 · mcsc · diff 0.5]

A invests Rs 16000 for 9 months, B invests Rs 24000 for 4 months and C invests Rs 12000 for 10 months in a business. If the total annual profit is Rs 9000, what is B's share?

- A) Rs 3600
- B) Rs 3000
- C) Rs 2400 ✓
- D) Rs 2000

Answer: C) Rs 2400

Solution: Capital-months are 144000, 96000 and 120000, which reduce to the ratio 6:4:5 summing to 15 parts. B holds 4 parts, so B earns 9000 times 4 over 15, equal to Rs 2400. The Rs 3000 trap splits equally, and Rs 3600 uses the wrong part count.

Q49. [QA-MCQ-3 · mcsc · diff 0.5]

A vessel holds 60 litres of a milk and water mixture in which milk and water are in the ratio 5:1. How much water must be added so that the ratio of milk to water becomes 5:2?

- A) 12 litres
- B) 10 litres ✓
- C) 15 litres
- D) 8 litres

Answer: B) 10 litres

Solution: The 60 litres contain 50 litres milk and 10 litres water. Milk stays fixed at 50, and for a 5:2 ratio the water needed is 50 times 2 over 5, which is 20 litres. Since 10 litres are already present, 10 more must be added. The 20 figure forgets the existing water.

Q50. [QA-MCQ-4 · mcsc · diff 0.5]

A boat covers 36 km downstream in 2 hours and the same 36 km upstream in 3 hours. What is the speed of the stream?

- A) 6 km/h
- B) 3 km/h ✓
- C) 2 km/h
- D) 15 km/h

Answer: B) 3 km/h

Solution: Downstream speed is 36 over 2, that is 18 km/h, and upstream speed is 36 over 3, that is 12 km/h. Stream speed is half the difference, so it equals half of 6, which is 3 km/h. The 15 value wrongly reports boat speed, and 6 forgets to halve the difference.

Q51. [QA-MCQ-5 · mcsc · diff 0.5]

A can finish a piece of work in 8 days and B can finish the same work in 12 days. They work on alternate days with A starting on the first day. In how many days will the work be completed?

- A) 9.5 days ✓
- B) 9 days
- C) 10 days
- D) 8.5 days

Answer: A) 9.5 days

Solution: With 24 total units, A does 3 units per day and B does 2 units per day. Over 9 alternating days starting with A the completed work is 23 units. One unit remains for B, who needs half a day at 2 units per day, giving 9.5 days in all. The 10 trap rounds up the final partial day.

Q52. [QA-MCQ-6 · mcsc · diff 0.5]

A sum of Rs 8000 is invested at 20% per annum compounded half-yearly for one year. What is the compound interest earned?

- A) Rs 1600
- B) Rs 1728
- C) Rs 1764
- D) Rs 1680 ✓

Answer: D) Rs 1680

Solution: Half-yearly the rate is 10% over two periods, so the amount is 8000 times 1.1 squared, that is 9680. The interest is 9680 minus 8000, which is Rs 1680. The Rs 1600 value uses simple interest, and Rs 1764 wrongly applies a 20% rate for two full periods.

Q53. [QA-MCQ-7 · mcsc · diff 0.5]

In how many ways can the letters of the word PENCIL be arranged so that the two vowels E and I are never together?

- A) 480 ✓
- B) 720
- C) 240
- D) 600

Answer: A) 480

Solution: All six distinct letters arrange in 720 ways. Treating the two vowels as one block gives 5 factorial times 2, that is 240 arrangements where they sit together. Subtracting leaves 720 minus 240, which is 480. The 240 value counts the together case instead of the apart case.

Q54. [QA-MCQ-8 · mcsc · diff 1]

A bag contains 5 red balls and 4 blue balls. Two balls are drawn at random without replacement. What is the probability that at least one of them is red?

A) $5/6$ ✓

B) $5/9$

C) $1/6$

D) $2/3$

Answer: A) $5/6$

Solution: The complement of at least one red is both balls blue, which has probability 6 over 36, that is $1/6$, using choose two from four over choose two from nine. Therefore at least one red is 1 minus $1/6$, which is $5/6$. The $1/6$ value reports the complement, and $5/9$ uses a single draw.

Q55. [QA-MCQ-9 · mcsc · diff 1]

A solid metallic sphere of radius 6 cm is melted and recast into small cones, each having base radius 2 cm and height 3 cm. How many such cones can be made?

A) 72 ✓

B) 36

C) 54

D) 108

Answer: A) 72

Solution: The sphere volume uses four-thirds times radius cubed, giving 288 times pi. Each cone volume uses one-third times base area times height, giving 4 times pi. Dividing 288 by 4 yields 72 cones. The 36 trap uses half the sphere volume, and 108 mishandles the cone factor.

Q56. [QA-MCQ-10 · mcsc · diff 1]

What is the remainder when 3 raised to the power 101 is divided by 100?

A) 9

B) 3 ✓

C) 1

D) 27

Answer: B) 3

Solution: Powers of 3 modulo 100 cycle with period 20, since 3 to the power 20 leaves remainder 1. As 101 leaves remainder 1 when divided by 20, the value equals 3 to the power 1 modulo 100, which is 3. The 1 trap forgets the leftover exponent, and 27 uses exponent 3.

Q57. [QA-MCQ-11 · mcsc · diff 1]

The sum of the first n terms of an arithmetic progression is given by $3n$ squared plus $5n$. What is the 12th term of this progression?

A) 68

B) 74 ✓

C) 80

D) 72

Answer: B) 74

Solution: The n th term equals the sum of n terms minus the sum of n minus one terms. Here that difference simplifies to $6n$ plus 2, so the 12th term is 72 plus 2, which is 74. The 72 trap drops the constant, and 80 mishandles the sum-difference step.

Q58. [QA-MCQ-12 · mcsc · diff 1]

If $a + b + c = 0$ and $a^2 + b^2 + c^2 = 14$, what is the value of $a^2b + b^2c + c^2a$?

A) 196

B) 49

C) 98 ✓

D) 72

Answer: C) 98

Solution: With $a+b+c=0$, we get $ab+bc+ca = -(a^2+b^2+c^2)/2 = -7$, so $(ab+bc+ca)^2$ gives $a^2b^2+b^2c^2+c^2a^2 = 49$ (the abc *sum term vanishes). Then $a^2b^2+b^2c^2+c^2a^2 = (a^2+b^2+c^2)^2 - 2(a^2b^2+b^2c^2+c^2a^2) = 196 - 98 = 98$. The 196 value forgets to subtract the cross terms, 49 stops at the cross-term sum, and 72 mishandles the squaring.

Q59. [QA-MCQ-13 · mcsc · diff 1]

For $x > 0$, what is the minimum value of $x^2 + 250/x$?

- A) 100
- B) 70
- C) 75 ✓
- D) 80

Answer: C) 75

Solution: Split the second term to apply AM-GM on three parts: $x^2 + 125/x + 125/x$ is at least 3 times the cube root of $(x^2 * 125/x * 125/x) = 3 * \text{cube root of } 15625 = 3 * 25 = 75$, with equality when $x^3 = 125$, so $x = 5$. Splitting into only two equal pieces gives the wrong 100 figure.

Q60. [QA-MCQ-14 · mcsc · diff 1]

What is the value of the product $(\log \text{ base 2 of 3})(\log \text{ base 3 of 4})(\log \text{ base 4 of 5}) \dots (\log \text{ base 63 of 64})$?

- A) 5
- B) 8
- C) 32
- D) 6 ✓

Answer: D) 6

Solution: Writing each factor as $\ln$ of the top over $\ln$ of the base makes the chain telescope, leaving $\ln 64$ divided by $\ln 2$, which equals $\log \text{ base 2 of } 64 = 6$ since $2^6 = 64$. The 32 trap divides 64 by 2, and 5 miscounts the exponent.

Q61. [QA-MCQ-15 · mcsc · diff 1]

Points A(2, 3) and B(8, 5) lie in the coordinate plane. A point P is chosen on the x-axis. What is the least possible value of $AP + PB$?

- A) 12
- B) $\sqrt{52}$
- C) $\sqrt{68}$
- D) 10 ✓

Answer: D) 10

Solution: Reflect A across the x-axis to A'(2, -3); then AP equals A'P, so $AP + PB$ is least when A', P, B are collinear, giving the straight distance $A'B = \sqrt{(8-2)^2 + (5+3)^2} = \sqrt{36+64} = 10$. Using the direct distance AB gives the $\sqrt{52}$ trap.

Q62. [QA-MCQ-16 · mcsc · diff 1]

Among 100 employees, 40 use Python, 48 use R, and 45 use SQL. Also 15 use both Python and R, 18 use both R and SQL, 12 use both Python and SQL, and 8 use all three tools. How many employees use exactly one of the three tools?

- A) 67 ✓
- B) 96
- C) 75
- D) 59

Answer: A) 67

Solution: The number in at least one is $40+48+45 - 15-18-12 + 8 = 96$. Those in exactly two pairs number $(15+18+12) - 3*8 = 21$, and 8 use all three, so exactly one is $96 - 21 - 8 = 67$. The 96 trap stops at the union total, and 59 subtracts the pairwise counts without adding back the triple.

Q63. [QA-MCQ-17 · mcsc · diff 1]

What is the sum $1/(1*3) + 1/(3*5) + 1/(5*7) + \dots + 1/(19*21)$?

- A) 9/19
- B) 10/21 ✓
- C) 20/21
- D) 11/23

Answer: B) 10/21

Solution: Each term $1/((2k-1)(2k+1))$ equals one half times $(1/(2k-1) - 1/(2k+1))$, so the ten terms telescope to one half times $(1 - 1/21) = \text{one half times } 20/21 = 10/21$. Forgetting the factor of one half gives the 20/21 trap.

Q64. [QA-TITA-1 · tita · diff 0.5]

Find the sum of all positive integer divisors of 360 (including 1 and 360 themselves).

Answer (TITA): 1170

Solution: Write 360 as 2^3 times 3^2 times 5. The sum of divisors is multiplicative, so it equals $(1+2+4+8)$ times $(1+3+9)$ times $(1+5)$, that is 15 times 13 times 6, which is 1170. A common slip is to forget the constant term 1 in one bracket, giving 14 times 13 times 6 or 1092, or to stop at 2^2 and use $(1+2+4)$ which yields 7 times 13 times 6 or 546. The full factorisation gives 1170.

Q65. [QA-TITA-2 · tita · diff 0.5]

An arithmetic progression has first term 7 and common difference 4. Find the sum of its first 30 terms.

Answer (TITA): 1950

Solution: The sum of the first n terms is $(n/2)$ times $(2a + (n-1)d)$. With a equal to 7, d equal to 4 and n equal to 30, this is 15 times $(14 + 29 \text{ times } 4)$, that is 15 times $(14 + 116)$, which is 15 times 130, giving 1950. Using $(n-1)$ as 30 instead of 29 gives 15 times 134 or 2010, and forgetting to double a gives 15 times $(7 + 116)$ or 1845. The correct sum is 1950.

Q66. [QA-TITA-3 · tita · diff 1]

What is the remainder when 17 raised to the power 256 is divided by 100? (In other words, give the last two digits of 17^{256} .)

Answer (TITA): 81

Solution: The last two digits of powers of 17 repeat with a cycle length of 20, since 17^{20} ends in 01 modulo 100. Reducing the exponent, 256 leaves remainder 16 when divided by 20, so 17^{256} has the same last two digits as 17^{16} . Squaring step by step modulo 100 gives 17^2 as 89, 17^4 as 21, 17^8 as 41, and 17^{16} as 81. Treating the cycle length as 4 (as if it were 7) mistakes it, and assuming 256 is a multiple of the period wrongly yields 01. The answer is 81.

Q67. [QA-TITA-4 · tita · diff 1]

In a group of 200 people, 100 use app A, 90 use app B and 80 use app C. Exactly 40 people use all three apps and exactly 20 people use none of the three. How many people use exactly two of the three apps?

Answer (TITA): 10

Solution: The number using at least one app is 200 minus 20, that is 180. By inclusion and exclusion, at least one equals the sum of singles minus the sum of the three pairwise overlaps plus the triple overlap, so 180 equals 270 minus (sum of pairwise) plus 40, giving the sum of pairwise overlaps as 130. Each pairwise overlap counts the all-three group as well, so the sum of pairwise overlaps equals (exactly two) plus three times 40. Thus exactly two equals 130 minus 120, which is 10. Forgetting to subtract the triple three times leaves 130, and ignoring the 20 with none gives a different single total. The answer is 10.

Q68. [QA-TITA-5 · tita · diff 1]

A positive integer 30 is to be written as a sum of one or more positive integers. Over all such ways, what is the maximum possible value of the product of those parts?

Answer (TITA): 59049

Solution: To maximise the product for a fixed sum, break the number into parts of size 3 as much as possible, avoiding parts of 1, and use a 2 only when the leftover after taking threes is 1 (by combining 3 plus 1 into 2 plus 2). Here 30 is exactly ten 3s with no remainder, so the best product is 3 raised to the power 10, which is 59049. Using all 2s gives 2 raised to the power 15 or 32768, which is smaller, and mixing in a part of 1 only lowers the product. The maximum is 59049.